

Statistics, Inference, Information

A collection of definitions, theorems, and formulae

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Part I Statistics and data analysis

1 Fundamentals

1.1 Numerical summaries of data

Given data $x_1, \dots, x_n \in \mathbb{R}$,

- The sample mean is

$$\bar{x} := \frac{1}{n} \sum_{i=1}^n x_i$$

- The (biased) sample variance is

$$\sigma_{\text{bias}}^2 := \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

- The (unbiased) sample variance is

$$\sigma_{\text{non-bias}}^2 := \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

- The sample median x_{median} depends on if n is even or odd

1.2 Empirical covariance of vectored data

Let $X \in \mathbb{R}^{n \times d}$ be a data matrix whose i th row $x_i^\top$ is an observation in $\mathbb{R}^d$. Define the sample mean

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \in \mathbb{R}^d, \quad \text{and the centering operator } H = I_n - \frac{1}{n} \mathbf{1}\mathbf{1}^\top,$$

therefore the centered data matrix is

$$\tilde{X} = HX \in \mathbb{R}^{n \times d}, \quad \text{whose rows are } (x_i - \bar{x})^\top.$$

If $x \in \mathbb{R}^d$ is a zero-mean random vector, then its population covariance is $\Sigma_X = \mathbb{E}[xx^\top]$ (if $x \in \mathbb{R}^d$ is not zero mean, this is simply the second moment).

From data iid centered data, the (biased) sample covariance is

$$S_{\text{bias}} = \frac{1}{n} \tilde{X}^\top \tilde{X} \in \mathbb{R}^{d \times d}, \quad \text{which satisfies} \quad \mathbb{E}[S_{\text{bias}}] = \frac{n-1}{n} \Sigma_X.$$

The unbiased version is

$$S_{\text{unbiased}} = \frac{1}{n-1} \tilde{X}^\top \tilde{X}, \quad \text{which satisfies} \quad \mathbb{E}[S] = \Sigma_X.$$

The scalar factor between S_{unbiased} and S_{bias} does not affect eigenvectors and only rescales eigenvalues.

1.3 Empirical pmf $\hat{\mathbb{P}}$ and cdf $\hat{\mathbb{F}}$

Given a collection of data $x_1, \dots, x_n$, the empirical distribution assigns equal mass to each observation:

$$\hat{\mathbb{P}}(x = x) = \frac{1}{n} \sum_{i=1}^n \delta(x_i = x)$$

Under this definition, we have that for any measurable set $\mathcal{A}$

$$\hat{\mathbb{P}}(\mathcal{A}) = \frac{1}{n} \# \{i : x_i \in \mathcal{A}\}.$$

The empirical cdf $\hat{\mathbb{F}}$ is the distribution function associated with $\hat{\mathbb{P}}$, namely

$$\hat{\mathbb{F}}(\alpha) := \frac{1}{n} \# \{i : x_i \leq \alpha\},$$

which is a right-continuous step function with jumps of size $1/n$ and values $\hat{\mathbb{F}} \in [0, 1]$.

1.4 Law of Large Numbers and Continuous Mapping Theorem

Let $x_1, x_2, \dots$ be i.i.d. with $\mathbb{E}[x_i] = \mu$, and define the sample mean

$$\bar{x}_n := \frac{1}{n} \sum_{i=1}^n x_i \rightarrow \mu$$

then the *Law of Large Numbers* states that

$$\bar{x}_n \xrightarrow[n \rightarrow \infty]{(\cdot)} \mu$$

in probability (weak LLN) and almost surely (strong LLN). More generally, for any integrable function $g(\cdot)$, the Continuous mapping theorem states that

$$\frac{1}{n} \sum_{i=1}^n g(x_i) \xrightarrow[n \rightarrow \infty]{(\cdot)} \mathbb{E}[g(x)].$$

By this result, the LLN has an impact at the level of distributions, namely

$$\hat{\mathbb{P}} \rightarrow \mathbb{P} \quad \text{as } n \rightarrow \infty,$$

and consequently $\hat{\mathbb{F}} \rightarrow \mathbb{F}$. Here $\mathbb{P}$ is the true (possibly continuous) distribution, whereas $\hat{\mathbb{P}}$ is discrete by construction.

1.5 Central Limit Theorem

Let $x_1, x_2, \dots$ be i.i.d. with mean μ and variance $\sigma^2 < \infty$. Then the Central Limit Theorem says that

$$\sqrt{n} \frac{\bar{x}_n - \mu}{\sigma} \xrightarrow[n \rightarrow \infty]{(d)} \mathcal{N}(0, 1),$$

In particular, the CLT says that at higher values of n , the random variable $\bar{x}_n$ behaves according to a standard normal distribution.

1.6 In what sense do we have $\hat{\mathbb{P}} \rightarrow \mathbb{P}$?

In what sense does the empirical distribution $\hat{\mathbb{P}}$ converge to the true distribution $\mathbb{P}$? Let $x_1, \dots, x_n \sim \mathbb{P}$ i.i.d. with the associated CDF $\mathbb{F}(x) := \mathbb{P}(x \leq x)$. Define the empirical PDF and empirical CDF using our data as

$$\hat{\mathbb{P}}_n(x = x) := \frac{1}{n} \sum_{i=1}^n \delta_{x_i=x}, \quad \hat{\mathbb{F}}_n(x) := \hat{\mathbb{P}}_n(x \in (-\infty, x]) = \frac{1}{n} \sum_{i=1}^n 1\{x_i \leq x\},$$

To answer the question of convergence of distributions, first define the distance between distributions using the Kolmogorov–Smirnov statistic (goodness-of-fit distance):

$$\begin{aligned} D_n &= \|\hat{\mathbb{F}}_n - \mathbb{F}\|_\infty \\ &= \sup_{x \in \mathbb{R}} |\hat{\mathbb{F}}_n(x) - \mathbb{F}(x)| \end{aligned}$$

A standard notion of convergence uses the Glivenko–Cantelli theorem, which says that

$$\|\hat{\mathbb{F}}_n - \mathbb{F}\|_\infty \xrightarrow[n \rightarrow \infty]{a.s.} 0$$

If we want a measure of convergence for a finite sample, the DKW inequality (finite-sample uniform deviation bound) says for all $\varepsilon > 0$,

$$\mathbb{P}\left(\|\hat{\mathbb{F}}_n - \mathbb{F}\|_\infty > \varepsilon\right) \leq 2e^{-2n\varepsilon^2}.$$

1.7 Empirical cross-entropy = negative log-likelihood

Let $\{(x_i, y_i)\}_{i=1}^n$ be data, where $x_i \in \mathcal{X}$ and $y_i \in \mathcal{Y}$.

- Let $p_\theta(y \mid x)$ be our parametric conditional model with parameter θ that we seek to optimize for θ .
- Let $\hat{p}_n(y \mid x)$ denote the empirical conditional distribution induced by the sample (i.e. it puts mass $1/n$ on each observed pair (x_i, y_i)).

Define the empirical cross-entropy between the empirical distribution $\hat{p}_n$ relative to p_θ by

$$\begin{aligned} H(\hat{p}_n, p_\theta^*) &:= \mathbb{E}_{(x,y) \sim \hat{p}_n} [-\log p_\theta^*(y \mid x)] \\ &= \sum_{i=1}^n \hat{p}_n(y_i \mid x_i) (-\log p_\theta(y_i \mid x_i)) \\ &= \sum_{i=1}^n \frac{1}{n} (-\log p_\theta(y_i \mid x_i)) \end{aligned}$$

Define the conditional log-likelihood by

$$\ell(\theta) := \sum_{i=1}^n \log p_\theta(y_i \mid x_i).$$

From the definitions above, it follows that

$$H(\hat{p}_n, p_\theta) = -\frac{1}{n} \ell(\theta),$$

and so

$$\arg \min_{\theta} H(\hat{p}_n, p_\theta) = \arg \max_{\theta} \ell(\theta).$$

In the above setting, we can interpret $-\log p_\theta(y_i | x_i)$ as the surprise assigned by our model p_θ to the event $(y_i | x_i)$. Therefore, the empirical cross entropy $H(\hat{p}_n, p_\theta^*)$ is the average surprisal of our model on the observed data.

In the idealized setting, if there is a true conditional distribution $p^*(y | x)$ and the expectation is taken under p^* , then the cross entropy between p^* and p_θ can be written as

$$H(p^*, p_\theta) := \mathbb{E}_{(x,y) \sim p^*} [-\log p_\theta(y | x)] = H(p^*) + \text{KL}(p^* \| p_\theta),$$

where the entropy of the true model p^* and the KL divergence are defined as

$$H(p^*) := \mathbb{E}_{(x,y) \sim p^*} [-\log p^*(y | x)], \quad \text{KL}(p^* \| p_\theta) := \mathbb{E}_{p^*} \left[\log \frac{p^*(y | x)}{p_\theta(y | x)} \right].$$

As a consequence of this definition, minimizing the cross entropy is equivalent to minimizing the KL between our model p_θ and the true p^* :

$$\min_{\theta} H(p^*, p_\theta) \iff \min_{\theta} \text{KL}(p^* \| p_\theta)$$

and

$$H(p^*, p_\theta) \geq H(p^*)$$

with equality if and only if $\text{KL}(p^* \| p_\theta) = 0$, hence if and only if $p_\theta(y | x) = p^*(y | x)$.

2 Function and parameter estimation using data

In this section, we describe how to compute a function of our distribution using the plug-in method, then how to estimate the parameter using feature matching and maximum likelihood estimation.

2.1 Feature of a distribution

A feature is a functional of a distribution:

$$h = h(\mathbb{P}),$$

- expectation $h(\mathbb{P}) = \mathbb{E}_{\mathbb{P}}[x]$
- variance $h(\mathbb{P}) = \text{Var}_{\mathbb{P}}(x)$
- quantile $h(\mathbb{P}) = \mathbb{F}^{-1}(0.75)$
- tail probability $h(\mathbb{P}) = \mathbb{P}(x \geq 1/2)$

2.2 Plug in estimators $\hat{h} = h(\hat{\mathbb{P}})$ to compute functions $h(\mathbb{P})$

Suppose we have a random variable $x \sim \mathbb{P}$ (non-parametric) and a feature $h = h(\mathbb{P})$. The plug-in method allows us to compute an estimate of this function $\hat{h}$ by replacing the true distribution $\mathbb{P}$ by the empirical distribution $\hat{\mathbb{P}}$, so that the estimate is $\hat{h} = h(\hat{\mathbb{P}})$. In particular

$$h(\mathbb{P}) := \mathbb{E}[g(x)] \quad \rightsquigarrow \quad \hat{h} = h(\hat{\mathbb{P}}) = \sum_{i=1}^n \hat{\mathbb{P}}(x = x_i) g(x_i).$$

For example,

$$h(\mathbb{P}) = \mathbb{E}[x^3] \quad \rightsquigarrow \quad \hat{h} = h(\hat{\mathbb{P}}) = \sum_{i=1}^n \hat{\mathbb{P}}(x = x_i) x_i^3$$

2.3 Feature matching to estimate θ

Suppose we are given data $x_1, \dots, x_n \sim \mathbb{P}^\theta$. Feature matching is a way to estimate the true parameter θ . The main idea is that since the model is parametric, $\{\mathbb{P}^\theta : \theta \in \Theta\}$, then the feature $h(\mathbb{P})$ becomes a function of the parameter θ underlying the distribution

$$h^\theta := h(\mathbb{P}^\theta).$$

The main method is:

1. From data we compute the empirical feature

$$\hat{h} := h(\hat{\mathbb{P}}_n),$$

2. Find analytically h^θ as a function of θ
3. Then estimate the distribution parameter θ by solving

$$h^\theta = \hat{h}.$$

to obtain the value of θ which produces a distribution that has the same feature h^θ we observe in our data $\hat{h}$.

A special case of feature matching is the method of moments, where we set up an system of independant equations to facilitate parameter estimation.

2.4 Method of moments on a Gaussian mixture

The method of moments is a way to do parameter estimation by setting up a system of equations involving the moments of the distribution. Consider the mixture model defined by

$$\mathbf{x} = z \mathbf{x}_1 + (1 - z) \mathbf{x}_2,$$

where $z \sim \text{Bern}(p)$ and, conditionally on z , we draw

$$\mathbf{x}_1 \sim \mathcal{N}(\mu_1, \sigma_1^2), \quad \mathbf{x}_2 \sim \mathcal{N}(\mu_2, \sigma_2^2),$$

with $z, \mathbf{x}_1, \mathbf{x}_2$ independent. Equivalently, $\mathbf{x}$ is drawn from $\mathcal{N}(\mu_1, \sigma_1^2)$ with probability p and from $\mathcal{N}(\mu_2, \sigma_2^2)$ with probability $(1 - p)$. For any integer $k \geq 1$, exchanging integrals gives the mixture-moment identity

$$\mathbb{E}[\mathbf{x}^k] = p \mathbb{E}[\mathbf{x}_1^k] + (1 - p) \mathbb{E}[\mathbf{x}_2^k].$$

If the parameters are unknown (e.g. $\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, p$), one can use the empirical PDF $\hat{\mathbb{P}}$ to estimate $\mathbb{E}[\mathbf{x}], \mathbb{E}[\mathbf{x}^2], \dots$ and thus obtain a system of equations in the unknown parameters.

2.5 Maximum likelihood

Assume a parametric model $\mathbf{x}_1, \dots, \mathbf{x}_n \sim \mathbb{P}^\theta$ i.i.d. with density or pmf $p_\theta(\cdot)$. Given observed data $\vec{x} = (x_1, \dots, x_n)$, define the likelihood

$$L(\theta; \vec{x}) := p_\theta(\vec{x}) = \prod_{i=1}^n p_\theta(x_i),$$

and the log-likelihood

$$\ell(\theta; \vec{x}) := \log L(\theta; \vec{x}) = \sum_{i=1}^n \log p_\theta(x_i).$$

The maximum likelihood estimator (MLE) is

$$\hat{\theta}_{\text{MLE}} \in \arg \max_{\theta \in \Theta} L(\theta; \vec{x}) \iff \hat{\theta}_{\text{MLE}} \in \arg \max_{\theta \in \Theta} \ell(\theta; \vec{x}).$$

Interpretation: $L(\theta; \vec{x})$ is the probability assigned by the model $\mathbb{P}^\theta$ to the observed sample; MLE chooses the parameter that makes the observed data most likely. To actually compute the MLE, we can sometimes solve first-order conditions

$$\nabla_\theta \ell(\theta; \vec{x}) = 0$$

3 Analysis of estimators

In the previous section, we discussed two methods for parameter estimation: method of moments and maximum likelihood. In this section, we consider a general estimation algorithm and model it as a random variable in an effort to answer questions regarding the properties of the estimator (such as its mean and variance) and questions regarding the level of confidence we should have in the estimator.

3.1 Estimators as random variables under the sampling distribution

An *estimation algorithm* $g(\cdot) : \mathcal{X}^n \rightarrow \Theta$, is a deterministic function which maps data $x^n = (x_1, \dots, x_n)$ to an estimate

$$\hat{\theta} = g(x^n).$$

Here $\hat{\theta}$ is a *number* because the input x^n is a realized dataset. To study the behavior of the estimator, we model the data as random:

$$x^n = (x_1, \dots, x_n) \sim \mathbb{P}^\theta.$$

Since the data is now modeled as random, the estimator becomes a random variable too:

$$\hat{\theta} = g(x^n).$$

The *sampling distribution* is the law of $\hat{\theta}$ when $x^n \sim \mathbb{P}^\theta$. We may denote this distribution by $\mathbb{Q}^\theta$, i.e.

$$\hat{\theta} \sim \mathbb{Q}^\theta \quad \text{when} \quad x^n \sim \mathbb{P}^\theta.$$

3.2 Explicit sampling distribution example (sum of Bernoulli $\rightarrow$ Binomial)

Under our assumption that the data $x \sim \mathbb{P}^\theta$, sometimes we can explicitly calculate the sampling distribution $\mathbb{Q}^\theta$ by analyzing the estimator $\hat{\theta} \sim \mathbb{Q}^\theta$. For example, let $x_i \stackrel{\text{iid}}{\sim} \text{Bern}(\theta)$ and define

$$g(x^n) = \frac{1}{n} \sum_{i=1}^n x_i.$$

Since the draws are i.i.d. Bernoulli, the sum is Binomial:

$$\sum_{i=1}^n x_i \sim \text{Bin}(n, \theta).$$

Hence the estimator

$$\hat{\theta} = g(x^n) = \frac{1}{n} \sum_{i=1}^n x_i$$

takes values in $\{0, \frac{1}{n}, \dots, 1\}$, and its sampling distribution is

$$\mathbb{P}\left(\hat{\theta} = \frac{k}{n}\right) = \binom{n}{k} \theta^k (1 - \theta)^{n-k}, \quad k = 0, 1, \dots, n.$$

Equivalently, $n\hat{\theta} \sim \text{Bin}(n, \theta)$.

3.3 Variance of an estimator

Recall that $\hat{\theta} = g(x^n) \sim \mathbb{Q}^\theta$ when $x^n \sim \mathbb{P}^\theta$. So under $\mathbb{P}^\theta$, the variance of the estimator $\hat{\theta}$ is

$$\text{Var}_{\mathbb{P}^\theta}(\hat{\theta}) = \mathbb{E}_{\mathbb{P}^\theta} \left[\left(\hat{\theta} - \mathbb{E}_{\mathbb{P}^\theta}[\hat{\theta}] \right)^2 \right] = v^\theta$$

3.4 Bias and MSE, and what b^θ means

Define the *bias* (at parameter value θ) by

$$b^\theta = \mathbb{E}_{\mathbb{P}^\theta}[\hat{\theta}] - \theta.$$

Note the types:

$$\hat{\theta} \text{ is random, } \quad \mathbb{E}_{\mathbb{P}^\theta}[\hat{\theta}] \text{ is deterministic, } \quad b^\theta \text{ is deterministic.}$$

The *mean squared error* (MSE) is

$$\text{MSE}^\theta = \mathbb{E}_{\mathbb{P}^\theta} \left[(\hat{\theta} - \theta)^2 \right].$$

Expanding $\hat{\theta} - \theta = (\hat{\theta} - \mathbb{E}[\hat{\theta}]) + (\mathbb{E}[\hat{\theta}] - \theta)$ yields the bias–variance decomposition

$$\text{MSE}^\theta = \underbrace{(b^\theta)^2}_{\text{squared bias}} + \underbrace{v^\theta}_{\text{variance}}.$$

This highlights a common tradeoff: procedures with smaller variance may have larger bias (and vice versa), and MSE captures both effects.

3.5 Parametric bootstrap sampling and re-estimation

We’ve established that when we treat the data as random $x \sim \mathbb{P}^\theta$ for some parametric model $\{\mathbb{P}^\theta : \theta \in \Theta\}$, the estimator $\hat{\theta} = g(x^n)$ is a random variable $\hat{\theta} \sim \mathbb{Q}^\theta$. Our goal now is to approximate this estimator distribution $\mathbb{Q}^\theta$.

Suppose we observe data $x^n = (x_1, \dots, x_n)$ and compute an estimate via an algorithm $g(\cdot)$:

$$\hat{\theta} = g(x^n).$$

Using this estimate, we then form the estimated distribution $\mathbb{P}^{\hat{\theta}}$. The idea of bootstrap is to then sample from $\mathbb{P}^{\hat{\theta}}$, obtain a new parameter estimate, then repeat:

$$x^n \xrightarrow{\text{use } g(\cdot)} \hat{\theta} \xrightarrow{\text{plug in}} \mathbb{P}^{\hat{\theta}} \xrightarrow{\text{resample}} x^{n,(b)} \xrightarrow{\text{use } g(\cdot)} \hat{\theta}^{(b)}.$$

In practice, we collect $\{\hat{\theta}^{(b)}\}_{b=1}^B$ and form a histogram or empirical CDF as an approximation of the sampling distribution $\mathbb{Q}^\theta$. From this estimate of the sampling distribution we can further estimate quantities such as variance, standard error, and confidence intervals.

3.6 Non-parametric bootstrap sampling and re-estimation

In non-parametric bootstrap, we do *not* assume a parametric model $\mathbb{P}^\theta$. To approximate the law of some function/estimator, we approximate the unknown data-generating law of $x \sim \mathbb{P}$ by

the empirical distribution

$$\hat{\mathbb{P}}_n := \frac{1}{n} \sum_{i=1}^n \delta_{x_i},$$

where δ_{x_i} is a point mass at x_i . Recall in the parametric setup we approximated $\mathbf{x} \sim \mathbb{P}^\theta$ using the distribution $\mathbb{P}^{\hat{\theta}}$.

Given observed data $x^n = (x_1, \dots, x_n)$ and an estimator computed by an algorithm $g(\cdot)$,

$$\hat{\theta} = g(x^n),$$

the bootstrap procedure resamples *from the data itself* according to $\hat{\mathbb{P}}_n$:

$$x^n \xrightarrow{\text{form } \hat{\mathbb{P}}_n} \hat{\mathbb{P}}_n \xrightarrow{\text{resample}} x^{n,(b)} \xrightarrow{\text{use } g(\cdot)} \hat{\theta}^{(b)} = g(x^{n,(b)}).$$

Concretely, we form the empirical distribution $\hat{\mathbb{P}}_n$ once, then for each $b = 1, \dots, B$:

1. Sample $x^{n,(b)} = (x_1^{(b)}, \dots, x_n^{(b)})$ i.i.d. according to $\hat{\mathbb{P}}_n$ (equivalently: sample with replacement from $\{x_1, \dots, x_n\}$).
2. Recompute the estimator $\hat{\theta}^{(b)} = g(x^{n,(b)})$.

We then use the empirical distribution of $\{\hat{\theta}^{(b)}\}_{b=1}^B$ (histogram or empirical CDF) as an approximation to the sampling distribution $\mathbb{Q}^\theta$ of $\hat{\theta} = g(x^n)$, and use it to estimate variance, standard error, and confidence intervals of this estimator $\hat{\theta} = g(x^n)$.

4 Frequentist confidence intervals

4.1 A confidence interval is defined as a random interval

Fix a parametric model $\{\mathbb{P}^\theta : \theta \in \Theta\}$ and data $x_i \sim \mathbb{P}^\theta$ for $i = 1, \dots, n$. In the frequentist setting, the *parameter* θ is a fixed (unknown) constant and the *data* $x^n = (x_1, \dots, x_n)$ is random. Therefore, any estimator

$$\hat{\theta} = g(x^n)$$

is a random variable with sampling distribution $\hat{\theta} \sim \mathbb{Q}^\theta$ (the law induced by $\mathbb{P}^\theta$ through g).

A $(1 - \alpha)$ confidence interval is a *random interval*

$$[\hat{\theta}^-, \hat{\theta}^+],$$

where the endpoints $\hat{\theta}^-, \hat{\theta}^+$ are random variables (functions of x^n), such that the *coverage* property holds:

$$\mathbb{P}^\theta(\hat{\theta}^- \leq \theta \leq \hat{\theta}^+) \geq 1 - \alpha, \quad \text{for all } \theta \in \Theta.$$

Interpretation: if we repeatedly sample a new dataset $x^n \sim \mathbb{P}^\theta$ (with the same fixed θ) and recompute the interval each time, then the random interval will contain the fixed truth θ at least a fraction $1 - \alpha$ of the time. It is *not* correct to say “ θ has probability $1 - \alpha$ of lying in this fixed interval.”

4.2 Designing a confidence interval of an estimator: CLT case

To construct a confidence interval, use the CLT to say that for large n ,

$$\frac{\hat{\theta} - \theta}{\text{se}_n} \approx \mathcal{N}(0, 1),$$

where se_n is the (possibly estimated) standard error of $\hat{\theta}$. Then letting $z_{1-\alpha/2}$ be the $(1 - \alpha/2)$ quantile of $\mathcal{N}(0, 1)$, we get that

$$\mathbb{P}^\theta(|\hat{\theta} - \theta| \leq z_{1-\alpha/2} \text{se}_n) \approx 1 - \alpha,$$

which is equivalent to the (approximate) $(1 - \alpha)$ CI

$$\left[\hat{\theta} - z_{1-\alpha/2} \text{se}_n, \hat{\theta} + z_{1-\alpha/2} \text{se}_n \right].$$

For example, when $\alpha = 0.05$, $z_{0.975} \approx 1.96$.

4.3 Using bootstrap to directly produce confidence intervals of an estimator

Designing confidence intervals as above requires knowing the standard deviation of the estimator $\hat{\theta} \sim \mathbb{Q}^\theta$. A different approach to produce confidence intervals is to use bootstrap to directly calculate an empirical distribution of $\hat{\theta}$ then use the quantiles. In particular:

- *Parametric bootstrap:*

$$x^n \xrightarrow{g(\cdot)} \hat{\theta} = g(x^n) \implies \text{sample } x^{n,(b)} \sim \mathbb{P}^{\hat{\theta}} \implies \hat{\theta}^{(b)} = g(x^{n,(b)}).$$

Then use the empirical quantiles of $\{\hat{\theta}^{(b)}\}_{b=1}^B$ to approximate quantiles of $\hat{\theta}$.

- *Non-parametric bootstrap:* First define the empirical distribution $\hat{\mathbb{P}}_n = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$ then for iteration $b = 1, \dots, B$

$$\text{resample } x^{n,(b)} \sim \hat{\mathbb{P}}_n \implies \hat{\theta}^{(b)} = g(x^{n,(b)}).$$

e.g. $x^{n,(b)} \sim \hat{\mathbb{P}}_n$ means we resample with replacement from $\{x_i\}$.

The output of either case is an empirical distribution of $\hat{\theta}$. We then use this to construct the *percentile interval*: if $\hat{q}_{\alpha/2}$ and $\hat{q}_{1-\alpha/2}$ are the empirical quantiles of $\{\hat{\theta}^{(b)}\}_{b=1}^B$, then take

$$\left[\hat{q}_{\alpha/2}, \hat{q}_{1-\alpha/2} \right].$$

5 Principal component analysis

5.1 Form the empirical covariance matrix

Given data points $x_1, \dots, x_n \in \mathbb{R}^d$, define the sample mean

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

and form the (empirical) covariance matrix

$$S = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})^\top \in \mathbb{R}^{d \times d}$$

Then compute the eigenvalues/eigenvectors of S :

$$Sv_j = \lambda_j v_j, \quad \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d,$$

where $v_1, \dots, v_d$ are orthonormal. Since this S is symmetric and positive semidefinite, its SVD coincides with its eigen-decomposition, since for a symmetric PSD matrix like S , the singular values σ_j equal the eigenvalues λ_j . Write the SVD as

$$S = U\Sigma U^\top,$$

where $U \in \mathbb{R}^{d \times d}$ is orthonormal ($U^\top U = I$) and

$$\Sigma = \text{diag}(\sigma_1, \dots, \sigma_d), \quad \sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_d \geq 0.$$

Writing u_j for the j th column of U , we can equivalently write that

$$Su_j = \sigma_j u_j, \quad j = 1, \dots, d,$$

and

$$S = \sum_{j=1}^d \sigma_j u_j u_j^\top.$$

5.2 k -dimensional representation of x_i

The k -dimensional representation of a data point x_i is the vector of projections onto the top- k eigen-directions $v_1, \dots, v_k$:

$$((x_i - \bar{x})^\top v_1, (x_i - \bar{x})^\top v_2, \dots, (x_i - \bar{x})^\top v_k) \in \mathbb{R}^k$$

More broadly, think of PCA as mapping each point x_i to a coordinate vector

$$x_1 \mapsto (\alpha_{11}, \dots, \alpha_{1k}), \quad x_2 \mapsto (\alpha_{21}, \dots, \alpha_{2k}), \quad \dots, \quad x_n \mapsto (\alpha_{n1}, \dots, \alpha_{nk}),$$

where these coefficients

$$\alpha_{ij} = (x_i - \bar{x})^\top v_j, \quad i \in [n], \quad j \in [k].$$

Using these coefficients, a rank- k approximation of x_i is

$$x_i \approx \bar{x} + \alpha_{i1}v_1 + \dots + \alpha_{ik}v_k.$$

5.3 Rank- r truncation of S and Frobenius error

Keeping the top k components gives the rank- k approximation

$$S_k = \sum_{\ell=1}^r \lambda_{\ell} v_{\ell} v_{\ell}^{\top}$$

then

$$S - S_k = \sum_{k=r+1}^d \lambda_k v_k v_k^{\top}.$$

Using $\|v_k v_k^{\top}\|_F^2 = 1$ and orthogonality across k , we obtain

$$\|S - S_k\|_F^2 = \sum_{\ell=k+1}^d \lambda_{\ell}^2, \quad \|S\|_F^2 = \sum_{\ell=1}^d \lambda_{\ell}^2,$$

and hence the relative (squared) Frobenius error is

$$\frac{\|S - S_r\|_F^2}{\|S\|_F^2} = \frac{\sum_{k=r+1}^d \lambda_k^2}{\sum_{k=1}^d \lambda_k^2}.$$

6 Akaike model selection

6.1 Model fit vs number of parameters

Suppose we are comparing two parametric models, e.g. Model 1 has p_1 parameters and Model 2 has p_2 parameters, with $p_2 > p_1$. Increasing the number of parameters often increases in-sample fit, but we need a way to express the price of higher model complexity

Suppose we observe the training dataset $x^n = (x_1, \dots, x_n)$, typically modeled as i.i.d. draws from an unknown true distribution $\mathbb{P}$. We consider a parametric model class

$$\mathcal{M} = \{p_{\theta}(\cdot) : \theta \in \Theta \subset \mathbb{R}^p\},$$

where p is the number of free parameters.

6.2 Fit the model by maximum likelihood

Assume the training dataset $x^n = (x_1, \dots, x_n)$ is generated as i.i.d. draws from an unknown true distribution $\mathbb{P}$, i.e.

$$x^n \sim \mathbb{P}^{\otimes n}.$$

For a parametric model class $\{p_{\theta} : \theta \in \Theta \subset \mathbb{R}^p\}$, the MLE is the (random) estimator

$$\hat{\theta}_{\text{MLE}}(x^n) \in \arg \max_{\theta \in \Theta} \sum_{i=1}^n \log p_{\theta}(x_i).$$

Here $\hat{\theta}_{\text{MLE}}$ is random because it is a function of the random sample x^n .

6.3 What AIC is trying to estimate (make the randomness explicit)

We care about predictive performance on new data. Let x' denote a fresh test draw. Given the fitted parameter $\hat{\theta}_{\text{MLE}}(x^n)$, the test log score is the random variable

$$\log p_{\hat{\theta}_{\text{MLE}}(x^n)}(x').$$

The target quantity AIC approximates is the expected test log score, with expectations taken over both sources of randomness:

$$\mathbb{E}_{x^n \sim \mathbb{P}^{\otimes n}, x' \sim \mathbb{P}} \left[\log p_{\hat{\theta}_{\text{MLE}}(x^n)}(x') \right].$$

6.4 Key intuition: in-sample fit is optimistically biased

Because $\hat{\theta}_{\text{MLE}}(x^n)$ is chosen to maximize the training log-likelihood, the in-sample score is typically larger than the out-of-sample score. Akaike's asymptotic bias correction states

$$\mathbb{E}_{x^n \sim \mathbb{P}^{\otimes n}, x' \sim \mathbb{P}} \left[\log p_{\hat{\theta}_{\text{MLE}}(x^n)}(x') \right] = \mathbb{E}_{x^n \sim \mathbb{P}^{\otimes n}} \left[\frac{1}{n} \sum_{i=1}^n \log p_{\hat{\theta}_{\text{MLE}}(x^n)}(x_i) \right] - \frac{p}{n} + o\left(\frac{1}{n}\right).$$

Interpretation: the expected training log score overestimates the expected test log score by about p/n .

6.5 Definition of AIC and how to use it in practice

Multiplying the preceding expansion by $-2n$ yields a criterion equivalent (up to asymptotically negligible terms) to maximizing expected test log score:

$$\text{AIC} = -2 \sum_{i=1}^n \log p_{\hat{\theta}_{\text{MLE}}(x^n)}(x_i) + 2p.$$

How to use AIC to compare models:

- For each candidate model class $\mathcal{M}_m = \{p_\theta : \theta \in \Theta_m \subset \mathbb{R}^{p_m}\}$:
 1. Fit by MLE on the observed training data x^n to obtain $\hat{\theta}_{m,\text{MLE}}(x^n)$.
 2. Compute

$$\text{AIC}_m = -2 \sum_{i=1}^n \log p_{\hat{\theta}_{m,\text{MLE}}(x^n)}(x_i) + 2p_m.$$

- Choose the model with the smallest AIC_m (best estimated out-of-sample performance).

7 Stein's paradox and the James–Stein estimator

7.1 Model setup and MLE benchmark

We observe a single random vector $\mathbf{z} \in \mathbb{R}^N$ drawn as

$$\mathbf{z} \sim \mathcal{N}(\boldsymbol{\mu}, I_N),$$

where the mean vector $\boldsymbol{\mu} \in \mathbb{R}^N$ is unknown and treated as a fixed (non-random) parameter. We evaluate an estimator $\hat{\boldsymbol{\mu}}(\cdot)$ under the squared-error loss

$$L(\boldsymbol{\mu}, \hat{\boldsymbol{\mu}}(\mathbf{z})) = \|\hat{\boldsymbol{\mu}}(\mathbf{z}) - \boldsymbol{\mu}\|_2^2,$$

and its risk at parameter value $\boldsymbol{\mu}$ is

$$R(\boldsymbol{\mu}, \hat{\boldsymbol{\mu}}) = \mathbb{E}_{\mathbf{z} \sim \mathcal{N}(\boldsymbol{\mu}, I_N)} [\|\hat{\boldsymbol{\mu}}(\mathbf{z}) - \boldsymbol{\mu}\|_2^2].$$

Since the coordinates are independent, the likelihood factorizes across coordinates,

$$p_{\mathbf{z}}(\mathbf{z}) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}(z_i - \mu_i)^2\right),$$

and therefore maximizing coordinatewise yields the MLE which is just the vector itself

$$\hat{\boldsymbol{\mu}}^{\text{MLE}}(\mathbf{z}) = \mathbf{z}.$$

7.2 Stein's paradox: the MLE is not optimal for $N \geq 3$

For $N \geq 3$, the estimator $\hat{\boldsymbol{\mu}}^{\text{MLE}}(\mathbf{z}) = \mathbf{z}$ is *inadmissible* under squared-error loss: there exists another estimator $\hat{\boldsymbol{\mu}}^*$ such that

$$R(\boldsymbol{\mu}, \hat{\boldsymbol{\mu}}^*) < R(\boldsymbol{\mu}, \hat{\boldsymbol{\mu}}^{\text{MLE}}) \quad \text{for all } \boldsymbol{\mu} \in \mathbb{R}^N.$$

What makes it feel paradoxical is the tension between:

- *Decoupled sampling model:* $p_{\mathbf{z}}(\mathbf{z})$ factorizes across coordinates, suggesting each μ_i can be estimated separately by z_i .
- *Coupled objective:* the loss $\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|_2^2$ aggregates error across all coordinates

7.3 James–Stein shrinkage estimator dominates MLE

A concrete estimator that dominates the MLE for $N \geq 3$ is the James–Stein (JS) estimator

$$\hat{\boldsymbol{\mu}}^{\text{JS}}(\mathbf{z}) = \left(1 - \frac{N-2}{\|\mathbf{z}\|_2^2}\right) \mathbf{z}.$$

which *shrinks* the entire vector toward 0 by a data-dependent factor.

7.4 The Bayesian shrinkage approach to hierarchical Gaussians

A hierarchical Gaussian model gives a simple Bayesian explanation for the shrinkage factor we observe in the James–Stein estimator. Suppose

$$\mu_i \sim \mathcal{N}(M, A), \quad \mathbf{z}_i \mid \mu_i = \mu_i \sim \mathcal{N}(\mu_i, 1), \quad i \in [N],$$

independently across coordinates. Here M and A are deterministic hyperparameters. The interpretation is:

- μ_i is the unknown true mean for coordinate i ;
- before seeing data, μ_i is random with prior distribution $\mathcal{N}(M, A)$;
- conditional on $\mu_i = \mu_i$, the observation z_i is noisy:

$$z_i = \mu_i + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, 1);$$

- after observing $z_i = z_i$, the posterior distribution of μ_i is

$$p_{\mu_i | z_i = z_i; M, A}.$$

7.5 Bayes estimator under squared-error loss

Under squared-error loss, the Bayes estimator is the posterior mean:

$$\hat{\mu}_i^{\text{Bayes}}(z_i) = \mathbb{E}_{p_{\mu_i | z_i = z_i; M, A}}[\mu_i], \quad i \in [N].$$

This expectation is taken over the posterior distribution of the random quantity μ_i , after conditioning on the deterministic observation $z_i = z_i$. By Bayes' rule,

$$p_{\mu_i | z_i = z_i; M, A}(\mu_i) \propto p_{z_i | \mu_i = \mu_i}(z_i) p_{\mu_i; M, A}(\mu_i).$$

The likelihood and prior are

$$p_{z_i | \mu_i = \mu_i}(z_i) \propto \exp\left(-\frac{1}{2}(z_i - \mu_i)^2\right),$$

and

$$p_{\mu_i; M, A}(\mu_i) \propto \exp\left(-\frac{1}{2A}(\mu_i - M)^2\right).$$

Multiplying these two Gaussian factors and completing the square gives

$$\mu_i | z_i = z_i \sim \mathcal{N}\left(\frac{A}{A+1}z_i + \frac{1}{A+1}M, \frac{A}{A+1}\right).$$

Therefore,

$$\hat{\mu}_i^{\text{Bayes}}(z_i) = \frac{A}{A+1}z_i + \frac{1}{A+1}M.$$

Thus the Bayes rule shrinks the noisy observation z_i toward the prior mean M . The shrinkage factor is

$$\frac{A}{A+1}.$$

When A is large, the prior distribution is diffuse, so we trust z_i more. When A is small, the prior says that the coordinates are tightly concentrated around M , so we shrink more strongly toward M .

7.6 Empirical Bayes: estimating the hyperparameters

The preceding Bayes rule assumes that M and A are known. In empirical Bayes, we estimate them from the data. Under the hierarchical model, the marginal distribution of each observation is

$$z_i \sim \mathcal{N}(M, A+1),$$

because

$$\mathbf{z}_i = \boldsymbol{\mu}_i + \boldsymbol{\epsilon}_i, \quad \boldsymbol{\mu}_i \sim \mathcal{N}(M, A), \quad \boldsymbol{\epsilon}_i \sim \mathcal{N}(0, 1).$$

Thus the observations $z_1, \dots, z_N$ can be used to estimate the common location M and the cross-coordinate variation. A natural estimate of M is the sample mean

$$\hat{M}(z) = \frac{1}{N} \sum_{i=1}^N z_i.$$

Since

$$\text{Var}(z_i) = A + 1,$$

one can estimate $A + 1$ by the sample variance of the z_i 's, and hence estimate A by subtracting the known noise variance 1. For example,

$$\widehat{A+1}(z) = \frac{1}{N-1} \sum_{i=1}^N (z_i - \hat{M}(z))^2, \quad \hat{A}(z) = \widehat{A+1}(z) - 1.$$

In practice, one often truncates at zero to enforce the constraint $A \geq 0$:

$$\hat{A}_+(z) = \max\{\hat{A}(z), 0\}.$$

Plugging the estimates $\hat{M}(z)$ and $\hat{A}_+(z)$ into the Bayes posterior mean gives the empirical-Bayes shrinkage rule

$$\hat{\mu}_i^{\text{EB}}(z) = \frac{\hat{A}_+(z)}{\hat{A}_+(z) + 1} z_i + \frac{1}{\hat{A}_+(z) + 1} \hat{M}(z), \quad i \in [N].$$

7.7 Connection between Bayesian shrinkage and James–Stein

Both James–Stein and empirical-Bayes estimators share the same structural idea:

$$\hat{\mu}(z) = (\text{shrinkage center}) + (\text{shrinkage factor}) \cdot (z - \text{shrinkage center}).$$

James–Stein uses center 0 (or, in variants, $\hat{M}(z)1$) and a data-dependent shrinkage factor that depends on $\|z\|_2^2$. The empirical-Bayes rule uses center $\hat{M}(z)$ and a shrinkage factor determined by an estimated signal variance $\hat{A}(z)$.

The key shared message is the same as Stein's paradox: when $N \geq 3$, appropriately shrinking a multivariate mean estimate can reduce total risk

$$\mathbb{E}_{\mathbf{z} \sim \mathcal{N}(\boldsymbol{\mu}, I_N)} \left[\sum_{i=1}^N (\hat{\mu}_i(\mathbf{z}) - \mu_i)^2 \right]$$

relative to the naive coordinatewise estimator $\hat{\mu}(\mathbf{z}) = \mathbf{z}$.

8 Linear regression and confidence intervals

8.1 Setup and notation

We observe training data as pairs

$$(x^{(i)}, y^{(i)}), \quad i = 1, \dots, n,$$

where $x^{(i)}$ are treated as deterministic inputs and $y^{(i)}$ are responses. Define the empirical quantities

$$\begin{aligned} \bar{x} &= \frac{1}{n} \sum_{i=1}^n x^{(i)}, & \bar{y} &= \frac{1}{n} \sum_{i=1}^n y^{(i)}, \\ s_X^2 &= \frac{1}{n} \sum_{i=1}^n (x^{(i)} - \bar{x})^2. \end{aligned}$$

8.2 Linear regression as prediction (ERM viewpoint)

We consider the linear predictor

$$\hat{Y}(x) = \beta_0 + \beta_1 x.$$

and seek to find it by minimizing empirical squared error:

$$\min_{\beta_0, \beta_1} \sum_{i=1}^n \left(\beta_0 + \beta_1 x^{(i)} - y^{(i)} \right)^2.$$

The solution is

$$\begin{aligned} \hat{\beta}_1 &= \frac{1}{ns_X^2} \sum_{i=1}^n (x^{(i)} - \bar{x})(y^{(i)} - \bar{y}) = \text{Corr}(X, Y) \frac{s_Y}{s_X} \\ \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x}. \end{aligned}$$

Practical recipe:

1. Compute $\bar{x}, \bar{y}$.
2. Compute s_X^2 .
3. Compute $\hat{\beta}_1$ using the covariance formula above.
4. Compute $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$.
5. Predict via $\hat{Y}(x) = \hat{\beta}_0 + \hat{\beta}_1 x$.

8.3 Goodness of fit: R^2

We now seek to answer how well our model fits the data. First define the Residual Sum of Squares and Total Sum of Squares as

$$\begin{aligned} \text{RSS} &= \sum_{i=1}^n \left(y^{(i)} - \hat{Y}(x^{(i)}) \right)^2, \\ \text{TSS} &= \sum_{i=1}^n (y^{(i)} - \bar{y})^2. \end{aligned}$$

The coefficient of determination is

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}}$$

which is the proportion of total variance (TSS) not explained by our model. Furthermore, in simple linear regression, it's easy to see from the definition of $\hat{\beta}_1$ that

$$R^2 = \text{corr}^2(X, Y),$$

the squared empirical correlation.

8.4 Randomness in the least-squares estimators

Assume the true linear model is

$$Y^{(i)} = \beta_0 + \beta_1 x^{(i)} + \epsilon_i, \quad i \in [n],$$

where $x^{(i)}$ is deterministic and the randomness comes only from the noise terms ϵ_i . Assume

$$\mathbb{E}_{p_{\epsilon_i}}[\epsilon_i] = 0, \quad \text{Var}_{p_{\epsilon_i}}(\epsilon_i) = \sigma^2,$$

and assume that $\epsilon_1, \dots, \epsilon_n$ are uncorrelated. As before, let

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x^{(i)}, \quad s_X^2 = \frac{1}{n} \sum_{i=1}^n (x^{(i)} - \bar{x})^2.$$

The least-squares estimators are random variables because they are functions of $Y^{(1)}, \dots, Y^{(n)}$. In particular, the least-squares estimators can be written as

$$\hat{\beta}_1 = \beta_1 + W_1, \quad \hat{\beta}_0 = \beta_0 + W_0,$$

where W_1 and W_0 are the (random) estimation errors.

8.5 Deriving the slope estimation error

Let

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y^{(i)}.$$

The random least-squares slope estimator is

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x^{(i)} - \bar{x})(Y^{(i)} - \bar{Y})}{\sum_{i=1}^n (x^{(i)} - \bar{x})^2}.$$

Since

$$\sum_{i=1}^n (x^{(i)} - \bar{x}) = 0,$$

the term involving $\bar{Y}$ disappears, so

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x^{(i)} - \bar{x})Y^{(i)}}{ns_X^2}.$$

Now substitute the true model

$$Y^{(i)} = \beta_0 + \beta_1 x^{(i)} + \epsilon_i.$$

Then

$$\begin{aligned}\hat{\beta}_1 &= \frac{1}{ns_X^2} \sum_{i=1}^n (x^{(i)} - \bar{x}) (\beta_0 + \beta_1 x^{(i)} + \epsilon_i) \\ &= \frac{\beta_0}{ns_X^2} \sum_{i=1}^n (x^{(i)} - \bar{x}) + \frac{\beta_1}{ns_X^2} \sum_{i=1}^n (x^{(i)} - \bar{x}) x^{(i)} + \frac{1}{ns_X^2} \sum_{i=1}^n (x^{(i)} - \bar{x}) \epsilon_i.\end{aligned}$$

The first term is zero because the centered $x^{(i)}$'s sum to zero. For the second term,

$$\sum_{i=1}^n (x^{(i)} - \bar{x}) x^{(i)} = \sum_{i=1}^n (x^{(i)} - \bar{x}) ((x^{(i)} - \bar{x}) + \bar{x}) = \sum_{i=1}^n (x^{(i)} - \bar{x})^2 = ns_X^2.$$

Therefore

$$\hat{\beta}_1 = \beta_1 + \frac{1}{ns_X^2} \sum_{i=1}^n (x^{(i)} - \bar{x}) \epsilon_i.$$

Thus the slope estimation error is

$$W_1 = \frac{1}{ns_X^2} \sum_{i=1}^n (x^{(i)} - \bar{x}) \epsilon_i.$$

8.6 Variance of the slope error

Since W_1 is a linear combination of the noise terms,

$$W_1 = \sum_{i=1}^n a_i \epsilon_i, \quad a_i = \frac{x^{(i)} - \bar{x}}{ns_X^2}.$$

Using the fact that the noise terms are uncorrelated and have variance σ^2 ,

$$\begin{aligned}\text{Var}_{p_{\epsilon_1, \dots, \epsilon_n}}(W_1) &= \text{Var}_{p_{\epsilon_1, \dots, \epsilon_n}} \left(\sum_{i=1}^n a_i \epsilon_i \right) \\ &= \sum_{i=1}^n a_i^2 \text{Var}_{p_{\epsilon_i}}(\epsilon_i) \\ &= \sigma^2 \sum_{i=1}^n \left(\frac{x^{(i)} - \bar{x}}{ns_X^2} \right)^2 \\ &= \frac{\sigma^2}{n^2 s_X^4} \sum_{i=1}^n (x^{(i)} - \bar{x})^2.\end{aligned}$$

Since

$$\sum_{i=1}^n (x^{(i)} - \bar{x})^2 = ns_X^2,$$

we get

$$\text{Var}_{p_{\epsilon_1, \dots, \epsilon_n}}(W_1) = \frac{\sigma^2}{ns_X^2}.$$

Therefore

$$\text{Var}_{p_{\epsilon_1, \dots, \epsilon_n}}(\hat{\beta}_1) = \frac{\sigma^2}{ns_X^2}.$$

8.7 Deriving the intercept estimation error

Recall the random least-squares intercept estimator is

$$\bar{Y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}.$$

From the true model,

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n (\beta_0 + \beta_1 x^{(i)} + \epsilon_i) = \beta_0 + \beta_1 \bar{x} + \bar{\epsilon},$$

where

$$\bar{\epsilon} = \frac{1}{n} \sum_{i=1}^n \epsilon_i.$$

Using

$$\hat{\beta}_1 = \beta_1 + W_1,$$

we obtain

$$\begin{aligned} \hat{\beta}_0 &= \bar{Y} - \hat{\beta}_1 \bar{x} \\ &= \beta_0 + \beta_1 \bar{x} + \bar{\epsilon} - (\beta_1 + W_1) \bar{x} \\ &= \beta_0 + (\bar{\epsilon} - \bar{x} W_1) \\ &= \beta_0 + W_0 \end{aligned}$$

for the intercept estimation error

$$W_0 := \bar{\epsilon} - \bar{x} W_1.$$

Substituting the expression for W_1 gives

$$W_0 = \frac{1}{n} \sum_{i=1}^n \epsilon_i - \frac{\bar{x}}{ns_X^2} \sum_{i=1}^n (x^{(i)} - \bar{x}) \epsilon_i.$$

Equivalently,

$$W_0 = \sum_{i=1}^n \left(\frac{1}{n} - \frac{\bar{x}(x^{(i)} - \bar{x})}{ns_X^2} \right) \epsilon_i.$$

8.8 Variance of the intercept error

Since we found that

$$W_0 := \bar{\epsilon} - \bar{x} W_1,$$

we compute

$$\text{Var}_{p_{\epsilon_1, \dots, \epsilon_n}}(W_0) = \text{Var}_{p_{\epsilon_1, \dots, \epsilon_n}}(\bar{\epsilon}) + \bar{x}^2 \text{Var}_{p_{\epsilon_1, \dots, \epsilon_n}}(W_1) - 2\bar{x} \text{Cov}_{p_{\epsilon_1, \dots, \epsilon_n}}(\bar{\epsilon}, W_1).$$

First,

$$\text{Var}_{p_{\epsilon_1, \dots, \epsilon_n}}(\bar{\epsilon}) = \text{Var}_{p_{\epsilon_1, \dots, \epsilon_n}}\left(\frac{1}{n} \sum_{i=1}^n \epsilon_i\right) = \frac{\sigma^2}{n}.$$

Next,

$$\text{Cov}_{p_{\epsilon_1, \dots, \epsilon_n}}(\bar{\epsilon}, W_1) = \text{Cov}_{p_{\epsilon_1, \dots, \epsilon_n}}\left(\frac{1}{n} \sum_{i=1}^n \epsilon_i, \frac{1}{ns_X^2} \sum_{j=1}^n (x^{(j)} - \bar{x}) \epsilon_j\right).$$

Because the noise terms are uncorrelated,

$$\text{Cov}_{p_{\epsilon_1, \dots, \epsilon_n}}(\bar{\epsilon}, W_1) = \frac{1}{n^2 s_X^2} \sum_{i=1}^n (x^{(i)} - \bar{x}) \sigma^2 = 0,$$

since

$$\sum_{i=1}^n (x^{(i)} - \bar{x}) = 0.$$

Therefore

$$\begin{aligned} \text{Var}_{p_{\epsilon_1, \dots, \epsilon_n}}(W_0) &= \frac{\sigma^2}{n} + \bar{x}^2 \frac{\sigma^2}{n s_X^2} \\ &= \frac{\sigma^2 (s_X^2 + \bar{x}^2)}{n s_X^2}. \end{aligned}$$

Thus

$$\text{Var}_{p_{\epsilon_1, \dots, \epsilon_n}}(\hat{\beta}_0) = \frac{\sigma^2 (s_X^2 + \bar{x}^2)}{n s_X^2}.$$

8.9 Unbiasedness and approximate normality

Both estimators are linear combinations of the noise terms. Therefore, if the noise terms are Gaussian, then the estimators are exactly Gaussian:

$$\hat{\beta}_1 \sim \mathcal{N}\left(\beta_1, \frac{\sigma^2}{n s_X^2}\right),$$

and

$$\hat{\beta}_0 \sim \mathcal{N}\left(\beta_0, \frac{\sigma^2 (s_X^2 + \bar{x}^2)}{n s_X^2}\right).$$

If the noise terms are not Gaussian, the same normal approximation is often used for moderate or large n , because $\hat{\beta}_0$ and $\hat{\beta}_1$ are weighted averages of many noise terms.

8.10 Estimating the noise variance

The variance σ^2 of our noise ϵ_i is unknown and must be estimated in order to construct confidence intervals for our points. As a first step, define the residuals

$$\hat{Z}^{(i)} = y^{(i)} - \hat{\beta}_0 - \hat{\beta}_1 x^{(i)}.$$

The unbiased estimator of σ^2 is

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n \left(y^{(i)} - \hat{\beta}_0 - \hat{\beta}_1 x^{(i)} \right)^2.$$

The denominator $n-2$ accounts for estimating two parameters.

8.11 Confidence intervals for $\hat{\beta}_0, \hat{\beta}_1$

Since our estimators $(\hat{\beta}_0, \hat{\beta}_1)$ are jointly Gaussian due to the noise in our data model, we can form a confidence interval for them. Approximate 95% confidence intervals:

$$\hat{\beta}_1 \pm 1.96 \sqrt{\frac{\hat{\sigma}^2}{n s_X^2}},$$

$$\hat{\beta}_0 \pm 1.96 \sqrt{\frac{\hat{\sigma}^2(s_X^2 + \bar{x}^2)}{ns_X^2}}.$$

Practical method:

1. Fit $\hat{\beta}_0, \hat{\beta}_1$.
2. Compute residuals and $\hat{\sigma}^2$.
3. Plug $\hat{\sigma}^2$ into the variance formulas.
4. Form the normal-based interval above.

8.12 Hypothesis testing for slope

To test

$$H : \beta_1 = 0 \quad \text{vs} \quad K : \beta_1 \neq 0,$$

compute the test statistic

$$T = \frac{\hat{\beta}_1}{\sqrt{\hat{\sigma}^2/(ns_X^2)}}.$$

For moderate/large n , compare $|T|$ to 1.96 for a 5% test.

8.13 Prediction error at a new point

For a new input x_{new} , prediction is

$$\hat{Y}_{\text{new}} = \hat{\beta}_0 + \hat{\beta}_1 x_{\text{new}}.$$

The conditional mean squared prediction error is

$$\mathbb{E} \left[\left(\hat{Y}_{\text{new}} - Y_{\text{new}} \right)^2 \mid X = x_{\text{new}} \right] = \sigma^2 + \frac{\sigma^2}{ns_X^2} \left(s_X^2 + (x_{\text{new}} - \bar{x})^2 \right).$$

or symbolically,

$$\text{prediction error} = \underbrace{\sigma^2}_{\text{irreducible noise}} + \underbrace{\text{Var}(\hat{\beta}_0 + \hat{\beta}_1 x_{\text{new}})}_{\text{estimation variance}}$$

8.14 Matrix derivation for multiple linear regression

Let

$$X \in \mathbb{R}^{n \times d}, \quad \mathbf{y} \in \mathbb{R}^n, \quad \mathbf{b} \in \mathbb{R}^d.$$

The matrix X contains the observed features, where the i th row is the feature vector for observation i . The vector $\mathbf{y}$ contains the observed responses. We consider the linear predictor

$$\hat{\mathbf{y}} = X\mathbf{b}.$$

Note that we can include an intercept by simply appending a column of 1's to X . The least-squares problem is to choose $\mathbf{b}$ to minimize the total squared prediction error:

$$\hat{\mathbf{b}} = \arg \min_{\mathbf{b} \in \mathbb{R}^d} \|X\mathbf{b} - \mathbf{y}\|_2^2.$$

Define the objective function

$$\begin{aligned} L(\mathbf{b}) &= \|X\mathbf{b} - \mathbf{y}\|_2^2 \\ &= (X\mathbf{b} - \mathbf{y})^\top (X\mathbf{b} - \mathbf{y}). \end{aligned}$$

Expanding,

$$L(\mathbf{b}) = (X\mathbf{b})^\top (X\mathbf{b}) - (X\mathbf{b})^\top \mathbf{y} - \mathbf{y}^\top (X\mathbf{b}) + \mathbf{y}^\top \mathbf{y}.$$

Since $(X\mathbf{b})^\top \mathbf{y}$ and $\mathbf{y}^\top X\mathbf{b}$ are the same scalar,

$$L(\mathbf{b}) = \mathbf{b}^\top X^\top X\mathbf{b} - 2\mathbf{b}^\top X^\top \mathbf{y} + \mathbf{y}^\top \mathbf{y}.$$

Now we must take the first-order conditions. In particular, to minimize $L(\mathbf{b})$, take the gradient with respect to $\mathbf{b}$. Since $X^\top X$ is symmetric,

$$\nabla_{\mathbf{b}} (\mathbf{b}^\top X^\top X\mathbf{b}) = 2X^\top X\mathbf{b},$$

and

$$\nabla_{\mathbf{b}} (-2\mathbf{b}^\top X^\top \mathbf{y}) = -2X^\top \mathbf{y}.$$

The term $\mathbf{y}^\top \mathbf{y}$ does not depend on $\mathbf{b}$, so its gradient is zero. Therefore

$$\nabla_{\mathbf{b}} L(\mathbf{b}) = 2X^\top X\mathbf{b} - 2X^\top \mathbf{y}.$$

At an optimum, the gradient must equal zero:

$$2X^\top X\hat{\mathbf{b}} - 2X^\top \mathbf{y} = 0$$

which gives the normal equations

$$X^\top X\hat{\mathbf{b}} = X^\top \mathbf{y}.$$

Lastly, we must solve the normal equations. If the columns of X are linearly independent, then $X^\top X$ is invertible. Therefore, from

$$X^\top X\hat{\mathbf{b}} = X^\top \mathbf{y},$$

we obtain the least-squares estimator

$$\boxed{\hat{\mathbf{b}} = (X^\top X)^{-1} X^\top \mathbf{y}}$$

9 Survival analysis

9.1 Time-to-event viewpoint of survival analysis

In survival analysis, the outcome of interest is a *time to event*. The *event* must be specified in advance by the study design. For example, in a drug-treatment study, the event might be:

- death,
- recovery,
- relapse,

It is important not to confuse the event with other ways observation may end. For instance, if the study is about *time to death*, then:

- a patient who dies has the event,
- and a patient who gets better is *not* counted as having the event
- a patient who is still alive but still sick when the study ends is censored,
- a patient who leaves the study early or is lost to follow-up is censored

9.2 Model, data, and survival function

For each individual $i = 1, \dots, n$, we observe deterministic covariates x_i , deterministic treatment status d_i , an observed time t_i , and a censoring indicator δ_i . Thus the dataset is

$$(t_i, x_i, d_i, \delta_i), \quad i = 1, \dots, n.$$

To formalize this, let the underlying random quantities be

$$\tau_i = \text{true event time}, \quad c_i = \text{censoring time}.$$

Here τ_i is the time at which the event occurs, while c_i is the time at which observation stops before the event is seen, for example because the study ends or the patient drops out. The observed time is the minimum of these two random times:

$$t_i = \min(\tau_i, c_i).$$

Its realized value is t_i . The censoring indicator is

$$\delta_i = \begin{cases} 1, & \text{if } \tau_i \leq c_i \text{ (the event is observed),} \\ 0, & \text{if } \tau_i > c_i \text{ (the observation is censored).} \end{cases}$$

Its realized value is $\delta_i = \delta_i$, thus:

- if $\delta_i = 1$, then the event occurred at time t_i ,
- if $\delta_i = 0$, then the event was not observed by time t_i , so we only know that $\tau_i > t_i$.

The main object of interest is the survival function

$$S(t \mid x, d) = \mathbb{P}(\tau > t \mid X = x, D = d), \quad t \geq 0,$$

which is the probability that the event has not yet occurred by time t , conditional on covariates and treatment.

9.3 Empirical survival estimator without censoring

As a warm-up, suppose there is no censoring, so $\delta_i = 1$ for all i . If we restrict attention to a fixed covariate-treatment group (x, d) , then the survival function can be estimated empirically

by

$$\hat{S}(t) = \frac{\#\{i : t_i > t\}}{n}.$$

This is just the fraction of individuals whose observed event time exceeds t .

9.4 Kaplan–Meier setup

Now suppose censoring is present, and again focus on a fixed group with common covariates and treatment:

$$x_i = x^*, \quad d_i = d^*, \quad i = 1, \dots, n.$$

We then estimate the group-specific survival function

$$S(t) = \mathbb{P}(\tau > t \mid X = x^*, D = d^*).$$

Among the observed times $t_1, \dots, t_n$, only some correspond to actual events, namely those with $\delta_i = 1$. Let the distinct observed event times be

$$t_{(1)} < t_{(2)} < \dots < t_{(K)}.$$

Here $K \leq n$ because several individuals may share the same event time, and some observed times may correspond only to censoring. For each event time $t_{(k)}$, define

$$n_{(k)} = \text{number of individuals at risk just before } t_{(k)},$$

meaning the number who have neither had the event nor been censored before $t_{(k)}$, and

$$q_{(k)} = \text{number of individuals whose event occurs at } t_{(k)}.$$

9.5 Product representation of survival

At the event times, the survival probability can be written as a product of conditional survival probabilities:

$$S(t_{(k)}) = \mathbb{P}(\tau > t_{(k)}) = \prod_{\ell=1}^k \mathbb{P}(\tau > t_{(\ell)} \mid \tau > t_{(\ell-1)}),$$

with the convention that the first factor is simply $\mathbb{P}(\tau > t_{(1)})$. This decomposition reflects the fact that surviving past $t_{(k)}$ requires surviving past each earlier event time.

9.6 Kaplan–Meier estimator

At time $t_{(\ell)}$, there are $n_{(\ell)}$ individuals at risk immediately before that time, and $q_{(\ell)}$ of them experience the event at that time. Thus the empirical conditional probability of surviving past $t_{(\ell)}$, given survival up to that point, is

$$\frac{n_{(\ell)} - q_{(\ell)}}{n_{(\ell)}}.$$

Multiplying these factors gives the Kaplan–Meier estimator:

$$\hat{S}(t_{(k)}) = \prod_{\ell=1}^k \frac{n_{(\ell)} - q_{(\ell)}}{n_{(\ell)}}.$$

More generally, for any $t \geq 0$,

$$\hat{S}(t) = \prod_{\ell: t_{(\ell)} \leq t} \frac{n_{(\ell)} - q_{(\ell)}}{n_{(\ell)}}.$$

So $\hat{S}(t)$ is a step function that changes only at observed event times.

9.7 Confidence interval and Greenwood's formula

A common large-sample confidence interval for $S(t)$ is

$$\hat{S}(t) \pm z_{1-\alpha/2} \hat{\sigma}(t),$$

where Greenwood's formula estimates the variance by

$$\hat{\sigma}^2(t) = \hat{S}^2(t) \sum_{j: t_{(j)} \leq t} \frac{q_{(j)}}{n_{(j)}(n_{(j)} - q_{(j)})}.$$

The sum is taken over distinct event times up to t , not over all observations.

Part II Inference and information

1 Preliminaries and introduction

1.1 Notation for common objects

- Random quantities are written as $\mathbf{x}$ (sans serif)
- Deterministic quantities, such as sample realizations, are x (serif)
- Randomized functions are written as $f(\cdot)$.
- Deterministic functions are written $f(\cdot)$
- Consequently, $f(\mathbf{x})$, $f(x)$, and $f(\mathbf{x})$ are random variables.
- Sets and events are written as $\mathcal{X}$. Sets that have structure are called ordinal, and sets without structure are called categorical. For an alphabet $\mathcal{X}$ and a positive integer N , we write

$$\mathcal{X}^N = \underbrace{\mathcal{X} \times \cdots \times \mathcal{X}}_{N\text{-fold}}$$

for the alphabet of N -tuples with entries in $\mathcal{X}$

1.2 PMF notation

For a discrete random variable $\mathbf{x} \in \mathcal{X}$, its PMF is $p_{\mathbf{x}}(\cdot)$ and we write (the slightly abused notation)

$$p_{\mathbf{x}}(x) = \mathbb{P}(\mathbf{x} = x),$$

where $\mathbb{P}$ denotes the underlying probability law, namely for an event $\mathcal{E} \subset \mathcal{X}$, we write $\mathbb{P}(\mathcal{E})$.

1.3 Expectation and conditioning operator

We use $\mathbb{E}[\cdot]$ to denote expectation. To avoid confusion, we make explicit the underlying probability law by adding a subscript. For example, if $\mathbf{x} \sim p_{\mathbf{x}}(\cdot)$, then

$$\mathbb{E}_p[f(\mathbf{x})] = \int f(x) p(x) dx.$$

For a conditional distribution $p_{\mathbf{x}|\mathbf{y}}(\cdot | \mathbf{y} = y)$, we write

$$\mathbb{E}_{p_{\mathbf{x}|\mathbf{y}}(\cdot|\mathbf{y}=y)}[f(\mathbf{x})] = \mathbb{E}[f(\mathbf{x}) | \mathbf{y} = y] = \int f(x) p_{\mathbf{x}|\mathbf{y}}(x | \mathbf{y} = y) dx.$$

In other words, for both probability and ex

1.4 Moment generating functions and characteristic functions

When well-defined, the moment-generating function of $\mathbf{x} \sim p(\cdot)$ is

$$M_{\mathbf{x}}(s) = \mathbb{E}_{p(\cdot)}[e^{s\mathbf{x}}],$$

and the characteristic function is

$$M_{\mathbf{x}}(j\omega) = \mathbb{E}_{p(\cdot)}[e^{j\omega\mathbf{x}}].$$

1.5 Big-O notation

For functions $f(\cdot)$ and $g(\cdot)$ defined on a suitable alphabet $\mathcal{U}$, we write

$$g(u) = O(f(u)) \quad \text{as } u \rightarrow a$$

to mean that there exist finite constants $c > 0$ and $\epsilon > 0$ such that

$$|g(u)| \leq c |f(u)| \quad \text{for all } u \text{ with } |u - a| < \epsilon.$$

1.6 Little-o notation

For functions $f(\cdot)$ and $g(\cdot)$ defined on a suitable alphabet $\mathcal{U}$, we write

$$g(u) = o(f(u)) \quad \text{as } u \rightarrow a$$

to mean that for every constant $c > 0$, there exists $\epsilon > 0$ such that

$$|g(u)| \leq c |f(u)| \quad \text{for all } u \text{ with } |u - a| < \epsilon.$$

1.7 Gaussian tail (Q-function)

The tail probability of a standard normal random variable is denoted by the Q -function,

$$Q(x) := \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-t^2/2} dt.$$

1.8 Gaussian random vectors

We write $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Lambda})$ to denote that $\mathbf{x}$ is a Gaussian random vector with mean

$$\mathbb{E}[\mathbf{x}] = \boldsymbol{\mu}$$

and covariance

$$\mathbb{E}[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^\top] = \boldsymbol{\Lambda}.$$

The explicit density of $\mathbf{x}$ is

$$\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Lambda}) := \frac{1}{|2\pi\boldsymbol{\Lambda}|^{1/2}} \exp\left\{-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Lambda}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right\}.$$

2 Bayesian hypothesis testing

2.1 The typical model of inference

We typically think of inference as the following process

$$x \longrightarrow \boxed{\text{model}} \longrightarrow y \longrightarrow \boxed{f(\cdot)} \longrightarrow \hat{x}$$

- We care about x , which is the ground truth, but we do not observe it
- However, we do see data y , which is x through some model that we assume. We often model this data y as being random
- $\hat{x}$ is the estimate of x , generated by applying our design $f(\cdot)$ over data y to obtain our prediction of the x that generated y . We will set this method $f(\cdot)$ based on desired performance criteria.

2.2 Bayesian hypothesis testing

Consider $M \geq 2$ hypotheses

$$\mathcal{H} = \{H_0, H_1, \dots, H_{M-1}\},$$

where each hypothesis specifies a different probabilistic model for the observed data. The observed data is a random vector y and the unknown hypothesis is a random variable $H \in \mathcal{H}$. A Bayesian specification consists of:

- prior (a priori) probabilities over hypotheses:

$$p_H(H_m) = \mathbb{P}_H(H = H_m), \quad m = 0, 1, \dots, M-1;$$

- conditional distributions (likelihood models) for the data under each hypothesis:

$$p_{y|H}(\cdot | H_m), \quad m = 0, 1, \dots, M-1.$$

After observing $y = y$, our knowledge of the true hypothesis is summarized by the posterior (a posteriori) probabilities

$$p_{H|y}(H_m | y) = \frac{p_{y|H}(y | H_m) p_H(H_m)}{\sum_{m'} p_{y|H}(y | H_{m'}) p_H(H_{m'})}.$$

for each $m = 0, 1, \dots, M-1$. This posterior $p_{H|y}(\cdot | y)$ is the revised belief over hypothesis after observing the data y . In many applications one subsequently makes a decision (i.e., selects a hypothesis) based on the posterior.

2.3 Binary hypothesis testing

In the binary case, the hypothesis set is

$$\mathcal{H} = \{H_0, H_1\},$$

and our Bayesian model has two components: prior probabilities on H and likelihood models for observation y .

- *Prior on each hypothesis in $\mathcal{H}$.* Define the priors

$$P_0 := \mathbb{P}_H(H = H_0), \quad P_1 := \mathbb{P}_H(H = H_1) = 1 - P_0.$$

- *Likelihoods.* The conditional distributions of y under each hypothesis in $\mathcal{H}$ are

$$p_{y|H}(y | H_0), \quad p_{y|H}(y | H_1).$$

2.4 The solution to hypothesis testing is a decision rule $\widehat{H}(y) \in \{H_0, H_1\}$

Decision rules decide over hypotheses. In particular, a deterministic decision rule is a function $\widehat{H}(\cdot)$ that maps each observation $y \in \mathcal{Y}$ to one of the two hypotheses:

$$\widehat{H} : \mathcal{Y} \rightarrow \mathcal{H}, \quad \mathcal{H} = \{H_0, H_1\}.$$

Choosing $\widehat{H}(\cdot)$ is equivalent to partitioning the observation space $\mathcal{Y}$ into two disjoint decision regions corresponding to the two possible decisions.

$$\begin{aligned} \mathcal{Y}_0 &= \{y \in \mathcal{Y} : \widehat{H}(y) = H_0\}, \\ \mathcal{Y}_1 &= \{y \in \mathcal{Y} : \widehat{H}(y) = H_1\}. \end{aligned}$$

2.5 Cost functions measure the performance of $\widehat{H}(y) \in \mathcal{H}$

We evaluate the quality of a decision rule via an expected cost. Let $C(H_j, H_i) = C_{ij}$ denote the cost of deciding $\widehat{H}(\cdot) = H_i$ when the true hypothesis is $H = H_j$. Equivalently, for a generic decision rule $f(\cdot) : \mathcal{Y} \rightarrow \mathcal{H}$, the incurred loss is $C(H, f(y))$, and we seek to minimize the the expected cost, known as Bayes risk

$$\varphi(f) := \mathbb{E}_{(H,y)}[C(H, f(y))],$$

In this setting, the Bayes optimal decision rule is

$$\begin{aligned} \widehat{H}(\cdot) &= \arg \min_{f(\cdot)} \mathbb{E}_{(H,y)}[C(H, f(y)).] \\ &= \arg \min_{f(\cdot)} \varphi(f) \end{aligned}$$

A common symmetric choice is the 0–1 loss function,

$$C_{00} = C_{11} = 0, \quad C_{01} = C_{10} = 1,$$

Note that this 0 – 1 loss function corresponds to minimizing the probability of classification error $\mathbb{P}_{\text{error}}$

2.6 Finding the Bayes optimum decision rule using likelihood ratio

We now need to find

$$\widehat{H}(\cdot) = \arg \min_{f(\cdot)} \mathbb{E}_{(H,y)}[C(H, f(y)).]$$

Fix an arbitrary (deterministic) decision rule $f(\cdot) : \mathcal{Y} \rightarrow \mathcal{H}$, where $\mathcal{H} = \{H_0, H_1\}$. And recall we call the Bayes risk

$$\varphi(f) := \mathbb{E}_{(H,y)}[C(H, f(y))].$$

By iterated expectation,

$$\varphi(f) = \mathbb{E}_y \left[\mathbb{E}_H [C(H, f(y)) | y = y] \right] = \int \tilde{\varphi}(f(y), y) p_y(y) dy,$$

where we define the expected conditional risk

$$\tilde{\varphi}(f(y), y) := \mathbb{E}_H[C(H, f(y)) \mid y = y].$$

Since $p_Y(y) \geq 0$, we minimize φ by minimizing $\tilde{\varphi}(f(y), y)$ pointwise in y . Thus, for each y we choose $f(y) \in \{H_0, H_1\}$ to minimize $\tilde{\varphi}(f(y), y)$. Writing $C(H_j, H_i) = C_{ji}$, we have

$$\begin{aligned}\tilde{\varphi}(H_0, y) &= C_{00} p_{H|Y}(H_0 \mid y) + C_{01} p_{H|Y}(H_1 \mid y) \\ \tilde{\varphi}(H_1, y) &= C_{10} p_{H|Y}(H_0 \mid y) + C_{11} p_{H|Y}(H_1 \mid y).\end{aligned}$$

Therefore, the Bayes optimal decision is to simply pick the hypothesis that corresponds to the lower expected conditional risk

$$C_{00} p_{H|Y}(H_0 \mid y) + C_{01} p_{H|Y}(H_1 \mid y) \underset{\hat{H}(y)=H_0}{\overset{\hat{H}(y)=H_1}{\gtrless}} C_{10} p_{H|Y}(H_0 \mid y) + C_{11} p_{H|Y}(H_1 \mid y).$$

Rearranging and assuming $C_{01} - C_{11} > 0$ and $C_{10} - C_{00} > 0$,

$$\frac{p_{H|Y}(H_1 \mid y)}{p_{H|Y}(H_0 \mid y)} \underset{\hat{H}(y)=H_0}{\overset{\hat{H}(y)=H_1}{\gtrless}} \frac{C_{10} - C_{00}}{C_{01} - C_{11}}.$$

Using Bayes' rule with priors $P_i := \mathbb{P}(H = H_i)$ and likelihoods $p_{Y|H}(y \mid H_i)$, we get that the posterior is

$$p_{H|Y}(H_i \mid y) = \frac{p_{Y|H}(y \mid H_i) P_i}{p_Y(y)}, \quad m \in \{0, 1\}.$$

Substituting gives the *likelihood ratio test*:

$$\mathcal{L}(\mathbf{y}) = \frac{p_{Y|H}(y \mid H_1)}{p_{Y|H}(y \mid H_0)} \underset{\hat{H}(y)=H_0}{\overset{\hat{H}(y)=H_1}{\gtrless}} \frac{C_{10} - C_{00}}{C_{01} - C_{11}} \cdot \frac{P_0}{P_1} = \eta.$$

2.7 The 0–1 loss $\rightarrow$ minimize the probability of error and obtain MAP rule

Assume the 0–1 loss

$$C_{ij} = 1 - 1_{i=j}, \quad C_{00} = C_{11} = 0, \quad C_{01} = C_{10} = 1$$

For a decision rule $f(\cdot)$, the Bayes risk equals the probability of error:

$$\begin{aligned}\varphi(f) &= \mathbb{E}_{H,Y}[C(H, f(Y))] \\ &= \mathbb{P}(f(Y) = H_0, H = H_1) + \mathbb{P}(f(Y) = H_1, H = H_0) \\ &= \mathbb{P}_{\text{error}}\end{aligned}$$

Moreover, the Bayes optimal decision rule for this setting of the 0–1 loss is the function that chooses the hypothesis with the larger posterior probability:

$$\hat{H}(y) \in \arg \max_{H \in \{H_0, H_1\}} p_{H|Y}(H \mid y).$$

This is the *maximum a posteriori* (MAP) decision rule.

3 Non-Bayesian hypothesis testing

We look at hypothesis testing when either priors or costs are unavailable.

3.1 Error probabilities for a binary decision rule

Consider a binary hypothesis test with $H \in \{H_0, H_1\}$ and a decision rule $\hat{H}(y) \in \{H_0, H_1\}$ based on observation y . The performance of any decision rule $\hat{H}(\cdot)$ is fully specified by the pair

$$P_F := \mathbb{P}(\hat{H}(y) = H_1 \mid H = H_0), \quad P_D := \mathbb{P}(\hat{H}(y) = H_1 \mid H = H_1).$$

which correspond to false alarm probability and true detection probability. To compute these, recall a deterministic decision induces a partition of $\mathcal{Y}$ into

$$\mathcal{Y}_0 := \{y \in \mathcal{Y} : \hat{H}(y) = H_0\}, \quad \mathcal{Y}_1 := \{y \in \mathcal{Y} : \hat{H}(y) = H_1\}.$$

Then,

$$P_F = \mathbb{P}(y \in \mathcal{Y}_1 \mid H = H_0) = \int_{\mathcal{Y}_1} p_{Y|H}(y \mid H_0) dy,$$

$$P_D = \mathbb{P}(y \in \mathcal{Y}_1 \mid H = H_1) = \int_{\mathcal{Y}_1} p_{Y|H}(y \mid H_1) dy.$$

Typically P_D and P_F are competing objectives: enlarging the acceptance region $\mathcal{Y}_1$ tends to increase both P_D and P_F , while shrinking $\mathcal{Y}_1$ tends to decrease both P_D and P_F .

3.2 Likelihood ratio test family indexed by η

Consider the family of likelihood ratio test (LRT) decision rules indexed by a threshold $\eta \in \mathbb{R}$. Equivalently,

$$\text{decide } \hat{H}(y) = \begin{cases} \geq \eta, & \hat{H}(y) = H_1, \\ < \eta, & \hat{H}(y) = H_0. \end{cases}$$

where

$$\mathcal{L}(y) := \frac{p_{Y|H}(y \mid H_1)}{p_{Y|H}(y \mid H_0)}.$$

3.3 Operating characteristic traced by varying η

Each threshold η uniquely specifies a decision rule, and hence an operating point $(P_F(\eta), P_D(\eta))$ in the (P_F, P_D) plane:

$$P_F(\eta) = \mathbb{P}(\mathcal{L}(y) \geq \eta \mid H = H_0), \quad P_D(\eta) = \mathbb{P}(\mathcal{L}(y) \geq \eta \mid H = H_1).$$

The operating characteristic of the likelihood ratio test (OC-LRT) is obtained by varying η and recording the point in the (P_F, P_D) plane:

$$\{(P_F(\eta), P_D(\eta)) : \eta \in (0, \infty)\}$$

The OC-LRT is non-decreasing, since for $\eta_2 > \eta_1$ we have that P_D is non-increasing and P_F is non-increasing.

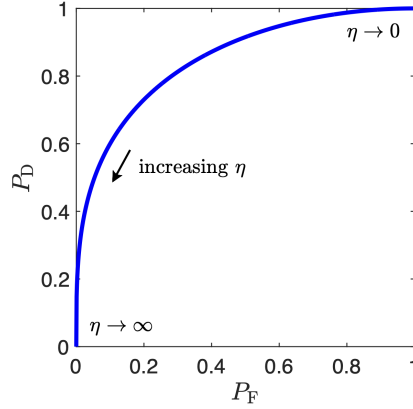


Figure 1: Operating characteristic of the likelihood ratio test.

3.4 The Bayes risk is a linear function in P_F, P_D

Assume prior probabilities

$$P_0 = \mathbb{P}(H = H_0), \quad P_1 = \mathbb{P}(H = H_1)$$

and costs C_{ij} of deciding H_i when the true $H = H_j$. For a deterministic rule $\hat{H}(\cdot)$, the Bayes risk is

$$\varphi(\hat{H}) = \mathbb{E}_{y,H} [C(H, \hat{H}(y))]$$

and using law of iterated expectation

$$= \sum_{i,j \in \{0,1\}} C_{ij} \mathbb{P}(\hat{H}(y) = H_i | H = H_j) P_j,$$

where probabilities are taken under the corresponding conditional law $y \sim p_{y|H}(\cdot | H_j)$.

Write the two key error probabilities:

$$P_F = \mathbb{P}(\hat{H}(y) = H_1 | H = H_0), \quad P_D = \mathbb{P}(\hat{H}(y) = H_1 | H = H_1).$$

Then

$$\mathbb{P}(\hat{H}(y) = H_0 | H = H_0) = 1 - P_F, \quad \mathbb{P}(\hat{H}(y) = H_0 | H = H_1) = 1 - P_D.$$

Substitute into $\varphi(\hat{H})$ and collect terms:

$$\begin{aligned} \varphi(\hat{H}) &= C_{00}(1 - P_F)P_0 + C_{10}P_F P_0 + C_{01}(1 - P_D)P_1 + C_{11}P_D P_1 \\ &= \underbrace{(C_{10} - C_{00})P_0}_{\alpha} P_F - \underbrace{(C_{01} - C_{11})P_1}_{\beta} P_D + \underbrace{(C_{00}P_0 + C_{01}P_1)}_{\gamma}. \end{aligned}$$

Hence the Bayes risk can be written as

$$\varphi(\hat{H}) = \alpha P_F - \beta P_D + \gamma,$$

for

$$\alpha = (C_{10} - C_{00})P_0, \quad \beta = (C_{01} - C_{11})P_1, \quad \gamma = C_{00}P_0 + C_{01}P_1.$$

3.5 Equi-Bayes-risk lines and the Bayesian operating point

From this linear equation for $\varphi(\hat{H})$, fix a constant c and consider the set of operating points (P_F, P_D) with the same Bayes risk $\varphi(\hat{H}) = c$

$$\alpha P_F - \beta P_D + \gamma = c$$

then solve for P_D in terms of P_F :

$$P_D = \frac{\alpha}{\beta} P_F + \frac{\gamma - c}{\beta},$$

which means that the set of points (P_F, P_D) with the same Bayes risk are straight lines, all with slope

$$\eta_B := \frac{\alpha}{\beta} = \frac{P_0(C_{10} - C_{00})}{P_1(C_{01} - C_{11})}.$$

Geometrically, the Bayesian operating point on the OC-ROC curve is obtained by finding the smallest value of c such that an equi-risk line is tangent to (or first touches) the achievable curve.

3.6 Connection to the Bayesian LRT threshold

The slope parameter above

$$\eta_B = \frac{\alpha}{\beta} = \frac{P_0(C_{10} - C_{00})}{P_1(C_{01} - C_{11})}$$

is exactly the likelihood-ratio threshold that appears in the optimal Bayesian LRT. Thus:

- choosing η_B selects a single LRT $\hat{H}_{\eta_B}(\cdot)$,
- and that LRT corresponds to a single operating point $(P_F(\eta_B), P_D(\eta_B))$ on the OC-ROC curve.

3.7 NP criterion: maximize P_D subject to a P_F constraint

When it is difficult or unnatural to assign costs, a common alternative is the Neyman–Pearson formulation:

$$\max_{\hat{H}(\cdot)} P_D \quad \text{subject to} \quad P_F \leq \alpha,$$

where $\alpha \in (0, 1)$ is the maximum allowable false-alarm probability.

3.8 Neyman–Pearson lemma (deterministic rules): pick η such that $P_F(\eta) = \alpha$

Assume the likelihood ratio $\mathcal{L}(y)$ is a purely continuous random variable under each hypothesis (i.e., $\mathbb{P}_{y \sim p_{y|H}(\cdot|H_j)}(\mathcal{L}(y) = \eta) = 0$ for all η and $j \in \{0, 1\}$). Then an optimal solution to the Neyman–Pearson problem among deterministic rules is an LRT:

$$\hat{H}(y) = H_1 1\{\mathcal{L}(y) \geq \eta\} + H_0 1\{\mathcal{L}(y) < \eta\},$$

where the threshold η is chosen so that the constraint is met with equality:

$$P_F = \mathbb{P}_{y \sim p_{y|H}(\cdot|H_0)}(\mathcal{L}(y) \geq \eta) = \alpha.$$

(Proof is non-obvious. We'll soon see a version of this theorem that applies to non-continuous likelihood ratios $\mathcal{L}(y)$). More importantly, recall varying η traces out the achievable operating characteristic curve of LRTs: each η produces a pair $(P_F(\eta), P_D(\eta))$. Under the Neyman–Pearson criterion, the operating point is found by selecting the η such that $P_F(\eta) = \alpha$ and reading off the corresponding $P_D(\eta)$.

4 Performance limits of hypothesis testing

4.1 Likelihood ratio discontinuous $\implies$ OC-LRT discontinuous

Consider binary-valued data $y \in \mathcal{Y} = \{0, 1\}$ with conditional PMFs

$$p_{y|H}(y | H_0) = \begin{cases} \frac{1}{2}, & y = 0, \\ \frac{1}{2}, & y = 1, \end{cases} \quad p_{y|H}(y | H_1) = \begin{cases} \frac{2}{3}, & y = 0, \\ \frac{1}{3}, & y = 1. \end{cases}$$

The likelihood ratio is discontinuous

$$\mathcal{L}(y) = \frac{p_{y|H}(y | H_1)}{p_{y|H}(y | H_0)} = \begin{cases} \frac{4}{3}, & y = 0, \\ \frac{2}{3}, & y = 1. \end{cases}$$

The likelihood ratio test (LRT) with threshold η decides H_1 iff $\mathcal{L}(y) \geq \eta$. The false-alarm and detection probabilities are

$$P_F(\eta) = \mathbb{P}(\mathcal{L}(y) \geq \eta | H = H_0) = \begin{cases} 0, & \text{if } \frac{4}{3} < \eta, \\ \frac{1}{2}, & \text{if } \frac{2}{3} < \eta \leq \frac{4}{3}, \\ 1, & \text{if } \eta \leq \frac{2}{3}, \end{cases}$$

$$P_D(\eta) = \mathbb{P}(\mathcal{L}(y) \geq \eta | H = H_1) = \begin{cases} 0, & \text{if } \frac{4}{3} < \eta, \\ \frac{2}{3}, & \text{if } \frac{2}{3} < \eta \leq \frac{4}{3}, \\ 1, & \text{if } \eta \leq \frac{2}{3}. \end{cases}$$

Hence the operating characteristic (ROC) consists of exactly three achievable (P_F, P_D) points:

$$(0, 0), \quad \left(\frac{1}{2}, \frac{2}{3}\right), \quad (1, 1).$$

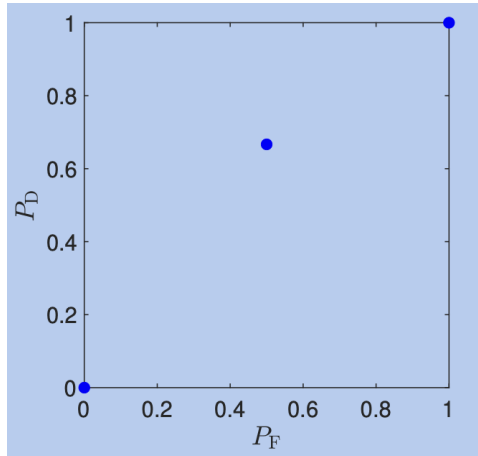


Figure 2: Operating characteristic may be discontinuous in η

It is important to emphasize that discontinuities in the operating characteristic $\{(P_F(\eta), P_D(\eta)) : \eta \geq 0\}$ arise due to discrete components in the distribution of the likelihood ratio $\mathcal{L}(y)$ under the constituent hypotheses (note definition of $P_F(\eta)$, $P_D(\eta)$ to see this). In particular, whether the OC is discontinuous is unrelated to whether there are discontinuities in the likelihood ratio $\mathcal{L}(y)$ as a function of the observed data y .

4.2 NP value $P_D^{\text{NP}}(\alpha)$ is discontinuous for discrete $\mathcal{L}(\cdot)$

Consider the Neyman–Pearson problem

$$P_D^{\text{NP}}(\alpha) := \max_{\hat{H}(\cdot)} P_D \quad \text{subject to} \quad P_F \leq \alpha,$$

and restrict $\hat{H}(\cdot)$ to be a (deterministic) likelihood-ratio test (LRT) with threshold η . If the likelihood ratio $\mathcal{L}(y)$ has discrete components, then as η varies the achievable operating points $(P_F(\eta), P_D(\eta))$ form a discrete set (the OC-LRT). As α increases continuously, the selected OC-LRT point stays fixed until α crosses the next achievable P_F level, at which time the optimizer jumps to a new point with a larger P_D . This implies that $P_D^{\text{NP}}(\alpha)$ (under the restriction to deterministic LRTs) is a step function of α .

4.3 Randomized decision rules

Fix two LRTs $\hat{H}'(\cdot)$ and $\hat{H}''(\cdot)$ corresponding to thresholds η' and η'' . Let $u \in \{\eta', \eta''\}$ be independent of y with

$$\mathbb{P}(u = \eta') = p, \quad \mathbb{P}(u = \eta'') = 1 - p, \quad p \in [0, 1].$$

Define the *randomized decision rule*

$$\hat{H}(y) = \begin{cases} \hat{H}'(y), & \text{if } u = \eta', \\ \hat{H}''(y), & \text{if } u = \eta''. \end{cases}$$

More generally, a randomized decision rule is characterized by a function $q(\cdot)$ with $q(y) \in [0, 1]$, where

$$q(y) := \mathbb{P}(\hat{H}(y) = H_1 \mid y = y), \quad \mathbb{P}(\hat{H}(y) = H_0 \mid y = y) = 1 - q(y).$$

4.4 Randomizing between two LRT thresholds yields convex combinations on the OC

Under the NP criterion, we can increase achievable operating points by *randomization* between likelihood thresholds. In particular, the detection and false-alarm probabilities of a randomized decision rule $\hat{H}(y)$ are the convex combinations

$$P_D = p P_D(\eta') + (1 - p) P_D(\eta''), \quad P_F = p P_F(\eta') + (1 - p) P_F(\eta'').$$

Equivalently, the operating point (P_F, P_D) lies on the line segment joining $(P_F(\eta''), P_D(\eta''))$ and $(P_F(\eta'), P_D(\eta'))$ and as p increases from 0 to 1, the operating point moves from the η'' -point to the η' -point.

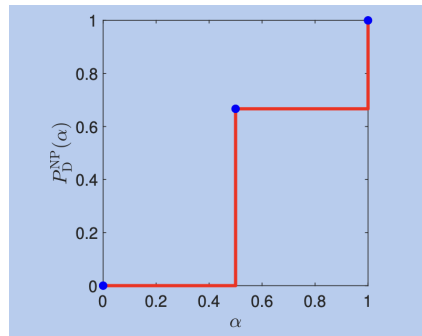


Figure 3: Oringal LRT-OC

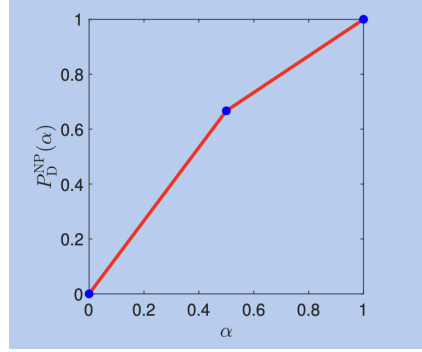


Figure 4: Randomization leads to convex combination

4.5 Randomization cannot beat the Bayes risk of the Bayes optimal LRT

In binary Bayesian hypothesis testing, allowing randomized decision rules does *not* reduce the minimum achievable Bayes risk: a deterministic rule already attains the Bayes-optimal risk among *all* (possibly randomized) tests. The Bayes risk of the randomized decision rule $q(\cdot)$ is

$$\varphi(q) := \mathbb{E}[C(H, \hat{H}(y))].$$

By iterated expectation,

$$\varphi(q) = \mathbb{E}[\mathbb{E}[C(H, \hat{H}(y)) | y]] = \mathbb{E}[\tilde{\varphi}(y)], \quad \tilde{\varphi}(y) := \mathbb{E}[C(H, \hat{H}(y)) | y = y].$$

Therefore, to minimize $\varphi(q)$ it suffices to minimize $\tilde{\varphi}(y)$ *pointwise* for each fixed y . Fix $y = y$. Under the posterior on H given $y = y$, define the two posterior expected costs

$$r_0(y) := \mathbb{E}[C(H, H_0) | y = y], \quad r_1(y) := \mathbb{E}[C(H, H_1) | y = y].$$

If we randomize at y with probability $q(y)$ of choosing H_1 , then

$$\tilde{\varphi}(y) = (1 - q(y)) r_0(y) + q(y) r_1(y),$$

which is a convex combination of $r_0(y)$ and $r_1(y)$. Hence the minimum over $q(y) \in [0, 1]$ is achieved at an endpoint:

$$\min_{q(y) \in [0, 1]} \tilde{\varphi}(y) = \min\{r_0(y), r_1(y)\},$$

so there exists an optimal rule that is *deterministic* at each y :

$$\hat{H}(y) \in \arg \min_{h \in \{H_0, H_1\}} \mathbb{E}[C(H, h) | y = y].$$

4.6 False positive and detection probabilities under randomized rules

A randomized binary decision rule is specified by

$$q(y) := p_{\hat{H}|y}(\hat{H} = H_1 | y) \in [0, 1],$$

therefore

$$P_F(q) = \mathbb{P}(\hat{H}(y) = H_1 | H = H_0) = \int q(y) p_{y|H}(y | H = H_0) dy = \mathbb{E}[q(y) | H = H_0],$$

$$P_D(q) = \mathbb{P}(\hat{H}(y) = H_1 \mid H = H_1) = \int q(y) p_{y|H}(y \mid H = H_1) dy = \mathbb{E}[q(y) \mid H = H_1].$$

4.7 Neyman–Pearson lemma, full version (with randomization)

Neyman–Pearson lemma: Given distributions of data under hypotheses

$$p_{y|H}(\cdot \mid H_0) \quad \text{and} \quad p_{y|H}(\cdot \mid H_1)$$

and any $\alpha \in [0, 1]$, there exist threshold η and $p \in [0, 1]$ such that the randomized rule

$$q_\star(y) = \begin{cases} 0, & \text{if } \mathcal{L}(y) < \eta, \\ p, & \text{if } \mathcal{L}(y) = \eta, \\ 1, & \text{if } \mathcal{L}(y) > \eta, \end{cases}$$

satisfies

$$P_F(q_\star) = \alpha, \quad P_D(q_\star) \geq P_D(q) \text{ for any } q(\cdot) \text{ with } P_F(q) \leq \alpha.$$

Earlier, when we restricted attention to *deterministic* LRTs, the set of achievable (P_F, P_D) points can be discrete when $\mathcal{L}(y)$ has atoms. Then for many values of α , no deterministic LRT achieves $P_F = \alpha$; the optimizer picks the largest achievable $P_F < \alpha$, making the NP value $P_D^{\text{NP}}(\alpha)$ a step function of α (hence discontinuous).

The full NP lemma resolves this by allowing *randomization on the boundary* $\{\mathcal{L}(y) = \eta\}$. By choosing the tie-breaking probability p , we can tune $P_F(q_\star)$ continuously to hit *any* target level $\alpha \in [0, 1]$ exactly.

4.8 The Neyman–Pearson function $\zeta_{\text{NP}}(\alpha)$

By the theorem above, the best performance curves are achievable *without* restricting the form of the test: optimizing over all tests is equivalent to optimizing over (possibly randomized) likelihood-ratio tests.

We define the *Neyman–Pearson function* of the model as the mapping $\zeta_{\text{NP}}(\cdot) : [0, 1] \rightarrow [0, 1]$ of false-alarm constraint levels to detection probability of the optimal randomized decision rule at false-alarm level $\alpha \in [0, 1]$,

$$\zeta_{\text{NP}}(\alpha) := P_D(q_\star), \quad \text{with } q_\star \text{ chosen so that } P_F(q_\star) = \alpha.$$

Equivalently, $\zeta_{\text{NP}}(\alpha)$ is the maximum achievable detection probability among all (possibly randomized) decision rules with false-alarm probability exactly α .

4.9 Efficient frontier and the Neyman–Pearson function

Given the Neyman–Pearson function

$$\zeta_{\text{NP}}(\alpha) := \max_{q(\cdot)} P_D(q) \quad \text{subject to} \quad P_F(q) \leq \alpha,$$

where $q(y) = p_{\hat{H}|y}(H_1 \mid y)$ parametrizes (possibly randomized) tests. Then the *efficient frontier* of achievable operating points is

$$\mathcal{F}_{p_{y|H}} := \{(P_F, P_D) : P_D = \zeta_{\text{NP}}(P_F)\},$$

because by optimality no test can achieve $P_D > \zeta_{\text{NP}}(P_F)$ at a given P_F .

4.10 ζ_{NP} Property I on endpoints

- $\zeta_{\text{NP}}(1) = 1$, i.e., $(1, 1) \in \mathcal{F}_{p_{y|H}}$ (declare H_1 always).
- $\zeta_{\text{NP}}(P_F) \geq P_F$, hence every frontier point satisfies $P_D \geq P_F$.

4.11 ζ_{NP} Property II on concavity

The function $\zeta_{\text{NP}}(\cdot)$ is concave. Indeed, if two tests achieve operating points $(P_F^{(1)}, P_D^{(1)})$ and $(P_F^{(2)}, P_D^{(2)})$, then randomizing between them with probability $\lambda \in [0, 1]$ yields

$$P_F = \lambda P_F^{(1)} + (1 - \lambda) P_F^{(2)}, \quad P_D = \lambda P_D^{(1)} + (1 - \lambda) P_D^{(2)},$$

so the achievable region is convex, and the upper boundary $P_D = \zeta_{\text{NP}}(P_F)$ is therefore concave.

4.12 ζ_{NP} Property III: slope interpretation

Let η_0 be an LRT threshold such that the likelihood ratio has no point mass at η_0 under either hypothesis:

$$\mathbb{P}(\mathcal{L}(y) = \eta_0 \mid H = H_0) = \mathbb{P}(\mathcal{L}(y) = \eta_0 \mid H = H_1) = 0.$$

Then the corresponding deterministic LRT operating point lies on the frontier:

$$P_D(\eta_0) = \zeta_{\text{NP}}(P_F(\eta_0)),$$

and the tail probabilities

$$P_F(\eta) = \mathbb{P}(\mathcal{L}(y) \geq \eta \mid H = H_0), \quad P_D(\eta) = \mathbb{P}(\mathcal{L}(y) \geq \eta \mid H = H_1)$$

are continuous in η at $\eta = \eta_0$. At such a “smooth” operating point (no atom at η_0), the Neyman–Pearson function is differentiable at $P_F(\eta_0)$ and its slope equals the corresponding LRT threshold:

$$\dot{\zeta}_{\text{NP}}(P_F(\eta_0)) = \eta_0.$$

Equivalently, the supporting line to the concave frontier at that point has slope η_0 , linking the local tradeoff $\frac{dP_D}{dP_F}$ to the LRT threshold.

4.13 OVERVIEW: Bayesian vs. Neyman–Pearson formulations for binary testing

We observe data $y \in \mathcal{Y}$ and must decide between H_0 and H_1 . A deterministic decision rule is a function $f(\cdot) : \mathcal{Y} \mapsto \{H_0, H_1\}$. On the other hand a randomized rule is specified by

$$r(y) := \mathbb{P}(\hat{H} = H_1 \mid y = y) \in [0, 1], \quad \mathbb{P}(\hat{H} = H_0 \mid y = y) = 1 - r(y).$$

- *Bayesian formulation (prior + costs)*: Model the hypothesis as a random variable $H \in \{H_0, H_1\}$ with prior $\pi_1 = \mathbb{P}(H = H_1)$ (and $\pi_0 = 1 - \pi_1$), and specify costs $C(\mathcal{H}_j, H_i)$ for deciding $\hat{H} = H_i$ when $H = H_j$. Choose a rule to minimize Bayes risk:

$$\hat{H}_{\text{B}}(\cdot) \in \arg \min_{f(\cdot)} \varphi_{\text{B}}(f), \quad \varphi_{\text{B}}(f) = \mathbb{E}[C(H, f(y))].$$

Allowing randomization does not improve the minimum attainable Bayes risk; a Bayes optimal rule can be taken deterministic and is an LRT with a threshold determined by priors and costs.

- *Neyman–Pearson formulation (no prior or costs, set error constraint):* Do *not* assume a prior or costs. Fix a false-alarm tolerance $\alpha \in (0, 1)$ and choose a rule to maximize detection subject to the constraint:

$$\hat{H}_{\text{NP}}(\cdot) \in \arg \max_{f(\cdot): P_{\text{F}}(f) \leq \alpha} P_{\text{D}}(f).$$

For a randomized rule $r(\cdot)$, the operating probabilities are linear in r :

$$P_{\text{F}}(r) = \mathbb{E}[r(y) \mid H = H_0], \quad P_{\text{D}}(r) = \mathbb{E}[r(y) \mid H = H_1].$$

Here randomization *can* matter: it may be needed to hit the constraint $P_{\text{F}} = \alpha$ exactly (especially with discrete components)

5 Minimax

5.1 Minimax formulation (costs known, prior unknown)

Sometimes we can meaningfully specify costs $C_{ij} = C(H_j, H_i)$ for $i, j \in \{0, 1\}$, but we do *not* want to commit to a prior $p = \mathbb{P}(H = H_1)$. The minimax approach models this as a zero sum game in which:

- We choose a possibly randomized decision rule $r(\cdot) : \mathcal{Y} \mapsto [0, 1]$, where

$$r(y) = \mathbb{P}(\hat{H} = H_1 \mid \mathcal{Y} = y)$$

- “Nature” chooses the most adverse prior $p \in [0, 1]$.

For a fixed prior p and rule r , the Bayes risk can be written as

$$\varphi(p, r) := \mathbb{E}[C(H, \hat{H})] = (1 - p) \mathbb{E}[C(H, \hat{H}) \mid H = H_0] + p \mathbb{E}[C(H, \hat{H}) \mid H = H_1],$$

and let us define the conditional risks as

$$\varphi_0(r) := \mathbb{E}[C(H, \hat{H}) \mid H = H_0], \quad \varphi_1(r) := \mathbb{E}[C(H, \hat{H}) \mid H = H_1].$$

The minimax decision rule is then

$$r_M(\cdot) \in \arg \min_{r: \mathcal{Y} \rightarrow [0, 1]} \varphi_M(r), \quad \varphi_M(r) := \max_{p \in [0, 1]} \varphi(p, r).$$

A decision rule r is called an *equalizer rule* if $\varphi_0(r) = \varphi_1(r)$, i.e., the risk $\varphi(p, r)$ does not depend on the prior p . In a minimax setting, equalizer rules are natural candidates because they “hedge” against the worst-case prior.

5.2 Matched Bayes LRT under prior p (0–1 costs)

For 0–1 costs and prior $p = \mathbb{P}(H = H_1)$, the Bayes decision rule is an LRT with threshold

$$\eta(p) = \frac{C_{10} - C_{00}}{C_{01} - C_{11}} \cdot \frac{P_0}{P_1} = \frac{1 - p}{p}.$$

Let $P_F(\eta(\cdot))$ and $P_D(\eta(\cdot))$ denote the false-alarm and detection probabilities achieved by the LRT with threshold $\eta(\cdot)$. The resulting probability of error is

$$P_e(p) = (1 - p) P_F(\eta(p)) + p(1 - P_D(\eta(p))).$$

5.3 Mismatched Bayes rule: design for q , operate under p

Suppose the true prior is still p , but we use the Bayes LRT designed for a different prior q . Then the rule uses threshold

$$\eta(q) = \frac{1 - q}{q},$$

and its (mismatched) probability of error is

$$P_e(p; q) = (1 - p) P_F(\eta(q)) + p(1 - P_D(\eta(q))).$$

For each fixed q , $P_e(p; q)$ is an affine (linear) function of p .

5.4 Minimax choice within the LRT family and equalizer condition

In a minimax setting where nature chooses p after we choose the (LRT) design parameter q , we want to minimize

$$\max_{p \in [0,1]} P_e(p; q)$$

which is the worst that nature can choose p to be given our choice of q . Since $P_e(p; q)$ is affine in p , the worst case occurs at an endpoint:

$$\max_{p \in [0,1]} P_e(p; q) = \max \{P_e(0; q), P_e(1; q)\} = \max \{P_F(\eta(q)), 1 - P_D(\eta(q))\}.$$

Therefore, our best choice of q equalizes these two endpoint errors: pick q such that

$$P_F(\eta(q)) = 1 - P_D(\eta(q)).$$

5.5 Randomized LRT with general costs: Bayes vs Neyman-Pearson

With general costs $C_{ij} = C(H_j, H_i)$ and design prior $p \in [0, 1]$, a randomized LRT can be written as

$$r_B(y; p, \lambda) := \begin{cases} 1, & \mathcal{L}(y) > \eta(p), \\ \lambda, & \mathcal{L}(y) = \eta(p), \\ 0, & \mathcal{L}(y) < \eta(p), \end{cases} \quad \eta(p) := \frac{1-p}{p} \cdot \frac{C_{10} - C_{00}}{C_{01} - C_{11}}, \quad \lambda \in [0, 1].$$

where p is the prior and $\lambda \in [0, 1]$ is the randomization parameter. Recall that randomization cannot improve aBayes decision rule, but nor does it incur additional cost. At the same time, recall from our development of Neyman-Pearson hypothesis testing that, in general, the associated false-alarm and detection probabilities

$$P_F^B(p, \lambda) = \mathbb{E}[r_B(y; p, \lambda) \mid H = H_0] = \mathbb{P}(\mathcal{L}(y) > \eta(p) \mid H = H_0) + \lambda \mathbb{P}(\mathcal{L}(y) = \eta(p) \mid H = H_0),$$

$$P_D^B(p, \lambda) = \mathbb{E}[r_B(y; p, \lambda) \mid H = H_1] = \mathbb{P}(\mathcal{L}(y) > \eta(p) \mid H = H_1) + \lambda \mathbb{P}(\mathcal{L}(y) = \eta(p) \mid H = H_1).$$

do depend on the randomization λ , and that every point on the efficient frontier is achievable by a randomized test of this form.

5.6 Notation for mismatched Bayes risk in randomized LRT

We define the *mismatched Bayes risk* as the risk under true prior p when using the Bayes rule designed for prior q :

$$\varphi_B(p, q, \lambda) := \varphi(p, r_B(\cdot; q, \lambda)).$$

The *matched* Bayes risk is

$$\varphi_B^*(p) := \varphi_B(p, p, \lambda),$$

and for the matched case the value does not depend on λ (randomization only affects tie-breaking on a set where both decisions have equal conditional Bayes cost).

5.7 Claim 5.1: key properties and brief proof sketches

Fix a randomized Bayes LRT $r_B(\cdot; q, \lambda)$ designed for prior $q \in [0, 1]$ (and tie-randomization $\lambda \in [0, 1]$). Let

$$\varphi_B(p, q, \lambda) := \varphi(p, r_B(\cdot; q, \lambda)), \quad \varphi_B^*(p) := \min_r \varphi(p, r).$$

Define the conditional Bayes risks of the rule $r_B(\cdot; q, \lambda)$ by

$$\varphi_B^0(q, \lambda) := \mathbb{E}[C(H, \hat{H}) \mid H = H_0], \quad \varphi_B^1(q, \lambda) := \mathbb{E}[C(H, \hat{H}) \mid H = H_1],$$

so that, in terms of $P_F^B(q, \lambda)$ and $P_D^B(q, \lambda)$,

$$\varphi_B^0(q, \lambda) = C_{00}(1 - P_F^B(q, \lambda)) + C_{10}P_F^B(q, \lambda), \quad \varphi_B^1(q, \lambda) = C_{01}(1 - P_D^B(q, \lambda)) + C_{11}P_D^B(q, \lambda).$$

P1. $\varphi_B(\cdot, q, \lambda)$ is affine in p .

Proof sketch. Use the law of total expectation under prior p :

$$\varphi_B(p, q, \lambda) = (1 - p) \varphi_B^0(q, \lambda) + p \varphi_B^1(q, \lambda) = \varphi_B^0(q, \lambda) + p(\varphi_B^1(q, \lambda) - \varphi_B^0(q, \lambda)),$$

which is linear in $p \in [0, 1]$.

P2. Lower bound: $\varphi_B(p, q, \lambda) \geq \varphi_B^*(p)$, with equality when $q = p$.

Proof sketch. By definition, $\varphi_B^*(p)$ is the minimum Bayes risk over *all* decision rules under prior p . Since $r_B(\cdot; q, \lambda)$ is just one feasible rule,

$$\varphi_B(p, q, \lambda) = \varphi(p, r_B(\cdot; q, \lambda)) \geq \min_r \varphi(p, r) = \varphi_B^*(p).$$

If $q = p$, then $r_B(\cdot; p, \lambda)$ is Bayes-optimal for that prior, so it achieves the minimum and equality holds.

P3. $\varphi_B^*(\cdot)$ is concave and continuous on $[0, 1]$.

Proof sketch (concavity). Fix $p_1, p_2 \in [0, 1]$ and $\theta \in [0, 1]$, and set $p = \theta p_1 + (1 - \theta)p_2$. For any fixed rule r , the Bayes risk is affine in p :

$$\varphi(p, r) = \theta \varphi(p_1, r) + (1 - \theta) \varphi(p_2, r).$$

Taking minima over r and using $\min_r(\theta a_r + (1 - \theta)b_r) \geq \theta \min_r a_r + (1 - \theta) \min_r b_r$ gives

$$\varphi_B^*(p) \geq \theta \varphi_B^*(p_1) + (1 - \theta) \varphi_B^*(p_2),$$

so φ_B^* is concave.

Proof sketch (continuity). A concave function on an interval is continuous on the interior $(0, 1)$. Continuity at the endpoints follows by bounding $\varphi_B^*(p)$ between endpoint risks and using that $\varphi(p, r)$ is affine in p for each fixed r , hence cannot jump at 0 or 1.

P4. Endpoint values: $\varphi_B^*(0) = C_{00}$ and $\varphi_B^*(1) = C_{11}$.

Proof sketch. If $p = 0$, then $H = H_0$ almost surely, so the Bayes risk reduces to the conditional risk under H_0 . The decision rule that always declares $\hat{H} = H_0$ achieves risk C_{00} , hence $\varphi_B^*(0) \leq C_{00}$. Conversely, no rule can achieve risk below C_{00} when $H = H_0$ is certain, because C_{00} is precisely the cost incurred when deciding H_0 under H_0 . Therefore $\varphi_B^*(0) = C_{00}$.

Similarly, if $p = 1$, then $H = H_1$ almost surely. The rule that always declares $\hat{H} = H_1$ achieves risk C_{11} , and no rule can do better when H_1 is certain. Thus $\varphi_B^*(1) = C_{11}$.

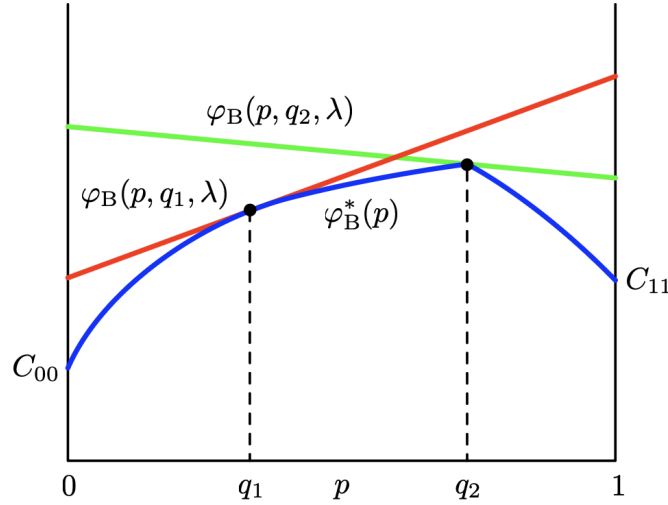


Figure 5.3: Mismatched and matched Bayes risk relationships. The blue curve indicates the Bayes risk as a function of the actual prior p for a Bayes decision rule that is correctly matched to p . The red and green lines indicate the mismatch Bayes risk when the Bayes decision rule is designed for prior q_1 and q_2 , respectively (but the actual prior is p).

5.8 Setup: what is given vs. what we choose

Given:

- Two hypotheses $H \in \{H_0, H_1\}$ and data models $p_{Y|H}(\cdot | H_i)$ for $i \in \{0, 1\}$.
- A cost matrix $C_{ij} = C(H_j, H_i)$ for $i, j \in \{0, 1\}$, where $C(H_j, H_i)$ is the cost of deciding $\hat{H} = H_i$ when the true hypothesis is $H = H_j$.

We choose: a (possibly randomized) decision rule $r : \mathcal{Y} \rightarrow [0, 1]$ with

$$r(y) := \mathbb{P}(\hat{H} = H_1 | y = y), \quad \mathbb{P}(\hat{H} = H_0 | y = y) = 1 - r(y).$$

Nature chooses (adversarially): an unknown prior

$$p \in [0, 1], \quad p := \mathbb{P}(H = H_1), \quad (1 - p) = \mathbb{P}(H = H_0).$$

For any rule r , define its conditional risks

$$\varphi_0(r) := \mathbb{E}[C(H, \hat{H}) | H = H_0], \quad \varphi_1(r) := \mathbb{E}[C(H, \hat{H}) | H = H_1],$$

and the Bayes risk under prior p as

$$\varphi(p, r) := (1 - p) \varphi_0(r) + p \varphi_1(r).$$

The *minimax* problem is

$$\min_{r: \mathcal{Y} \rightarrow [0, 1]} \max_{p \in [0, 1]} \varphi(p, r)$$

where we use the notation $r_M(\cdot) = r_*(\cdot)$ such that

$$r_* \in \arg \min_{r: \mathcal{Y} \rightarrow [0, 1]} \max_{p \in [0, 1]} \varphi(p, r),$$

and r_M is simply a name for such an optimizer.

5.9 Theorem 5.1 (characterization of the minimax rule)

Theorem. There exist $(p_*, \lambda_*) \in [0, 1]^2$ such that a minimax-optimal rule is a *randomized Bayes likelihood-ratio test* (LRT)

$$r_M(\cdot) = r_B(\cdot; p_*, \lambda_*),$$

where $r_B(\cdot; p, \lambda)$ denotes the Bayes LRT designed for prior p (with tie-randomization parameter λ on the threshold set). Moreover, (p_*, r_*) form a saddle point, meaning that

$$\min_r \max_{p \in [0,1]} \varphi(p, r) = \max_{p \in [0,1]} \min_r \varphi(p, r) = \varphi(p_*, r_*).$$

The prior p_* is called the *least-favorable (minimax) prior*.

5.10 How to find (p_*, λ_*) (what you compute)

Define the *matched Bayes risk* (Bayes-optimal value if the prior were p) by

$$\varphi_B^*(p) := \min_{r: \mathcal{Y} \rightarrow [0,1]} \varphi(p, r).$$

where we recall that $\varphi(p, r)$ is the Bayes risk under prior p is

$$\varphi(p, r) := (1 - p) \varphi_0(r) + p \varphi_1(r).$$

Also define the Neyman–Pearson efficient frontier

$$\zeta_{\text{NP}}(P_F) := \sup \left\{ P_D(r) : r : \mathcal{Y} \rightarrow [0, 1], P_F(r) \leq P_F \right\},$$

where, for any randomized rule r ,

$$P_F(r) := \mathbb{E}[r(y) \mid H = H_0], \quad P_D(r) := \mathbb{E}[r(y) \mid H = H_1].$$

From the costs, form the *equalizer line* (a linear function of P_F)

$$g_M(P_F) := \frac{C_{01} - C_{00}}{C_{01} - C_{11}} - \frac{C_{10} - C_{00}}{C_{01} - C_{11}} P_F.$$

- *Interior case (solve an intersection):* if there exists $P_F^* \in [0, 1]$ such that

$$\zeta_{\text{NP}}(P_F^*) = g_M(P_F^*),$$

then one can choose (p_*, λ_*) so that the Bayes LRT $r_B(\cdot; p_*, \lambda_*)$ attains the operating point

$$P_F(r_B(\cdot; p_*, \lambda_*)) = P_F^*, \quad P_D(r_B(\cdot; p_*, \lambda_*)) = \zeta_{\text{NP}}(P_F^*).$$

Geometrically: the minimax operating point is where the NP frontier intersects the equalizer line.

- *Endpoint case (no intersection):* if $\zeta_{\text{NP}}(P_F) > g_M(P_F)$ for all $P_F \in [0, 1]$, then $p_* = 0$. If on the other hand $\zeta_{\text{NP}}(P_F) < g_M(P_F)$ for all $P_F \in [0, 1]$, then $p_* = 1$.

5.11 Two exam-useful corollaries (quick checks)

- *Least-favorable prior maximizes matched Bayes risk:* if $r_B(\cdot; p_*, \lambda_*)$ is minimax, then

$$p_* \in \arg \max_{p \in [0,1]} \varphi_B^*(p).$$

- *Equalizer condition (interior solution)*: if the least-favorable prior satisfies $p_\star \in (0, 1)$, then the corresponding minimax Bayes LRT equalizes conditional risks:

$$\varphi_0(r_B(\cdot; p_\star, \lambda_\star)) = \varphi_1(r_B(\cdot; p_\star, \lambda_\star)).$$

(If $p_\star \in \{0, 1\}$, the minimax solution is at an endpoint and tie-randomization is immaterial.)

5.12 How to plot and find the intersection in the (P_F, P_D) plane

1. *Trace the NP frontier* For each threshold η of the likelihood ratio $\mathcal{L}(y) = \frac{p_{y|H}(y|H_1)}{p_{y|H}(y|H_0)}$ (and tie-randomization λ on $\{\mathcal{L}(y) = \eta\}$), form the randomized LRT

$$r_\eta(y) = \begin{cases} 1, & \mathcal{L}(y) > \eta, \\ \lambda, & \mathcal{L}(y) = \eta, \\ 0, & \mathcal{L}(y) < \eta, \end{cases}$$

and compute

$$P_F(\eta, \lambda) = \mathbb{E}[r_\eta(y) \mid H = H_0], \quad P_D(\eta, \lambda) = \mathbb{E}[r_\eta(y) \mid H = H_1].$$

Plot the resulting points $(P_F(\eta, \lambda), P_D(\eta, \lambda))$; as η varies you trace the ROC frontier (randomization fills in gaps if there are discrete components).

2. *Plot the equalizer line.* On the same axes, plot

$$P_D = g_M(P_F) = \frac{C_{01} - C_{00}}{C_{01} - C_{11}} - \frac{C_{10} - C_{00}}{C_{01} - C_{11}} P_F, \quad P_F \in [0, 1].$$

3. *Read off the minimax operating point.* If the line intersects the ROC frontier, the intersection point $(P_F^\star, P_D^\star)$ is the minimax operating point:

$$P_D^\star = \zeta_{\text{NP}}(P_F^\star) = g_M(P_F^\star).$$

Choose (η, λ) that achieves $(P_F^\star, P_D^\star)$; that LRT is minimax.

4. *If there is no intersection.* If $\zeta_{\text{NP}}(P_F) > g_M(P_F)$ for all P_F , the minimax prior is $p_\star = 0$ (nature concentrates on H_0); if $\zeta_{\text{NP}}(P_F) < g_M(P_F)$ for all P_F , then $p_\star = 1$.

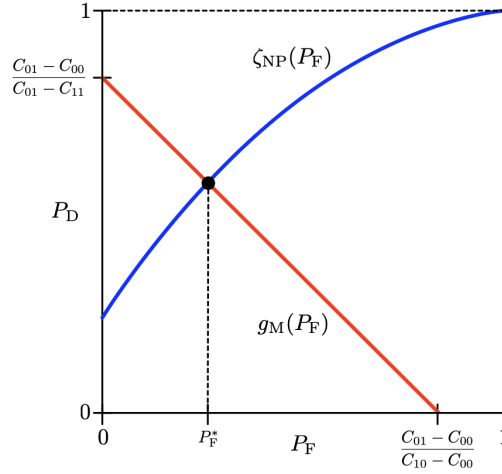


Figure 5: The intersection (P_F^*, P_D^*) is the minimax operating point: $P_D^* = \zeta_{\text{NP}}(P_F^*) = g_M(P_F^*)$.

5.13 From an intersection point P_F^* to (p_*, λ_*)

Assume we have identified an intersection point $P_F^* \in [0, 1]$ such that

$$\zeta_{\text{NP}}(P_F^*) = g_M(P_F^*).$$

By definition of the NP frontier, there exists a (possibly randomized) likelihood-ratio test (LRT) with some threshold η_* and tie-randomization $\lambda_* \in [0, 1]$ whose operating point equals

$$P_F(\eta_*, \lambda_*) = P_F^*, \quad P_D(\eta_*, \lambda_*) = \zeta_{\text{NP}}(P_F^*).$$

Concretely, take η_* so that

$$P_F(\eta_*, 0) \leq P_F^* \leq P_F(\eta_*, 1),$$

where $P_F(\eta, 0) = \mathbb{P}(\mathcal{L}(y) > \eta \mid H = H_0)$ and $P_F(\eta, 1) = \mathbb{P}(\mathcal{L}(y) \geq \eta \mid H = H_0)$, and then set

$$\lambda_* = \frac{P_F^* - \mathbb{P}(\mathcal{L}(y) > \eta_* \mid H = H_0)}{\mathbb{P}(\mathcal{L}(y) = \eta_* \mid H = H_0)} \in [0, 1],$$

(with the convention that if $\mathbb{P}(\mathcal{L}(y) = \eta_* \mid H = H_0) = 0$, then no randomization is needed and any λ_* works).

It remains to choose p_* so that the Bayes LRT threshold induced by p_* equals η_* . Recall the Bayes LRT threshold for prior p and costs C_{ij} is

$$\eta(p) = \frac{1-p}{p} \cdot \frac{C_{10} - C_{00}}{C_{01} - C_{11}}.$$

Thus we choose p_* by solving $\eta(p_*) = \eta_*$, i.e.,

$$\frac{1-p_*}{p_*} = \eta_* \cdot \frac{C_{01} - C_{11}}{C_{10} - C_{00}}, \quad \implies \quad p_* = \frac{1}{1 + \eta_* \frac{C_{01} - C_{11}}{C_{10} - C_{00}}}.$$

With this choice, the Bayes LRT $r_B(\cdot; p_*, \lambda_*)$ uses threshold η_* and (by the construction of λ_*) attains the desired operating point

$$P_F(r_B(\cdot; p_*, \lambda_*)) = P_F^*, \quad P_D(r_B(\cdot; p_*, \lambda_*)) = \zeta_{\text{NP}}(P_F^*).$$

6 Bayesian parameter estimation

6.1 Bayesian estimation: setup and posterior

In Bayesian estimation, the unknown parameter is modeled as a random variable $x \in \mathcal{X}$, and the observation is a random variable $y \in \mathcal{Y}$. The Bayesian model is specified by:

- a prior distribution $p_x(\cdot)$ on the unknown parameter x
- a likelihood (data model) $p_{y|x}(\cdot | x)$ describing how the data y depends on the parameter x

After observing $y = y$, we update our belief about x via Bayes' rule:

$$p_{x|y}(x | y) = \frac{p_{y|x}(y | x) p_x(x)}{\int_{\mathcal{X}} p_{y|x}(y | x') p_x(x') dx'}.$$

The posterior $p_{x|y}(\cdot | y)$ summarizes all information about x contained in the prior and the observed data.

6.2 Estimators and Bayes risk

An *estimator* is a deterministic function $\hat{x}(\cdot) : \mathcal{Y} \rightarrow \mathcal{X}$; given observation y , it outputs the estimate $\hat{x}(y)$. To choose an estimator, specify a loss function $C(x, \hat{x})$ (cost of estimating x by $\hat{x}$), and minimize the expected loss (Bayes risk):

$$\hat{x}(\cdot) \in \arg \min_{f(\cdot)} \mathbb{E}[C(x, f(y))],$$

where the expectation is with respect to the joint law $p_{x,y}(x, y) = p_x(x) p_{y|x}(y | x)$. Equivalently (pointwise form), the Bayes-optimal estimator minimizes posterior expected loss for each realized y :

$$\hat{x}(y) \in \arg \min_{\hat{x} \in \mathcal{X}} \mathbb{E}[C(x, \hat{x}) | y = y].$$

6.3 Pointwise Bayes optimality

Let $x \in \mathcal{X}$ be the unknown parameter with prior $p_x(\cdot)$, and let $y \in \mathcal{Y}$ be the observation with likelihood $p_{y|x}(\cdot | x)$. An estimator is a deterministic function $\hat{x}(\cdot) : \mathcal{Y} \rightarrow \mathcal{X}$, and given a loss function $C(x, \hat{x})$ the Bayes risk is

$$R(\hat{x}) := \mathbb{E}[C(x, \hat{x}(y))].$$

Using $p_{x,y}(x, y) = p_{x|y}(x | y) p_y(y)$,

$$R(\hat{x}) = \int_{\mathcal{Y}} \left(\int_{\mathcal{X}} C(x, \hat{x}(y)) p_{x|y}(x | y) dx \right) p_y(y) dy.$$

Since $p_y(y) \geq 0$, minimizing $R(\hat{x})$ is equivalent to minimizing the inner (posterior) expected loss for each fixed y :

$$\hat{x}(y) \in \arg \min_{a \in \mathcal{X}} \int_{\mathcal{X}} C(x, a) p_{x|y}(x | y) dx,$$

with the integral replaced by a sum if $\mathcal{X}$ is discrete.

6.4 Three canonical Bayes estimators

These follow by plugging different losses into the pointwise rule above (assume scalar x for MAE/MAP; the MSE form holds for vectors).

- *MAE (using absolute error loss):* If $C(x, a) = |x - a|$, then the Bayes estimator is any posterior median:

$$\hat{x}_{\text{MAE}}(y) \in \text{median}\left(p_{x|y}(\cdot | y)\right).$$

- *MAP (using MUC loss):* For $\epsilon > 0$, define the minimum uniform cost (MUC) loss

$$C_\epsilon(x, a) = \begin{cases} 1, & |x - a| > \epsilon, \\ 0, & \text{otherwise.} \end{cases}$$

As $\epsilon \rightarrow 0$, minimizing the posterior expected C_ϵ selects a posterior mode, i.e., the MAP estimator:

$$\hat{x}_{\text{MAP}}(y) \in \arg \max_{a \in \mathcal{X}} p_{x|y}(a | y).$$

- *BLS (using MSE loss):* If $C(x, a) = \|x - a\|^2$, then the Bayes least-squares (BLS), also called the MMSE estimator, is the posterior mean:

$$\hat{x}_{\text{BLS}}(y) = \mathbb{E}[x | y = y].$$

6.5 Bias–variance tradeoff

Let $x \in \mathbb{R}^N$ be the parameter and let y be the observation. For an estimator $\hat{x}(y)$, define the estimation error

$$e := \hat{x}(y) - x.$$

Now define the mean error as

$$b := \mathbb{E}[e],$$

and the error covariance as

$$\Lambda_e := \mathbb{E}[(e - b)(e - b)^\top].$$

Then the error second moment (error correlation) decomposes as

$$\mathbb{E}[ee^\top] = \Lambda_e + bb^\top,$$

since the cross terms vanish under $\mathbb{E}[e] = b$. Taking the trace gives the mean-squared error:

$$\text{MSE}(\hat{x}) := \mathbb{E}[\|\hat{x}(y) - x\|^2] = \text{tr}(\mathbb{E}[ee^\top]) = \text{tr}(\Lambda_e) + \|b\|^2.$$

Thus MSE splits into a *variance term* $\text{tr}(\Lambda_e)$ and a *bias term* $\|b\|^2$.

6.6 BLS is unbiased

Let $\hat{x}_{\text{BLS}}(y) := \mathbb{E}[x | y]$ and define the error $e := \hat{x}_{\text{BLS}}(y) - x$. The bias is

$$b_{\text{BLS}} := \mathbb{E}[e] = \mathbb{E}[\mathbb{E}[x | y] - x] = \mathbb{E}[\mathbb{E}[x | y]] - \mathbb{E}[x] = \mathbb{E}[x] - \mathbb{E}[x] = 0,$$

where we used the law of iterated expectation $\mathbb{E}[\mathbb{E}[x | y]] = \mathbb{E}[x]$. Hence the BLS estimator is unbiased:

$$\mathbb{E}[\hat{x}_{\text{BLS}}(y) - x] = 0.$$

6.7 Error covariance equals expected posterior covariance

Let (x, y) be jointly distributed. Define the (random) BLS/MMSE estimator

$$\hat{x}(y) := \mathbb{E}_{p_{x|y}}[x | y],$$

i.e. $\hat{x}(\cdot)$ is a deterministic function, and $\hat{x}(y)$ is random only through y . For each fixed y , define the posterior covariance matrix

$$\Lambda_{x|y}(y) := \mathbb{E}_{p_{x|y}(\cdot|y)} \left[(\hat{x}(y) - x)(\hat{x}(y) - x)^\top \right].$$

Define the estimation error random vector

$$\mathbf{e} := \hat{x}(y) - x.$$

Then the (unconditional) BLS error covariance is

$$\Lambda_{\text{BLS}} := \mathbb{E}_{p_{x,y}}[\mathbf{e}\mathbf{e}^\top] = \mathbb{E}_{p_{x,y}} \left[(\hat{x}(y) - x)(\hat{x}(y) - x)^\top \right].$$

Applying iterated expectation (conditioning on y) gives

$$\Lambda_{\text{BLS}} = \mathbb{E}_{p_y} \left[\mathbb{E}_{p_{x|y}(\cdot|y)} \left[(\hat{x}(y) - x)(\hat{x}(y) - x)^\top \right] \right] = \mathbb{E}_{p_y}[\Lambda_{x|y}(y)].$$

7 NonBayesian parameter estimation

7.1 Nonrandom parameter estimation and parameterized observation model

In many estimation problems it is unnatural to place a prior on the unknown quantity of interest. In such settings we treat the parameter as *deterministic but unknown*. We write

$$x \in \mathcal{X} \quad (\text{unknown, deterministic parameter}), \quad y \in \mathcal{Y} \quad (\text{random observation}).$$

Since the parameter is fixed, the observation model is *not* a conditional law $p_{y|x}(\cdot | \cdot)$; instead, it is a *family of distributions* on $\mathcal{Y}$ parameterized by x , written

$$p_y(y; x), \quad \mathcal{P} := \{ p_y(\cdot; x) \in \mathcal{P}^{\mathcal{Y}} : x \in \mathcal{X} \}.$$

7.2 Why the naive Bayesian MSE criterion fails without a prior

An estimator is a function $\hat{x} : \mathcal{Y} \rightarrow \mathcal{X}$, and $\hat{x}(y)$ is random only through y . Recall that in Bayesian parameter estimation we found the estimator to minimize Bayes risk. A tempting but incorrect adaptation of the Bayesian framework is

$$\hat{x}(\cdot) \stackrel{?}{=} \arg \min_{f: \mathcal{Y} \rightarrow \mathcal{X}} \mathbb{E}_{p_y(\cdot; x)} [(x - f(y))^2].$$

Here the expectation is *only over* $y \sim p_y(\cdot; x)$ (since x is not random). But then the minimizer is

$$\hat{x}(\cdot) = f(y) = x,$$

which is useless because it depends on the very unknown parameter we are trying to estimate.

7.3 Valid estimators are statistics (must not depend on the parameter)

The failure above shows that in nonrandom parameter estimation we must restrict attention to estimators $\hat{x}(\cdot)$ that do not depend on the true parameter x

A valid estimator has the form $\hat{x}(\cdot) = f(y)$ where $f(\cdot)$ does not depend on x .

7.4 Bias, error covariance, and MSE for nonrandom parameters

As in the case of random parameters, two important quantities that impact the mean-square error performance of an estimator for nonrandom parameters are the bias and error covariance. However, there are some significant distinctions between these quantities in the nonrandom case, which we emphasize.

Fix an unknown deterministic parameter $x \in \mathcal{X}$, and let $y \in \mathcal{Y}$ be the random observation with law $p_y(\cdot; x)$. Let a (valid) estimator be a deterministic function $\hat{x} : \mathcal{Y} \rightarrow \mathcal{X}$, so that the random estimate is $\hat{x}(y)$ (random only through y).

Define the (random) estimation error

$$e(y; x) := \hat{x}(y) - x.$$

The bias function is

$$b_x(x) := \mathbb{E}_{p_y(\cdot; x)}[e(y; x)] = \mathbb{E}_{p_y(\cdot; x)}[\hat{x}(y)] - x,$$

and the error covariance matrix is

$$\Lambda_e(x) := \text{Cov}_{p_y(\cdot; x)}(e(y; x), e(y; x)) = \mathbb{E}_{p_y(\cdot; x)}[(e - b_{\hat{x}}(x))(e - b_{\hat{x}}(x))^\top].$$

Both $b_{\hat{x}}(x)$ and $\Lambda_e(x)$ generally depend on x . The mean-square error (MSE) is

$$\text{MSE}(x) := \mathbb{E}_{p_y(\cdot; x)}[e^\top e] = \text{tr}(\mathbb{E}_{p_y(\cdot; x)}[ee^\top]),$$

with the bias–covariance decomposition

$$\mathbb{E}_{p_y(\cdot; x)}[ee^\top] = b_{\hat{x}}(x) b_{\hat{x}}(x)^\top + \Lambda_e(x).$$

7.5 Unbiasedness of an estimator

An estimator $\hat{x}(\cdot)$ is *unbiased* if

$$b_{\hat{x}}(x) = 0, \quad \forall x \in \mathcal{X}.$$

7.6 $\Lambda_e(x) = \Lambda_{\hat{x}}(x)$

In the nonrandom-parameter setting, the error covariance equals the covariance of the estimator itself. Fix a deterministic $x \in \mathcal{X}$ and let $y \sim p_y(\cdot; x)$. Let $\hat{x} : \mathcal{Y} \rightarrow \mathcal{X}$ be an estimator, and recall the random error

$$e := \hat{x}(y) - x, \quad b_{\hat{x}}(x) := \mathbb{E}_{p_y(\cdot; x)}[e] = \mathbb{E}_{p_y(\cdot; x)}[\hat{x}(y)] - x.$$

Then

$$\begin{aligned} \Lambda_e(x) &:= \mathbb{E}_{p_y(\cdot; x)}[(e - b_{\hat{x}}(x))(e - b_{\hat{x}}(x))^\top] \\ &= \mathbb{E}_{p_y(\cdot; x)}\left[\left((\hat{x}(y) - x) - (\mathbb{E}_{p_y(\cdot; x)}[\hat{x}(y)] - x)\right)\left((\hat{x}(y) - x) - (\mathbb{E}_{p_y(\cdot; x)}[\hat{x}(y)] - x)\right)^\top\right] \\ &= \mathbb{E}_{p_y(\cdot; x)}\left[(\hat{x}(y) - \mathbb{E}_{p_y(\cdot; x)}[\hat{x}(y)])(\hat{x}(y) - \mathbb{E}_{p_y(\cdot; x)}[\hat{x}(y)])^\top\right] \\ &=: \Lambda_{\hat{x}}(x). \end{aligned}$$

7.7 Minimum-variance unbiased estimator

Let $\mathcal{A}$ denote the class of valid unbiased estimators:

$$\mathcal{A} := \left\{ \hat{x} : \mathcal{Y} \rightarrow \mathcal{X} : \hat{x}(\cdot) \text{ does not depend on } x, b_{\hat{x}}(x) = 0 \forall x \in \mathcal{X} \right\}.$$

If there exists $\hat{x}_\star(\cdot) \in \mathcal{A}$ such that for every $\hat{x}(\cdot) \in \mathcal{A}$,

$$\Lambda_{\hat{x}}(x) \succeq \Lambda_{\hat{x}_\star}(x), \quad \forall x \in \mathcal{X},$$

then $\hat{x}_\star(\cdot)$ is the *minimum-variance unbiased (MVU) estimator*, denoted $\hat{x}_{\text{MVU}}(\cdot)$.

7.8 Why MVU estimators may fail to exist and why we use lower bounds

The MVU estimator $\hat{x}_{\text{MVU}}(\cdot)$ (if it exists) is an element of the class $\mathcal{A}$ of valid unbiased estimators that has the smallest variance for *every* $x \in \mathcal{X}$. There are two basic obstructions:

- $\mathcal{A}$ may be empty (there may be no valid unbiased estimators).
- Even if $\mathcal{A} \neq \emptyset$, there may be no single estimator whose variance is uniformly smallest over

all $x \in \mathcal{X}$ (different unbiased estimators can dominate at different parameter values). Moreover, even when an MVU estimator exists, it can be difficult to identify or compute. A common approach is therefore to derive a universal *lower bound* on the variance $\lambda_{\hat{x}}(x)$ (or on the covariance in the vector case) that applies to *all* unbiased estimators; such a bound can be used both to rule out overly-optimistic performance claims and, when achieved, to certify optimality. This is why Cramér-Rao is useful.

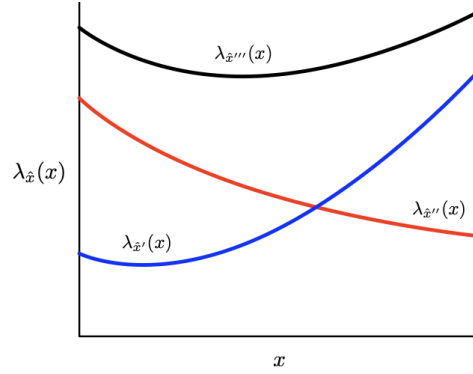


Figure 6: The variances of unbiased estimators $\hat{x}'(\cdot)$, $\hat{x}''(\cdot)$, $\hat{x}'''(\cdot)$. No estimator achieves lower variances over all possible parameters $x \in \mathcal{X}$.

7.9 Preliminary of Cramér-Rao: score function and Fisher information

Fix $x \in \mathcal{X}$ and let $y \sim p_y(\cdot; x)$ with $p_y(y; x) > 0$ and differentiable in x for each $y \in \mathcal{Y}$. The *score* is the random function of y ,

$$S(y; x) := \frac{\partial}{\partial x} \ln p_y(y; x).$$

The *Fisher information in y about x* is

$$J_y(x) := \mathbb{E}_{p_y(\cdot; x)}[S(y; x)^2].$$

7.10 Cramér–Rao bound

Theorem (Cramér–Rao bound). Assume $p_y(y; x)$ is positive and differentiable in $x \in \mathcal{X} \subset \mathbb{R}$ and satisfies the regularity condition

$$\mathbb{E}_{p_y(\cdot; x)}[S(y; x)] = 0, \quad \forall x \in \mathcal{X}.$$

Let $\hat{x} : \mathcal{Y} \rightarrow \mathbb{R}$ be a valid unbiased estimator, i.e.

$$\mathbb{E}_{p_y(\cdot; x)}[\hat{x}(y)] = x, \quad \forall x \in \mathcal{X},$$

and suppose the further regularity condition

$$\mathbb{E}_{p_y(\cdot; x)}[\hat{x}(y) S(y; x)] = 1, \quad \forall x \in \mathcal{X},$$

holds. Define the variance of the estimator $\hat{x}(y)$ as

$$\lambda_{\hat{x}}(x) := \text{Var}_{p_y(\cdot; x)}(\hat{x}(y)).$$

then the Cramér Rao bound says that

$$\lambda_{\hat{x}}(x) \geq \frac{1}{J_y(x)}, \quad \forall x \in \mathcal{X},$$

so we have a lower bound on the variance of the estimator. A few notes

- The Fisher information cannot be computed in all problems, i.e., the regularity condition may not be satisfied, in which case no Cramér-Rao bound exists. For example, $p_y(y; x)$ is not positive for all $x \in \mathcal{X}$.
- Any estimator that satisfies the Cramér-Rao bound with equality must be a MVU estimator.

7.11 Definition of an efficient estimator using the Cramér–Rao bound

Fix a deterministic $x \in \mathcal{X} \subset \mathbb{R}$ and let $y \sim p_y(\cdot; x)$. Assume the conditions of the scalar Cramér–Rao theorem hold, and let

$$S(y; x) := \frac{\partial}{\partial x} \ln p_y(y; x), \quad J_y(x) := \mathbb{E}_{p_y(\cdot; x)}[S(y; x)^2].$$

We say that a valid unbiased estimator $\hat{x}(\cdot)$ is *efficient* if it attains the Cramér–Rao bound with equality for all $x \in \mathcal{X}$, i.e.

$$\lambda_{\hat{x}}(x) = \frac{1}{J_y(x)}, \quad \forall x \in \mathcal{X}.$$

7.12 Characterization of efficient estimators

Corollary of Cramér Rao: A valid unbiased estimator $\hat{x}(\cdot)$ is efficient if and only if it can be written as

$$\hat{x}(y) = x + \frac{1}{J_y(x)} S(y; x),$$

where the right-hand side, viewed as a function of y , must be independent of x (so that $\hat{x}(\cdot)$ is a valid estimator).

7.13 Maximum likelihood estimator and connections to efficiency

Fix a model family

$$\mathcal{P} := \{ p_y(\cdot; x) \in \mathcal{P}^{\mathcal{Y}} : x \in \mathcal{X} \},$$

where $x \in \mathcal{X}$ is an unknown deterministic parameter and $y \sim p_y(\cdot; x)$ is the random observation.

Definition (Maximum likelihood estimator): The maximum likelihood (ML) estimate based on observation $y \in \mathcal{Y}$ is

$$\hat{x}_{\text{ML}}(y) := \arg \max_{x \in \mathcal{X}} p_y(y; x),$$

It is often convenient to define the (random) likelihood function of x given y by

$$L_y(x) := p_y(y; x),$$

and, for a fixed realized observation y , its deterministic version

$$L_y(x) := p_y(y; x),$$

which emphasizes the dependence on x for a fixed y .

Connection to efficiency

Corollary: Suppose, for the model family above, that an efficient estimator $\hat{x}_{\text{eff}}(\cdot)$ and an ML estimator $\hat{x}_{\text{ML}}(\cdot)$ both exist (and are unique). Then they coincide:

$$\hat{x}_{\text{eff}}(\cdot) = \hat{x}_{\text{ML}}(\cdot).$$

For convenience, we write

$$\lambda_{\text{ML}}(x) := \lambda_{\hat{x}_{\text{ML}}}(x) = \text{Var}_{p_{\mathbf{y}}(\cdot|x)}(\hat{x}_{\text{ML}}(\mathbf{y})).$$

This does *not* mean ML estimators are always efficient: if an efficient estimator does not exist, the ML estimator need not be unbiased or have good finite-sample variance. Nonetheless, ML estimators often enjoy favorable *asymptotic* properties when many independent observations are available.

7.14 ML estimate for the linear-Gaussian model and least squares connection

Let $\mathbf{x} \in \mathcal{X} = \mathbb{R}^K$ denote an unknown deterministic parameter vector, and let $\mathbf{y} \in \mathbb{R}^N$ be the random observation vector. A *linear-Gaussian model* is

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{w}, \quad \mathbf{w} \sim \mathcal{N}(0, \mathbf{\Lambda}_{\mathbf{w}}),$$

where $\mathbf{H} \in \mathbb{R}^{N \times K}$ is known and $\mathbf{\Lambda}_{\mathbf{w}} \in \mathbb{R}^{N \times N}$ is a known noise covariance matrix. Assume $\mathbf{\Lambda}_{\mathbf{w}} \succ 0$ and $\mathbf{H}$ has full column rank. Then the maximum likelihood estimator $\hat{\mathbf{x}}_{\text{ML}} : \mathbb{R}^N \rightarrow \mathbb{R}^K$ is

$$\hat{\mathbf{x}}_{\text{ML}}(\mathbf{y}) = \left(\mathbf{H}^\top \mathbf{\Lambda}_{\mathbf{w}}^{-1} \mathbf{H} \right)^{-1} \mathbf{H}^\top \mathbf{\Lambda}_{\mathbf{w}}^{-1} \mathbf{y},$$

and it is MVU.

Gauss–Markov theorem as weighted least squares: For the same model, the ML estimate is also the solution to a weighted least-squares problem:

$$\hat{\mathbf{x}}_{\text{ML}}(\mathbf{y}) = \arg \min_{\mathbf{x} \in \mathbb{R}^K} \varphi_{\text{WLS}}(\mathbf{x}),$$

where

$$\varphi_{\text{WLS}}(\mathbf{x}) := (\mathbf{y} - \mathbf{H}\mathbf{x})^\top \mathbf{\Lambda}_{\mathbf{w}}^{-1} (\mathbf{y} - \mathbf{H}\mathbf{x}) = \|\mathbf{\Lambda}_{\mathbf{w}}^{-1/2} (\mathbf{y} - \mathbf{H}\mathbf{x})\|_2^2.$$

Additionally, the ordinary least squares can be obtained as a special case: If $\mathbf{\Lambda}_{\mathbf{w}} \propto \mathbf{I}$, then $\hat{\mathbf{x}}_{\text{ML}}(\mathbf{y})$ solves the ordinary least-squares problem

$$\hat{\mathbf{x}}_{\text{ML}}(\mathbf{y}) = \arg \min_{\mathbf{x} \in \mathbb{R}^K} \|\mathbf{y} - \mathbf{H}\mathbf{x}\|_2^2.$$

8 Exponential families

8.1 Single-parameter exponential family

Let $\mathcal{X} \subseteq \mathbb{R}$ contain an open, nonempty set which we think of as a parameter. A parameterized family of distributions

$$\{p_y(\cdot; x) : x \in \mathcal{X}\}$$

over the outcome set $\mathcal{Y}$ is called a *single-parameter exponential family* if it can be written as

$$p_y(y; x) = \exp\{\lambda(x)t(y) - \alpha(x) + \beta(y)\}, \quad x \in \mathcal{X}, y \in \mathcal{Y},$$

for some functions

$$\lambda(\cdot) : \mathcal{X} \rightarrow \mathbb{R}, \quad t(\cdot) : \mathcal{Y} \rightarrow \mathbb{R}, \quad \beta(\cdot) : \mathcal{Y} \rightarrow \mathbb{R}.$$

Here:

- $\lambda(\cdot)$ is the *natural parameter*,
- $t(\cdot)$ is the *natural statistic*,
- $\beta(\cdot)$ is the *log base function*,
- $\alpha(x)$ is the partition function.

Together with $\mathcal{X}$ and $\mathcal{Y}$, these specify the family.

8.2 Normalization and partition function

The function $\alpha(x)$ is *not* officially part of the specification of the family, since its role is only to normalize each distribution. Therefore, it is fully determined by the other three functions. Writing

$$p_y(y; x) = \frac{1}{Z(x)} \exp\{\lambda(x)t(y) + \beta(y)\},$$

the normalization constant is

$$Z(x) = e^{\alpha(x)}.$$

It is chosen so that $p_y(\cdot; x)$ sums or integrates to one. In the discrete case,

$$Z(x) = e^{\alpha(x)} := \sum_y \exp\{\lambda(x)t(y) + \beta(y)\}.$$

8.3 Notation for membership in an family

We write

$$y \sim p \in \mathcal{E}(\mathcal{X}, \mathcal{Y}; \lambda(\cdot), t(\cdot), \beta(\cdot))$$

to indicate that the distribution of y has the above form for some $x \in \mathcal{X}$. Thus, one may view the parameter x as an index into the family.

8.4 Support condition

A family of distributions whose support depends on the parameter cannot be an exponential family: the support of the random variable y must be independent of the parameter x .

8.5 Useful properties of exponential families

1. Derivative of the log-partition function: for a general single-parameter exponential family, the log-partition function satisfies

$$\dot{\alpha}(x) = \dot{\lambda}(x) \mathbb{E}[t(y)].$$

2. Second derivative of the log-partition function: for a general single-parameter exponential family, the log-partition function satisfies

$$\ddot{\alpha}(x) = \dot{\lambda}(x)^2 \text{var}[t(y)] + \ddot{\lambda}(x) \mathbb{E}[t(y)].$$

3. Score function: for a general single-parameter exponential family, the score function satisfies

$$S(y; x) = \dot{\lambda}(x) (t(y) - \mathbb{E}[t(y)]).$$

4. Fisher information: for a general single-parameter exponential family, the Fisher information satisfies

$$J_y(x) = \dot{\lambda}(x)^2 \text{var}[t(y)].$$

5. Derivative identity for the mean of the natural statistic: for a general single-parameter exponential family, we have

$$\frac{d}{dx} \mathbb{E}[t(y)] = \dot{\lambda}(x) \text{var}[t(y)].$$

8.6 Canonical single-parameter exponential family

A *canonical* single-parameter exponential family is an exponential family for which

$$\lambda(x) = x.$$

Hence its distributions have the form

$$p_y(y; x) = \exp\{x t(y) - \alpha(x) + \beta(y)\}, \quad x \in \mathcal{X}, y \in \mathcal{Y},$$

with partition specializing in this case to

$$Z(x) = e^{\alpha(x)} = \sum_y \exp\{x t(y) + \beta(y)\}.$$

8.7 Natural parameter space of a canonical exponential family

For fixed $t(\cdot)$ and $\beta(\cdot)$, it is often convenient to consider the largest canonical family that can be normalized. The *natural parameter space* $\mathcal{X}$ of a canonical exponential family is

$$\mathcal{X} = \{x \in \mathbb{R} : \alpha(x) < \infty\},$$

which is the set of all parameters x for which the associated distribution can be normalized.

8.8 Geometric means of any distributions are exponential families

Let $p_1(\cdot)$ and $p_2(\cdot)$ be two strictly positive probability distributions on the alphabet $\mathcal{Y}$. Define their normalized weighted geometric mean by

$$p_y(y; x) = \frac{p_1(y)^x p_2(y)^{1-x}}{Z(x)}, \quad x \in \mathcal{X} = [0, 1].$$

Since

$$\ln p_y(y; x) = x \ln \frac{p_1(y)}{p_2(y)} + \ln p_2(y) - \ln Z(x),$$

this is an exponential family with

$$\lambda(x) = x, \quad t(y) = \ln \frac{p_1(y)}{p_2(y)}, \quad \beta(y) = \ln p_2(y), \quad \alpha(x) = \ln Z(x).$$

8.9 Geometric mean characterization of canonical exponential families

Define a one-dimensional family of distributions over $\mathcal{Y}$ and convex $\mathcal{X}$ as

$$\mathcal{P} = \{p_y(\cdot; x) : x \in \mathcal{X} \subseteq \mathbb{R}\}.$$

Then $\mathcal{P}$ is a canonical exponential family if and only if for every $p_1(\cdot), p_2(\cdot), p_3(\cdot) \in \mathcal{P}$ there exists some $\lambda \in \mathbb{R}$ such that

$$p_3(y) = \frac{p_1(y)^\lambda p_2(y)^{1-\lambda}}{Z(\lambda)}.$$

Therefore, any canonical exponential family is a geometric mean, and any geometric mean is a canonical exponential family.

8.10 Tilting distributions

Another natural construction is *tilting*. Given a base distribution $q(\cdot)$ over alphabet $\mathcal{Y} \subseteq \mathbb{R}$, define

$$p_y(y; x) = \frac{q(y)e^{xt(y)}}{Z(x)}, \quad y \in \mathcal{Y}, \quad x \in \mathbb{R}.$$

This is called a tilted distribution because:

- when $x > 0$, it exponentially upweights larger values of $t(y)$ relative to $q(\cdot)$,
- when $x < 0$, it does the opposite.

Then $p_y(y; x)$ is a member of the single-parameter exponential family with

$$\lambda(x) = x, \quad t(y) = t(y), \quad \beta(y) = \ln q(y), \quad \alpha(x) = \ln Z(x).$$

8.11 Example: tilting a Bernoulli distribution

Let

$$\mathcal{Y} = \{0, 1\}, \quad t(y) = y, \quad q \sim \text{B}(1/2).$$

Then the tilted distribution satisfies

$$p_y(\cdot; x) \sim \text{B}\left(\frac{1}{1 + e^{-x}}\right),$$

because

$$\frac{p_y(1; x)}{p_y(0; x)} = \frac{e^{x/2}}{1/2} = e^x.$$

Thus, when $x > 0$, tilting makes $y = 1$ more likely.

8.12 Multi-parameter exponential families

The single-parameter exponential families generalize naturally to families with multiple parameters. Let $\mathcal{X} \subseteq \mathbb{R}^L$ contain an open, nonempty set. A parameterized family of distributions

$$\{p_y(\cdot; x) : x \in \mathcal{X}\}$$

is called a *K-parameter exponential family* if there exist

$$\lambda(\cdot) = \begin{bmatrix} \lambda_1(\cdot) \\ \vdots \\ \lambda_K(\cdot) \end{bmatrix} : \mathcal{X} \rightarrow \mathbb{R}^K, \quad t(\cdot) = \begin{bmatrix} t_1(\cdot) \\ \vdots \\ t_K(\cdot) \end{bmatrix} : \mathcal{Y} \rightarrow \mathbb{R}^K, \quad \beta(\cdot) : \mathcal{Y} \rightarrow \mathbb{R},$$

such that each member of the family has the form

$$p_y(y; x) = \exp\left\{\lambda(x)^\top t(y) - \alpha(x) + \beta(y)\right\}.$$

Here:

- $x \in \mathbb{R}^L$ is a deterministic parameter vector,
- y is the observed data
- $\lambda(x) \in \mathbb{R}^K$ is the natural parameter vector,
- $t(y) \in \mathbb{R}^K$ is the natural statistic evaluated at y .
- $\beta(y)$ is the log base function

As before, $\alpha(x)$ normalizes the distribution.

8.13 Canonical multi-parameter exponential families

A *canonical multi-parameter exponential family* is an exponential family for which

$$\lambda(x) = x.$$

Thus its members have the form

$$p_y(y; x) = \exp\left\{x^\top t(y) - \alpha(x) + \beta(y)\right\}, \quad x \in \mathcal{X}, \quad y \in \mathcal{Y}.$$

8.14 Identifiable families

For a family of distributions

$$\{p_y(\cdot; \mathbf{x}), \mathbf{x} \in \mathcal{X}\},$$

the parameter $\mathbf{x}$ is *identifiable* from y if and only if for all $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{X}$,

$$\mathbf{x}_1 = \mathbf{x}_2 \implies p_y(\cdot; \mathbf{x}_1) = p_y(\cdot; \mathbf{x}_2).$$

8.15 Identifiable if and only if no affine constraint on natural statistic

The parameter of a canonical exponential family is identifiable if and only if the natural statistics satisfy no affine constraint. That is,

$$\sum_{k=1}^K b_k t_k(\cdot) = c$$

can hold only if

$$b_1 = \cdots = b_K = 0, \quad c = 0.$$

8.16 Minimal exponential family

An identifiable exponential family with natural parameters

$$\lambda_1(\cdot), \dots, \lambda_K(\cdot)$$

is called *minimal* when the natural parameters satisfy no affine constraint. That is,

$$\sum_{k=1}^K b_k \lambda_k(\cdot) = c$$

can hold only if

$$b_1 = \cdots = b_K = 0, \quad c = 0.$$

All identifiable canonical exponential families are minimal.

8.17 Reducibility of non-minimal exponential families

An exponential family of dimension $K \geq 2$ that is not minimal can be equivalently re-expressed as an exponential family of dimension $K - 1$.

9 Sufficient statistics

9.1 Sufficient statistics arise from the need to extract features of data

In many inference problems, it is useful to first preprocess the raw data into a more compact collection of features, and then perform inference using those features rather than the full data. Such a collection of features is called a *representation* of the data. This raises two natural questions:

- When is the preprocessing *lossless* for the inference task?
- Among lossless preprocessings, how should we quantify *compactness*?

The key idea is that the extracted features are deterministic functions of the data, and are known as *statistics*. In the non-Bayesian setting, lossless preprocessing corresponds to extracting a *sufficient statistic*. Among sufficient statistics, the most compact ones are the *minimal sufficient statistics*.

9.2 Model family and statistics in the non-Bayesian setting

We work with a model family of the form

$$\mathcal{P} = \{p_y(\cdot; x) \in \mathcal{P}^{\mathcal{Y}} : x \in \mathcal{X}\},$$

where:

- $y \in \mathcal{Y}$ is the random data, where the realization y should be understood as the entire observed data object
- $x \in \mathcal{X}$ is an unknown but nonrandom parameter,
- $p_y(\cdot; x)$ is the distribution of the dataset y under parameter value x

A statistic is a deterministic function of the data,

$$t : \mathcal{Y} \rightarrow \mathbb{R}^K,$$

for some $K \in \mathbb{N}$, and the induced random quantity is

$$t = t(y).$$

Thus, preprocessing replaces the full data y by the lower-dimensional feature vector t .

9.3 Distribution induced by a statistic

Given a statistic $t(\cdot)$, the model can be decomposed as

$$p_y(y; x) = p_{y|t}(y | t(y); x) p_t(t(y); x), \quad y \in \mathcal{Y}, x \in \mathcal{X}.$$

where the marginal distribution of the statistic is

$$p_t(t'; x) = \sum_{\{y \in \mathcal{Y} : t(y) = t'\}} p_y(y; x).$$

So once a statistic is chosen, the model decomposes into

- the remaining uncertainty in y after t is known, and
- the law of the extracted feature t .

9.4 Definition of sufficient statistic

A statistic $t(\cdot)$ is *sufficient* for the model family

$$\mathcal{P} = \{p_y(\cdot; x) : x \in \mathcal{X}\}$$

if the conditional distribution

$$p_{y|t}(\cdot | t'; x)$$

does not depend on x for every attainable value t' . Equivalently, once we know the value of the statistic $t(y)$, the remaining randomness in y carries no further information about the parameter x

9.5 Likelihood characterization of sufficiency

Given observed data y , define

$$\mathcal{L}_y(x) \triangleq p_y(y; x).$$

Likewise, if we only observe the statistic value $t(y)$, the corresponding likelihood is

$$\mathcal{L}_{t(y)}(x) \triangleq p_t(t(y); x).$$

A statistic $t(\cdot)$ is sufficient for the model family if and only if

$$\frac{\mathcal{L}_y(x)}{\mathcal{L}_{t(y)}(x)} = \frac{p_y(y; x)}{p_t(t(y); x)} = p_{y|t}(y | t(y); x)$$

is not a function of x for every $y \in \mathcal{Y}$, where we used the decomposition

$$p_y(y; x) = p_{y|t}(y | t(y); x) p_t(t(y); x).$$

9.6 Neyman factorization theorem allows a check on if a statistic is sufficient

The definition of sufficiency is conceptually important, but it is not always the easiest way to verify that a proposed statistic is sufficient. A more practical characterization is given by the Neyman factorization theorem. Let

$$\mathcal{P} = \{p_y(\cdot; x) : x \in \mathcal{X}\}.$$

A statistic $t(\cdot)$ is sufficient with respect to this model family if and only if there exist functions

$$a(\cdot, \cdot) \quad \text{and} \quad b(\cdot)$$

such that

$$p_y(y; x) = a(t(y), x) b(y)$$

for all $x \in \mathcal{X}$ and $y \in \mathcal{Y}$. The interpretation is that all dependence on the parameter x must pass through the statistic $t(y)$. The remaining factor $b(y)$ may depend on the data y , but it must not depend on x .

9.7 Definition of minimal sufficiency

When seeking minimality of statistics, one could encode an entire vector of real numbers into a single real number by interleaving binary digits. This would reduce the formal dimension to one, but it would not produce meaningful compaction since it would preserve essentially all of the original data without any useful simplification. So minimality should not be defined in

terms of raw dimension alone.

Instead of measuring dimension, a sufficient statistic $s(\cdot)$ is called *minimal sufficient* with respect to the model family

$$\mathcal{P} = \{p_Y(\cdot; x) : x \in \mathcal{X}\}$$

if for every other sufficient statistic $t(\cdot)$, there exists a deterministic function $g(\cdot)$ such that

$$s(y) = g(t(y)) \quad \text{for all } y \in \mathcal{Y}.$$

Equivalently, a minimal sufficient statistic can be computed from any other sufficient statistic.

9.8 Nonuniqueness of minimal sufficient statistics

A minimal sufficient statistic is not unique as a literal function. If $s(\cdot)$ is minimal sufficient and $h(\cdot)$ is one-to-one, then

$$h(s(\cdot))$$

is also minimal sufficient. Thus minimal sufficient statistics are unique only up to invertible reparameterizations.

9.9 Bayesian setting and joint model

In the Bayesian setting, the unknown parameter is itself modeled as a random quantity. Let

$$\mathbf{x} \in \mathcal{X}$$

be a random parameter with prior distribution

$$p_{\mathbf{x}}(\cdot) \in \mathcal{P}^{\mathcal{X}}.$$

The data are represented by a random quantity

$$y \in \mathcal{Y},$$

and the conditional model for the data given the parameter is

$$p_{Y|\mathbf{x}}(\cdot | x) \in \mathcal{P}^{\mathcal{Y}}, \quad x \in \mathcal{X}.$$

Thus the model family is

$$\mathcal{P} = \{p_{Y|\mathbf{x}}(\cdot | x) \in \mathcal{P}^{\mathcal{Y}} : x \in \mathcal{X}\}.$$

Together with the prior, this yields the joint distribution

$$p_{\mathbf{x},Y}(x, y) = p_{\mathbf{x}}(x) p_{Y|\mathbf{x}}(y | x), \quad x \in \mathcal{X}, y \in \mathcal{Y}.$$

9.10 Sufficiency in Bayesian form

Let

$$t : \mathcal{Y} \rightarrow \mathbb{R}^K$$

be a deterministic statistic, and define

$$t = t(y).$$

In the Bayesian setting, the statistic $t(\cdot)$ is sufficient with respect to the joint model

$$p_{\mathbf{x},Y}(x, y) = p_{\mathbf{x}}(x) p_{Y|\mathbf{x}}(y | x)$$

if

$$p_{y|t,x}(y | t(y), x) = p_{y|t}(y | t(y)), \quad x \in \mathcal{X}, y \in \mathcal{Y}.$$

9.11 Belief characterization of Bayesian sufficiency

In the Bayesian setting, the central inferential object is the posterior distribution of the parameter after observing data. By starting at the original definition of sufficiency, one can show that a statistic $t(\cdot)$ is sufficient if and only if

$$p_{x|y}(\cdot | y) = p_{x|t}(\cdot | t(y)), \quad y \in \mathcal{Y}.$$

which means that observing the full data y gives the same posterior belief about x as observing only the statistic value $t(y)$.

9.12 Bayesian Neyman factorization

There is also a Bayesian analogue of the Neyman factorization theorem. A statistic $t(\cdot)$ is sufficient with respect to the joint model

$$p_{x,y}(x, y) = p_x(x) p_{y|x}(y | x)$$

if and only if

$$p_{y|x}(y | x) = p_{t|x}(t(y) | x) p_{y|t}(y | t(y)), \quad x \in \mathcal{X}, y \in \mathcal{Y}.$$

The key point is that the second factor

$$p_{y|t}(y | t(y))$$

does not depend on x . Hence all dependence of the data distribution on the parameter passes through the statistic $t(y)$.

9.13 Main takeaway for the Bayesian setting

In the Bayesian setting, a sufficient statistic is again a lossless representation of the data for inference about the parameter. The lossless nature can be expressed in two equivalent ways:

- the conditional law of y given t and x does not depend on x , and
- the posterior distribution of x given y is the same as the posterior distribution of x given t .

So sufficient statistics remain the correct notion of lossless preprocessing, now interpreted through posterior beliefs rather than likelihoods alone.

10 The expectation-maximization (EM) algorithm

10.1 Why EM is needed

Suppose we observe data y from a model

$$\mathcal{P} = \{p_Y(\cdot; x), x \in \mathcal{X}\},$$

and we want the maximum-likelihood estimate

$$\hat{x}(y) \in \arg \max_{a \in \mathcal{X}} \log p_Y(y; a).$$

The difficulty is that, in many models, the observed-data likelihood is hard to maximize directly. EM is designed for exactly this situation: the model becomes easy to handle if we imagine some missing or latent data that would make the likelihood simpler.

10.2 Latent-data viewpoint

Introduce an unobserved random quantity s and regard the *complete data* as

$$z = (y, s), \quad z = (y, s).$$

We then work with a complete-data model is

$$p_{Y,s}(\cdot, \cdot; x),$$

whose observed-data model is obtained by marginalizing out the latent part:

$$p_Y(y; x) = \sum_{s \in \mathcal{S}} p_{Y,s}(y, s; x)$$

in the discrete case, or

$$p_Y(y; x) = \int p_{Y,s}(y, s; x) ds$$

in the continuous case. The key modeling idea is:

complete-data likelihood easy, observed-data likelihood hard.

where the complete-data likelihood is simply

$$\log p_{Y,s}(y, s; x).$$

10.3 The EM surrogate objective

Given a current parameter value x' , define

$$U(x; x') \triangleq \mathbb{E}_{p_{s|y}(\cdot|y; x')} [\log p_{Y,s}(y, s; x)].$$

This is the expected complete-data log-likelihood, where the expectation is taken with respect to the current conditional law of the latent variable.

Interpretation:

- x' tells us how plausible each latent configuration s is, given the observed data y ;
- $U(x; x')$ measures how good a candidate parameter x would be for the complete-data model after averaging over those plausible latent configurations.

10.4 The EM algorithm

Starting from an initial value $\hat{x}^{(0)}$, repeat:

E-step: compute $U(x; \hat{x}^{(l)}) = \mathbb{E}_{p_{s|y}(\cdot|y; \hat{x}^{(l)})} [\log p_{y,s}(y, s; x)]$.

M-step: set $\hat{x}^{(l+1)} \in \arg \max_{x \in \mathcal{X}} U(x; \hat{x}^{(l)})$.

So:

- the E-step computes conditional expectations under the current parameter;
- the M-step maximizes the resulting surrogate objective.

A useful slogan is:

E-step = fill in the missing data probabilistically,

M-step = maximize as if those filled-in values were observed.

10.5 Why EM increases the likelihood

EM is built from a lower-bound argument. For any distribution $q_{s|y}(\cdot | y)$,

$$\log p_Y(y; x) \geq \mathbb{E}_{q_{s|y}(\cdot|y)} [\log p_{Y,s}(y, s; x)] - \mathbb{E}_{q_{s|y}(\cdot|y)} [\log q_{s|y}(s | y)].$$

This is the key lower bound behind EM.

Moreover, equality holds when

$$q_{s|y}(\cdot | y) = p_{s|y}(\cdot | y; x).$$

So at the current iterate $\hat{x}^{(l)}$, the E-step chooses the conditional distribution that makes the bound tight, and the M-step maximizes that bound over x . Therefore,

$$\log p_Y(y; \hat{x}^{(l+1)}) \geq \log p_Y(y; \hat{x}^{(l)}).$$

This monotonicity is the main theoretical guarantee of EM. However, EM typically converges to a *stationary point*, not necessarily the global maximum.

10.6 What you should remember for exams

The most important formulas are:

$$U(x; x') = \mathbb{E}_{p_{s|y}(\cdot|y; x')} [\log p_{Y,s}(y, s; x)].$$

$$\hat{x}^{(l+1)} \in \arg \max_{x \in \mathcal{X}} U(x; \hat{x}^{(l)}).$$

10.7 EM with exponential-family complete data

EM becomes especially clean when the complete-data model belongs to an exponential family. Suppose

$$p_{Y,s}(y, s; x) = \exp\{x^\top t(y, s) - \alpha(x) + \beta(y, s)\},$$

where $x \in \mathcal{X} \subseteq \mathbb{R}^K$ and

$$t(y, s) \in \mathbb{R}^K.$$

Then

$$U(x; x') = x^\top \mathbb{E}_{p_{s|y}(\cdot|y; x')} [t(y, s)] - \alpha(x) + \text{constant in } x.$$

So the E-step reduces to computing

$$\mathbb{E}_{p_{s|y}(\cdot|y;x')} [t(y, \mathbf{s})],$$

the conditional expectation of the complete-data sufficient statistic.

The M-step then chooses x to maximize

$$x^\top \mathbb{E}_{p_{s|y}(\cdot|y;x')} [t(y, \mathbf{s})] - \alpha(x).$$

This is the version of EM that appears most often in examples because it produces simple updates.

11 Inference as decision-making

11.1 Generalized Bayesian decision theory

In standard Bayesian decision theory, the decision device takes as input observed data y , a cost function $C(\cdot, \cdot)$, an observation model $p_{y|x}(\cdot | \cdot)$, and a prior belief $p_x(\cdot)$ over the unknown state x . It then outputs a hard decision $\hat{x}(y) \in \mathcal{X}$. A generalized decision device instead outputs a distribution $q(\cdot)$ on $\mathcal{X}$ based on the observed data $y = y$. Ideally, one would like

$$q(a) = 1_{a=x},$$

since this identifies the realized x exactly, but this is impossible in general because $q(\cdot)$ can depend only on the observed data and not directly on the unknown realization x .

11.2 Evaluate $q(\cdot)$ using cost function $C(x, q)$

A loss function takes in the true parameter x and a distribution $q(\cdot)$. The log-loss is

$$C(x, q) = -A \log q(x) + B(x),$$

where $A > 0$ and $B(\cdot)$ are arbitrary. The log-loss favors distributions $q(\cdot)$ that place high mass on the true parameter x .

11.3 Proper cost functions are maximized by the posterior

A cost function $C(\cdot, \cdot)$ is called proper if

$$p_{x|y}(\cdot | y = y) = \arg \min_{\{q(\cdot) \geq 0: \sum_{a \in \mathcal{X}} q(a) = 1\}} \mathbb{E}_{p_{x|y=y}}[C(x, q)] \quad \text{for all } y \in \mathcal{Y}.$$

Thus, a proper cost function makes it optimal to report the true posterior belief. Note that the log-loss is proper.

11.4 Local cost functions

A cost function $C(\cdot, \cdot)$ is called local if there exists a function

$$\phi : \mathcal{X} \times \mathbb{R} \rightarrow \mathbb{R}$$

such that we can write the cost function as

$$C(x, q) = \phi(x, q(x)) \quad \text{for all } x \in \mathcal{X}.$$

Therefore, a local cost depends only on the chosen $q(\cdot)$ and the true x .

11.5 The log-loss is the only smooth, local, and proper cost function

When the alphabet $|\mathcal{X}| \geq 3$, the log-loss is the only smooth, local, proper cost function. Therefore, without loss of generality, one often restricts attention to the normalized log-loss

$$C(x, q) = -\log q(x),$$

corresponding to $A = 1$ and $B(\cdot) = 0$.

11.6 Minimum log-loss without observations leads to entropy

We now study the minimum expected cost induced by the log-loss criterion. First consider the case with no observations, so the posterior reduces to the prior:

$$p_{x|y}(\cdot | y = \emptyset) = p_x(\cdot).$$

Since log-loss is proper, the minimizing reported distribution is the true prior itself. Hence the minimum expected cost is

$$\min_{\{q(\cdot) \geq 0: \sum_{a \in \mathcal{X}} q(a) = 1\}} \mathbb{E}_{p_{x|y=\emptyset}}[C(x, q)] = \mathbb{E}_{p_x}[C(x, p_x)] = -\mathbb{E}_{p_x}[\log p_x(x)].$$

This quantity is called the *entropy* of x

$$H(x) \triangleq - \sum_{a \in \mathcal{X}} p_x(a) \log p_x(a).$$

It measures the average uncertainty or randomness of the discrete random variable x . If x be a discrete random variable taking values in the alphabet $\mathcal{X}$. Then

$$0 \leq H(x) \leq \log |\mathcal{X}|.$$

The lower bound is attained if and only if x is deterministic. The upper bound is attained when x is uniformly distributed over $\mathcal{X}$.

11.7 Binary entropy function

If $x \sim \text{Bernoulli}(p)$, then its entropy depends only on the parameter p and is written as

$$H(x) = H_B(p) := -p \log p - (1-p) \log(1-p)$$

The binary entropy function is symmetric:

$$H_B(p) = H_B(1-p).$$

The derivative of the binary entropy function is

$$\dot{H}_B(p) = \log \frac{1-p}{p}.$$

11.8 Relation of the binary entropy to the logit function

The derivative of binary entropy is the negative of the log-odds, or logit, function:

$$\text{logit}(p) := \log \frac{p}{1-p}.$$

Therefore,

$$\dot{H}_B(p) = -\text{logit}(p).$$

This relation is useful in many calculations involving Bernoulli models and binary inference.

11.9 Minimizing log-loss given observations leads to conditional entropy

After observing $y = y$, properness of the log-loss implies

$$\min_{\{q(\cdot) \geq 0: \sum_{a \in \mathcal{X}} q(a) = 1\}} \mathbb{E}_{p_{x|y=y}}[C(x, q)] = \mathbb{E}_{p_{x|y=y}}[C(x, p_{x|y=y})] = -\mathbb{E}_{p_{x|y=y}}[\log p_{x|y}(x | y)].$$

Which defines the conditional entropy of x given the realized observation $y = y$.

$$H(x | y = y) := - \sum_{a \in \mathcal{X}} p_{x|y}(a | y) \log p_{x|y}(a | y).$$

11.10 Randomized conditional entropy over data

Averaging over all possible values of y gives the average posterior cost

$$\mathbb{E}_{p_y}[\mathbb{E}_{p_{x|y}}[C(x, p_{x|y}) | y]] = \sum_{y \in \mathcal{Y}} p_y(y) H(x | y = y) \triangleq H(x | y).$$

Equivalently,

$$H(x | y) \triangleq - \sum_{a \in \mathcal{X}, b \in \mathcal{Y}} p_{x,y}(a, b) \log p_{x|y}(a | b).$$

Conditional entropy measures the remaining uncertainty in x after observing y . It always satisfies

$$H(x | y) \geq 0.$$

Equality holds if and only if x is a deterministic function of y .

11.11 Conditioning cannot increase entropy

Entropy decreases, or at least does not increase, after conditioning:

$$H(x | y) \leq H(x).$$

Equality holds if and only if $x \perp\!\!\!\perp y$, that is, if x and y are independent. Thus observing y cannot increase uncertainty about x as measured by entropy.

11.12 Joint entropy and the chain rule

Consider the pair (x, y) as a single random quantity. Its entropy is

$$H(x, y) = - \sum_{a \in \mathcal{X}, b \in \mathcal{Y}} p_{x,y}(a, b) \log p_{x,y}(a, b).$$

Using

$$p_{x,y}(a, b) = p_{x|y}(a | b) p_y(b),$$

we obtain

$$H(x, y) = H(x | y) + H(y).$$

This is known as the chain rule of entropy.

11.13 Mutual information is the cost reduction under log-loss

Under the log-loss cost function, the average reduction in expected cost from observing y is called the mutual information between x and y :

$$I(x; y) := H(x) - H(x | y) = \text{“}\Delta \mathbb{E}[C(x, q)]\text{”}$$

Mutual information $I(x; y)$ measures how much uncertainty about x is reduced on average after observing y . It also has some nice properties: For discrete random variables $x \in \mathcal{X}$ and $y \in \mathcal{Y}$,

$$I(x; y) = \sum_{a \in \mathcal{X}, b \in \mathcal{Y}} p_{x,y}(a, b) \log \frac{p_{x,y}(a, b)}{p_x(a) p_y(b)}.$$

Mutual information is symmetric:

$$I(x; y) = I(y; x).$$

It is also always nonnegative:

$$I(x; y) \geq 0,$$

with equality if and only if $x \perp y$.

11.14 Chain rule of mutual information

For random quantities x, y, z , the chain rule is

$$I(x; y, z) = I(x; z) + I(x; y | z).$$

Which decomposes information that (y, z) provides about x into information from z + additional information from y once z is known.

11.15 Conditional mutual information

The conditional mutual information is defined by

$$I(x; y | z) := H(x | z) - H(x | y, z).$$

It measures how much uncertainty about x is reduced by observing y when z is already known. It satisfies $I(x; y | z) \geq 0$, with equality if and only if x and y are conditionally independent given z , namely $x \leftrightarrow z \leftrightarrow y$.

11.16 Conditioning does not always reduce mutual information

It is not true in general that conditioning must decrease mutual information. In other words, one cannot conclude in general that

$$I(x; y | z) \leq I(x; y).$$

This requires extra structure. A key sufficient condition is the Markov chain $x \leftrightarrow y \leftrightarrow z$.

11.17 Ven diagram for mutual information and entropies

A useful identity relating joint entropy and mutual information is

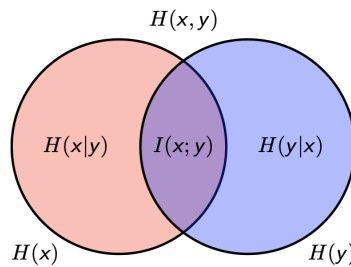


Figure 7: Venn diagram of relationships between mutual information and entropies.

For example, we can see that

$$H(x, y) = H(x) + H(y) - I(x; y).$$

the joint uncertainty equals the sum of the separate uncertainties minus the overlap.

11.18 Data processing inequality

The data processing inequality connects mutual information with sufficient statistics. In particular, if

$$x \leftrightarrow y \leftrightarrow t$$

where t is any statistic constructed from y , then

$$I(x; y) \geq I(x; t).$$

Thus, processing y into t cannot increase the amount of information about x .

11.19 Data processing equality condition and relation to sufficiency

From the chain rule we see that equality

$$I(x; t) = I(x; y)$$

holds if and only if

$$I(x; y \mid t) = 0.$$

This is equivalent to the conditional independence relation

$$x \leftrightarrow t \leftrightarrow y.$$

Hence, a statistic $t = t(y)$ is sufficient if and only if it preserves all the mutual information that y has about x .

11.20 Inference interpretation of the data processing inequality

The data processing inequality says that preprocessing data in the sense of

$$\text{full observation } y \longrightarrow \text{statistic } t$$

cannot improve inference about the target random quantity x . Any statistic extracted from y can only lose information, unless it is sufficient, in which case it maintains information.

11.21 KL divergence is the approximation loss under log-loss

In practice,

- before observing any data, the true belief is the prior $p_x(\cdot)$,
- after observing $y = y$, the true belief is the posterior $p_{x|y}(\cdot \mid y = y)$.

but either of these may be expensive, so we approximate it with $q(\cdot)$. Let's work in the case where p_x is difficult. Under log-loss, the excess expected cost of using another distribution $q(\cdot)$ instead of p_x is

$$“\Delta \mathbb{E}[C(x, q)]” = -\mathbb{E}_{p_x}[\log q(x)] + \mathbb{E}_{p_x}[\log p_x(x)] = \mathbb{E}_{p_x} \left[\log \frac{p_x(x)}{q(x)} \right].$$

This quantity is defined as the information divergence, or Kullback–Leibler divergence, from q to p_x :

$$D(p_x \parallel q) := “\Delta \mathbb{E}[C(x, q)]” := \mathbb{E}_{p_x} \left[\log \frac{p_x(x)}{q(x)} \right].$$

11.22 KL divergence: non-symmetric, non-negative, support mismatch

For a discrete random quantity $x \in \mathcal{X}$,

$$D(p_x \| q) := \sum_{a \in \mathcal{X}} p_x(a) \log \frac{p_x(a)}{q(a)}.$$

Thus, KL divergence measures the penalty incurred by using q in place of the true law p_x . It also has some nice properties: Divergence is not a true distance because it is not symmetric:

$$D(p_x \| q) \neq D(q \| p_x).$$

A basic consequence of Gibbs' inequality is that $D(p_x \| q) \geq 0$. Where equality holds if and only if $p_x(a) = q(a)$ for all $a \in \mathcal{X}$.

If $\text{supp}(p_x) \not\subseteq \text{supp}(q)$, then $D(p_x \| q) = \infty$. This occurs because $q(a) = 0$ at some value where $p_x(a) > 0$, so the log-ratio becomes infinite.

11.23 Entropy measures KL divergence from uniform distribution

Let $\mathcal{U}(\mathcal{X})$ denote the uniform distribution on $\mathcal{X}$, so

$$\mathcal{U}(\mathcal{X})(x) = \frac{1}{|\mathcal{X}|}.$$

Then

$$D(p_x \| \mathcal{U}(\mathcal{X})) = \sum_{x \in \mathcal{X}} p_x(x) \log \frac{p_x(x)}{1/|\mathcal{X}|},$$

so

$$D(p_x \| \mathcal{U}(\mathcal{X})) = \log |\mathcal{X}| - H(x).$$

Therefore, entropy measures how close a distribution is to being uniform: the larger $H(x)$, the smaller the divergence to the uniform distribution.

11.24 Mutual information as a KL divergence

Mutual information can be written as a divergence between the joint law and the product of marginals:

$$I(x; y) = D(p_{x,y} \| p_x p_y).$$

So mutual information measures how far the joint distribution is from the independence model

$$p_{x,y}(x, y) = p_x(x) p_y(y).$$

Hence, $I(x; y) = 0$ if and only if x and y are independent.

11.25 Convexity of divergence

For finite $\mathcal{Y}$ and distributions $p, q \in \mathcal{P}^{\mathcal{Y}}$, the information divergence

$$D(p \| q)$$

is convex jointly in (p, q) . More precisely, for $0 \leq \lambda \leq 1$,

$$D(\lambda p_1 + (1 - \lambda) p_2 \| \lambda q_1 + (1 - \lambda) q_2) \leq \lambda D(p_1 \| q_1) + (1 - \lambda) D(p_2 \| q_2).$$

12 Information Geometry

12.1 Inference can be viewed in different spaces

Inference tasks can be understood in more than one kind of space. A common viewpoint is the *data space*, where points in the space are observed data values themselves, either in raw form or through features such as sufficient statistics. However, another equally important viewpoint is the *distribution space*, where the objects of interest are probability distributions rather than individual data points. A key tool in distribution space is the information divergence

$$D(p||q) = \sum_y p(y) \log \frac{p(y)}{q(y)}.$$

Where $p(\cdot)$ and $q(\cdot)$ are two distributions on the same space and $D(p||q)$ measures how different these distributions are.

12.2 Probability simplex for a finite alphabet

Let y be a random variable taking values in a finite alphabet

$$\mathcal{Y} = \{1, 2, \dots, M\}, \quad M < \infty.$$

A distribution $p_y(\cdot)$ can be represented by the vector

$$(p_y(1), p_y(2), \dots, p_y(M)) \in \mathbb{R}^M.$$

Because probabilities satisfy the following affine constraints

$$p_y(y) \geq 0 \quad \text{for all } y \in \mathcal{Y}, \quad \sum_{y \in \mathcal{Y}} p_y(y) = 1,$$

they exist in a probability simplex. In general we denote the set of all distributions on $\mathcal{Y}$ by

$$\mathcal{P}^{\mathcal{Y}} = \left\{ p_y(\cdot) : p_y(y) \geq 0, \sum_{y \in \mathcal{Y}} p_y(y) = 1 \right\}.$$

This set is called the *probability simplex*. Since one coordinate is determined by the normalization constraint, the simplex has dimension

$$|\mathcal{Y}| - 1 = M - 1.$$

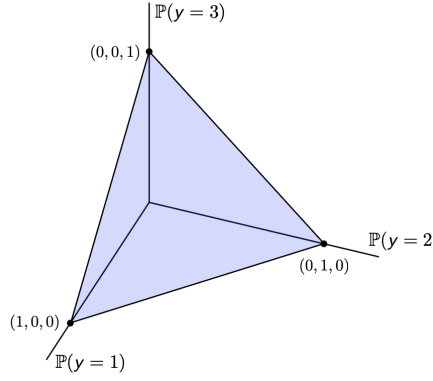


Figure 8: The probability simplex for distributions defined on the alphabet $\mathcal{Y} = \{1, 2, 3\}$, which lies in a plane of dimension $|\mathcal{Y}| - 1 = 2$ within the space of dimension $|\mathcal{Y}| = 3$.

12.3 Interior and boundary of the simplex

A distribution is in the interior of the simplex exactly when all coordinates are strictly positive:

$$p_{\mathcal{Y}}(\cdot) \in \text{int}(\mathcal{P}^{\mathcal{Y}}) \iff p_{\mathcal{Y}}(y) > 0 \text{ for all } y \in \mathcal{Y}.$$

A distribution is on the boundary exactly when at least one coordinate is zero:

$$p_{\mathcal{Y}}(\cdot) \in \partial(\mathcal{P}^{\mathcal{Y}}) \iff p_{\mathcal{Y}}(y') = 0 \text{ for some } y' \in \mathcal{Y}.$$

Each boundary component corresponds to setting one or more coordinates equal to zero, so it is itself a lower-dimensional simplex.

12.4 Divergence finite in interior, infinite on boundry

For two distributions $p_{\mathcal{Y}}(\cdot)$ and $q_{\mathcal{Y}}(\cdot)$,

$$D(p_{\mathcal{Y}} \| q_{\mathcal{Y}}) = \sum_{y \in \mathcal{Y}} p_{\mathcal{Y}}(y) \log \frac{p_{\mathcal{Y}}(y)}{q_{\mathcal{Y}}(y)}.$$

As long as $q_{\mathcal{Y}}(\cdot)$ is in the interior of the simplex, then

$$D(p_{\mathcal{Y}} \| q_{\mathcal{Y}}) < \infty,$$

because every denominator $q_{\mathcal{Y}}(y)$ is strictly positive. On the other hand, if $p_{\mathcal{Y}}(\cdot)$ is in the interior but $q_{\mathcal{Y}}(\cdot)$ is on the boundary, then

$$D(p_{\mathcal{Y}} \| q_{\mathcal{Y}}) = \infty.$$

Notice the asymmetry between two directions: one quantity can be finite but the reverse is infinite.

12.5 I-projection onto a set of distributions

Let $\mathcal{P}$ be a nonempty, closed set of distributions. The *information projection* or *I-projection* of a distribution $q_{\mathcal{Y}}(\cdot)$ onto $\mathcal{P}$ is a distribution

$$p_{\mathcal{Y}}^* = \arg \min_{p_{\mathcal{Y}}(\cdot) \in \mathcal{P}} D(p_{\mathcal{Y}} \| q_{\mathcal{Y}}).$$

So, among all distributions in $\mathcal{P}$, the I-projection is closest to $q_{\mathcal{Y}}(\cdot)$ in information divergence.

12.6 Existence and uniqueness of the I-projection

In general, an I-projection exists because

$$D(p_Y \| q_Y)$$

is nonnegative and continuous in $p_Y(\cdot)$, and $\mathcal{P}$ is closed and nonempty. Uniqueness need not hold for an arbitrary set $\mathcal{P}$. However, if $\mathcal{P}$ is convex, then the I-projection is unique.

12.7 Pythagorean theorem for information divergence

Let p_Y^* be the I-projection of $q_Y(\cdot)$ onto a nonempty, closed, convex set $\mathcal{P}$. Then for every

$$p_Y(\cdot) \in \mathcal{P},$$

we have

$$D(p_Y \| q_Y) \geq D(p_Y \| p_Y^*) + D(p_Y^* \| q_Y).$$

This is the information-geometry analogue of Pythagoras' theorem.

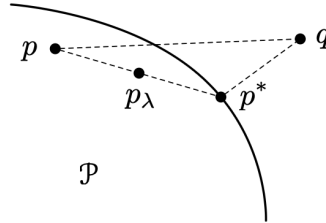


Figure 9: Divergence relationships in the information version of Pythagoras' theorem.

12.8 Corollary: simplex interior projection $\neq$ isolated boundary component

Assume we have a distribution in the interior of the probability simplex

$$q_Y(\cdot) \in \text{int}(\mathcal{P}^{\mathcal{Y}}),$$

and let p_Y^* be its I-projection onto some family of distributions $\mathcal{P}$. Then p_Y^* cannot lie on a boundary of the simplex unless the entire family $\mathcal{P}$ is contained in that same boundary component. Recall that a boundary is a wall of the simplex where at least one coordinate is zero; landing on one such wall is only possible if the whole constraint set already lies within that wall. The reason is that for any $p_Y(\cdot) \in \mathcal{P}$,

$$D(p_Y \| q_Y) < \infty$$

since $q_Y(\cdot)$ is strictly positive. By the Pythagorean inequality, this forces

$$D(p_Y \| p_Y^*) < \infty.$$

But if p_Y^* were on the boundary, finiteness of

$$D(p_Y \| p_Y^*)$$

would require every $p_Y(\cdot) \in \mathcal{P}$ to lie on the same boundary component. Hence, unless all of $\mathcal{P}$ is confined there, the projection cannot land on the boundary of the simplex.

12.9 Definition of a linear family

For outcome space $\mathcal{Y} = \{1, \dots, M\}$, a set of distributions $\mathcal{P} \subset \mathcal{P}^{\mathcal{Y}}$ is called a *linear family* if there exist a function vector

$$\mathbf{t}(y) = \begin{bmatrix} t_1(y) \\ \vdots \\ t_K(y) \end{bmatrix}, \quad K < M,$$

and a constant vector

$$\bar{\mathbf{t}} = \begin{bmatrix} \bar{t}_1 \\ \vdots \\ \bar{t}_K \end{bmatrix},$$

such that every distribution in this set $p_Y(\cdot) \in \mathcal{P}$ satisfies

$$\mathbb{E}_{p_Y}[t_i(y)] = \bar{t}_i, \quad i = 1, \dots, K.$$

So a linear family is the set of all distributions whose expectations of certain statistics are fixed.

12.10 Matrix form of the constraints

If

$$\mathcal{Y} = \{1, \dots, M\},$$

and we define the matrix

$$\tilde{\mathbf{T}} = \begin{bmatrix} t_1(1) - \bar{t}_1 & \cdots & t_1(M) - \bar{t}_1 \\ \vdots & \ddots & \vdots \\ t_K(1) - \bar{t}_K & \cdots & t_K(M) - \bar{t}_K \end{bmatrix}.$$

Then an equivalent formulation of a linear family $\mathcal{P}$ is as the intersection of the simplex with the null space of $\tilde{\mathbf{T}}$

$$\tilde{\mathbf{T}}\mathbf{p} = 0.$$

Each linearly independent constraint reduces the dimension by one, so geometrically we get a lower-dimensional flat set inside the simplex. Since the simplex has dimension

$$M - 1,$$

we get

$$\dim \mathcal{P} = M - \text{rank}(\tilde{\mathbf{T}}) - 1.$$

12.11 Minimal representation

The representation $(\mathbf{t}, \bar{\mathbf{t}})$ is called *minimal* when the rows of $\tilde{\mathbf{T}}$ are linearly independent. In that case, all listed constraints are genuinely distinct, and

$$\text{rank}(\tilde{\mathbf{T}}) = K.$$

Hence,

$$\dim \mathcal{P} = M - K - 1.$$

12.12 Convexity and closedness

A linear family is convex. If

$$p_{y,1}(\cdot), p_{y,2}(\cdot) \in \mathcal{P},$$

and

$$p_y(\cdot) = \lambda p_{y,1}(\cdot) + (1 - \lambda) p_{y,2}(\cdot), \quad \lambda \in [0, 1],$$

then

$$p_y(\cdot) \in \mathcal{P}.$$

which follows by checking this distribution satisfies the equalities that define the linear family. A linear family is also closed.

12.13 Convexity and affine-line property

A linear family is convex. Indeed, if $p_1, p_2 \in \mathcal{P}$ and

$$p = \lambda p_1 + (1 - \lambda) p_2 \in \mathcal{P}^{\mathcal{Y}} \in \mathcal{P}$$

for some $\lambda \in \mathbb{R}$, then still $p \in \mathcal{P}$. So the whole line through two points of the family remains in the family, as long as the resulting point is still a valid distribution.

12.14 Pythagorean identity for I-projections onto a linear family

Let $\mathcal{L}$ be a linear family on the outcome space $\mathcal{Y}$, and let q be a distribution in the interior of the simplex on $\mathcal{Y}$. Letting

$$p^* = \arg \min_{p \in \mathcal{L}} D(p \| q)$$

be the I-projection of q onto the linear family $\mathcal{L}$, then for every $p \in \mathcal{L}$,

$$D(p \| q) = D(p \| p^*) + D(p^* \| q).$$

So the KL divergence decomposes exactly into the divergence from p to the projection point plus the divergence from the projection point to q . More broadly, this says that for linear families, the Pythagorean inequality becomes an equality.

12.15 Linear family through a given distribution

Fix a minimal representation with statistic vector

$$\mathbf{t} = [t_1, \dots, t_K]^\top.$$

Given any distribution $p^* \in \mathcal{P}^{\mathcal{Y}}$, there is a unique vector of constants

$$\bar{\mathbf{t}} = [\bar{t}_1, \dots, \bar{t}_K]^\top$$

such that p^* belongs to the corresponding linear family. Indeed, the only possible choice is

$$\bar{\mathbf{t}} = \mathbb{E}_{p^*}[\mathbf{t}(y)].$$

Thus the linear family through p^* is

$$\mathcal{L}_{\mathbf{t}}(p^*) = \left\{ p \in \mathcal{P}^{\mathcal{Y}} : \mathbb{E}_p[\mathbf{t}(y)] = \bar{\mathbf{t}} \right\}, \quad \bar{\mathbf{t}} = \mathbb{E}_{p^*}[\mathbf{t}(y)].$$

However, this does not mean p^* uniquely determines the statistics. In fact, if

$$p \in \mathcal{L}_{\mathbf{t}}(p^*) \quad \text{then} \quad \mathcal{L}_{\mathbf{t}}(p^*) = \mathcal{L}_{\mathbf{t}}(p)$$

12.16 Orthogonality of linear and exponential families

Once $\mathcal{L}_{\mathbf{t}}(p^*)$ is fixed, a natural question is:

Which distributions $q \in \mathcal{P}^{\mathcal{Y}}$ have I-projection onto $\mathcal{L}_{\mathbf{t}}(p^*)$ equal to p^* ?

Thrm: Let $p^* \in \mathcal{P}^{\mathcal{Y}}$ be arbitrary, and define the linear family

$$\mathcal{L}_{\mathbf{t}}(p^*) = \left\{ p \in \mathcal{P}^{\mathcal{Y}} : \mathbb{E}_p[\mathbf{t}(y)] = \bar{\mathbf{t}} \triangleq \mathbb{E}_{p^*}[\mathbf{t}(y)] \right\}.$$

Then p^* is the I-projection of a distribution $q \in \text{int}(\mathcal{P}^{\mathcal{Y}})$ onto $\mathcal{L}_{\mathbf{t}}(p^*)$ if and only if q belongs to the exponential family

$$\mathcal{E}_{\mathbf{t}}(p^*) = \left\{ r \in \mathcal{P}^{\mathcal{Y}} : r(y) = p^*(y) \exp\{\mathbf{x}^\top \mathbf{t}(y) - \alpha(\mathbf{x})\}, \quad y \in \mathcal{Y}, \mathbf{x} \in \mathcal{X} \right\},$$

where $\mathcal{X} \subset \mathbb{R}^K$ is the natural parameter space.

Equivalently,

$$p^* = \arg \min_{p \in \mathcal{L}_{\mathbf{t}}(p^*)} D(p||q) \quad \Longleftrightarrow \quad q \in \mathcal{E}_{\mathbf{t}}(p^*).$$

This theorem says that the set of all distributions whose I-projection onto a fixed linear family equals p^* is exactly an exponential family through p^* . Hence linear families and exponential families form orthogonal pairs in information geometry.

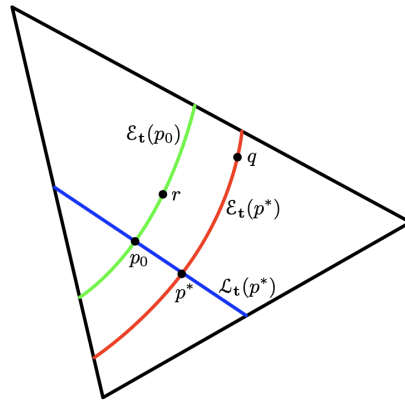


Figure 10: Figure 14.6: Orthogonality relationships in the simplex. The members of exponential family $\mathcal{E}_{\mathbf{t}}(p^*) = \mathcal{E}_{\mathbf{t}}(q)$, denoted by the red curve, all have an I-projection of p^* onto the linear family $\mathcal{L}_{\mathbf{t}}(p^*) = \mathcal{L}_{\mathbf{t}}(p_0)$, denoted by the blue line. The members of the exponential family $\mathcal{E}_{\mathbf{t}}(r) = \mathcal{E}_{\mathbf{t}}(p_0)$, denoted by the green curve, all have an I-projection of p_0 onto this linear family.

12.17 Dimensions of the orthogonal pair

When the components of $\mathbf{t}$ are linearly independent, the linear family $\mathcal{L}_{\mathbf{t}}(p^*)$ is a minimal linear family of dimension

$$M - K - 1,$$

while the associated exponential family $\mathcal{E}_{\mathbf{t}}(p^*)$ is a K -dimensional canonical exponential family. So the simplex is decomposed locally into two complementary directions:

linear-family directions $(M - K - 1)$ and exponential-family directions K .

12.18 How to compute the I-projection

If p^* is the I-projection of q onto the linear family determined by $\mathbf{t}$ and $\bar{\mathbf{t}}$, then p^* lies both in the linear family and in the corresponding orthogonal exponential family. Hence for some $\mathbf{x} \in \mathbb{R}^K$,

$$p^*(y) = q(y) \exp\{\mathbf{x}^\top \mathbf{t}(y) - \alpha(\mathbf{x})\},$$

and simultaneously

$$\sum_{y \in \mathcal{Y}} \mathbf{t}(y) p^*(y) = \bar{\mathbf{t}}.$$

Substituting the exponential-family form into the moment constraint gives

$$\sum_{y \in \mathcal{Y}} \mathbf{t}(y) q(y) \exp\{\mathbf{x}^\top \mathbf{t}(y) - \alpha(\mathbf{x})\} = \bar{\mathbf{t}}.$$

This is a nonlinear system of K equations in the K unknown components of $\mathbf{x}$. Solving for $\mathbf{x}$ determines p^* .

12.19 Main takeaways on linear families

A linear family is the set of distributions satisfying fixed expectation constraints. It is an affine slice of the simplex, closed under affine combinations that remain valid distributions. Its I-projection geometry is especially clean:

- the KL Pythagorean identity holds with equality;
- the distributions projecting to a fixed point p^* form an exponential family;
- computing the I-projection reduces to solving moment-matching equations in exponential-family coordinates.

12.20 Canonical exponential family and the base distribution

Consider the one-dimensional canonical exponential family on a finite alphabet $\mathcal{Y}$:

$$p_y(y; x) = \exp\{xt(y) - \alpha(x) + \beta(y)\}.$$

It is convenient to choose

$$\beta(y) = \ln q(y)$$

for some reference distribution $q \in \mathcal{P}^{\mathcal{Y}}$. Then the family becomes

$$p_y(y; x) = q(y) \exp\{xt(y) - \alpha(x)\}.$$

and evaluating at parameter $x = 0$ gives

$$p_y(y; 0) = q(y) \exp\{-\alpha(0)\}.$$

Since $p_y(\cdot; 0)$ must be a distribution, normalization implies

$$\alpha(0) = 0, \quad p_y(y; 0) = q(y).$$

Thus $p_y(\cdot; 0)$ is the distinguished reference, or base, distribution in the family, and the rest of the family is obtained by exponentially tilting this base distribution. We've previously seen that the first derivative of the log-partition function satisfies

$$\eta(x) := \dot{\alpha}(x) = \mathbb{E}_{p_y(\cdot; x)}[t(y)].$$

So $\eta(x)$ is the mean parameter, namely the expectation of the sufficient statistic under the distribution indexed by x . Differentiating once more yields

$$\dot{\eta}(x) = \ddot{\alpha}(x) = J_y(x) \geq 0,$$

where $J_y(x)$ is the Fisher information for the natural parameter x . Hence Fisher information measures how quickly the mean parameter changes as we move along the exponential family.

We now compare each distribution $p_y(\cdot; x)$ to the base distribution $p_y(\cdot; 0)$. Using

$$\log \frac{p_y(y; x)}{p_y(y; 0)} = xt(y) - \alpha(x),$$

we obtain

$$D(p_y(\cdot; x) \| p_y(\cdot; 0)) = \sum_y p_y(y; x) (xt(y) - \alpha(x)) = x \mathbb{E}_{p_y(\cdot; x)}[t(y)] - \alpha(x).$$

Since $\mathbb{E}_{p_y(\cdot; x)}[t(y)] = \eta(x)$, this becomes

$$D(p_y(\cdot; x) \| p_y(\cdot; 0)) = x\eta(x) - \alpha(x).$$

This becomes especially informative when we regard divergence as a function of the mean parameter η rather than of x . Differentiate with respect to x :

$$\frac{d}{dx} D(p_y(\cdot; x) \| p_y(\cdot; 0)) = \eta(x) + x\dot{\eta}(x) - \dot{\alpha}(x) = x\dot{\eta}(x),$$

because $\dot{\alpha}(x) = \eta(x)$. Applying the chain rule,

$$\frac{d}{d\eta} D(p_y(\cdot; x) \| p_y(\cdot; 0)) = \frac{d}{dx} D(p_y(\cdot; x) \| p_y(\cdot; 0)) \cdot \frac{dx}{d\eta} = x\dot{\eta}(x) \cdot \frac{1}{\dot{\eta}(x)} = x.$$

This shows that η is a natural coordinate for measuring divergence along the family. Differentiating once more with respect to η gives

$$\frac{d^2}{d\eta^2} D(p_y(\cdot; x) \| p_y(\cdot; 0)) = \frac{dx}{d\eta} = \frac{1}{\dot{\eta}(x)} = \frac{1}{J_y(x)} \geq 0.$$

Thus the curvature of divergence in the mean-parameter coordinate is the reciprocal of the Fisher information. This is the key interpretation of the previous subsection: the mean parameter $\eta(x)$ tells us where we are along the family, while the Fisher information $J_y(x)$ tells us how sharply the divergence bends there.

13 Modeling as inference

13.1 Modeling versus estimation

In the nonBayesian modeling setup, we observe samples of a random variable y taking values in an alphabet $\mathcal{Y}$, and we know that its distribution belongs to a parametric family

$$\mathcal{P} = \{p_y(\cdot; x) \in \mathcal{P}^{\mathcal{Y}} : x \in \mathcal{X}\},$$

where x is an unknown deterministic parameter. The key distinction is:

- estimation aims to learn the unknown parameter x ;
- modeling aims to choose a distribution $q(\cdot)$ that well approximates the true distribution of y .

Thus, even when the model class is parametric, the modeling problem is generally not the same as the estimation problem since we desire a whole distribution, not just a parameter.

13.2 A Bernoulli example and the plug-in heuristic

Here is a case where modeling the distribution by choosing a $q(\cdot)$ actually *is equivariant* as parametric estimation. Suppose $y_1, y_2, \dots, y_{n-1}$ are i.i.d. Bernoulli random variables with unknown parameter $x \in [0, 1]$, so $y_i \sim \text{Bernoulli}(x)$. After observing realizations $y_1, \dots, y_{n-1}$, we want a good model for y_n .

A natural heuristic is the plug-in approach: first estimate x , then substitute this estimate into the parametric model. The maximum likelihood estimator is

$$\hat{x}_{\text{ML}}^{(n-1)} = \frac{1}{n-1} \sum_{i=1}^{n-1} y_i$$

The plug-in estimated model $q(\cdot)$ is then

$$q(y_n) = p_y(y_n; \hat{x}_{\text{ML}}^{(n-1)}).$$

This can be asymptotically justified as $n \rightarrow \infty$, but it is generally not optimal for finite n .

13.3 Modeling loss from $q(\cdot)$ is measured by KL divergence

To judge whether a candidate model q is good, we measure the loss from using q instead of the true distribution $p_y(\cdot; x)$ by the information divergence

$$D(p_y(\cdot; x) \| q(\cdot)).$$

So the modeling problem is to choose a distribution $q \in \mathcal{P}^{\mathcal{Y}}$ that does not depend on the unknown parameter x , the divergence small for all $x \in \mathcal{X}$. A central message is that a good model need not arise from explicitly estimating x first.

13.4 Modeling via mixtures (an alternative to directly estimating x)

The fundamental idea is to model y by a mixture of the candidate distributions in the class:

$$q_w(y) = \sum_{x \in \mathcal{X}} w(x) p_y(y; x),$$

where the weights satisfy

$$w(x) \geq 0 \quad \text{for all } x \in \mathcal{X}, \quad \sum_{x \in \mathcal{X}} w(x) = 1.$$

Thus w is itself a distribution on $\mathcal{X}$. From a Bayesian viewpoint, w looks like a prior, but here the mixture construction is introduced purely as a modeling device.

So the proposal is:

- do not pick one fitted model $p_Y(\cdot; \hat{x})$;
- instead, form our modeling distribution $q(\cdot)$ by averaging over the candidate models using weights w

13.5 Mixtures can be simultaneously close to distant candidate models

Consider the extreme case where $x \in \{0, 1\}$ and

$$y_n \sim \text{Bernoulli}(x).$$

Then the two candidate models are very far apart:

$$D(p_Y(\cdot; 0) \| p_Y(\cdot; 1)) = \infty, \quad D(p_Y(\cdot; 1) \| p_Y(\cdot; 0)) = \infty.$$

However, the equal-weight mixture

$$q(y) = \frac{1}{2}p_Y(y; 0) + \frac{1}{2}p_Y(y; 1)$$

is simultaneously close to both:

$$D(p_Y(\cdot; 0) \| q) = D(p_Y(\cdot; 1) \| q) = 1 \text{ bit}.$$

So even though the candidate models are infinitely far from each other in KL divergence, their mixture can still provide a good approximation to each one, defying our intuition from Euclidean geometry.

13.6 Mixtures are always at least as good as any fixed model

Let $q \in \mathcal{P}^{\mathcal{Y}}$ be any candidate distribution. Then there exists a choice of weights $w(\cdot)$ such that the mixture

$$q_w(\cdot) = \sum_{x \in \mathcal{X}} w(x) p_Y(\cdot; x)$$

satisfies

$$D(p_Y(\cdot; x) \| q_w) \leq D(p_Y(\cdot; x) \| q) \quad \text{for all } x \in \mathcal{X}.$$

In words: for every fixed model q , there is a mixture model that performs at least as well against every member of the class. To prove this, define the set of all mixtures of the candidate models as

$$\mathcal{P}_{\text{mix}} = \left\{ \sum_{x \in \mathcal{X}} w(x) p_Y(\cdot; x) : w \in \mathcal{P}^{\mathcal{X}} \right\}.$$

This set is closed and convex. Given any distribution q , define q_w as the I-projection of q onto the mixture family $\mathcal{P}_{\text{mix}}$

$$q_w = \arg \min_{p \in \mathcal{P}_{\text{mix}}} D(p \| q).$$

By the Pythagorean theorem of information geometry,

$$D(p\|q) \geq D(p\|q_w) + D(q_w\|q) \geq D(p\|q_w) \quad \text{for all } p \in \mathcal{P}_{\text{mix}}.$$

Since each candidate distribution $p_y(\cdot; x)$ lies in $\mathcal{P} \subset \mathcal{P}_{\text{mix}}$, we obtain

$$D(p_y(\cdot; x) \| q) \geq D(p_y(\cdot; x) \| q_w) \quad \text{for all } x \in \mathcal{X}.$$

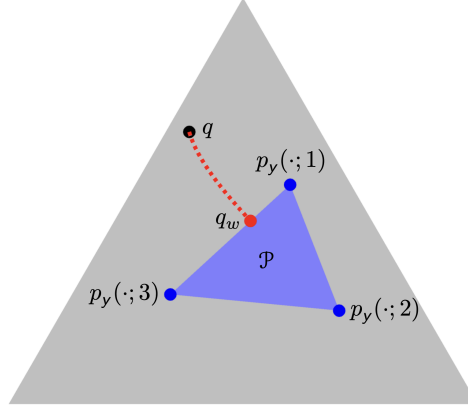


Figure 11: The I -projection of a non-mixture model q onto $\mathcal{P}$ produces a mixture model q_w , which is closer (in a divergence sense) to each of the possible models than q . In this case $\mathcal{X} = \{1, 2, 3\}$

Main takeaway:

For modeling, the right object is often not a plug-in estimate based on first estimating x , but rather a mixture over the model class. Mixtures are fundamentally good because:

- they can stay close to many candidate models at once;
- they dominate arbitrary fixed choices of q in KL divergence;
- they connect nonBayesian modeling and Bayesian averaging in a natural way.

So the Bayesian and nonBayesian viewpoints are deeply intertwined: even in a nonBayesian modeling problem, mixture distributions play the central role.

Note: From here on out, we use the mixture model $q = q_w \in \mathcal{P}^{\mathcal{Y}}$.

13.7 Minimax formulation of the modeling problem

A natural way to choose a model $q \in \mathcal{P}^{\mathcal{Y}}$ is by minimizing the worst-case KL loss over the class

$$\mathcal{P} = \{p_y(\cdot; x) \in \mathcal{P}^{\mathcal{Y}} : x \in \mathcal{X}\}.$$

This leads to the minimax redundancy

$$R^+ \triangleq \min_{q \in \mathcal{P}^{\mathcal{Y}}} \max_{x \in \mathcal{X}} D(p_y(\cdot; x) \| q).$$

Interpretation:

- we choose $q \in \mathcal{P}^{\mathcal{Y}}$ to make the modeling loss small;
- nature then chooses $x \in \mathcal{X}$ to make the loss as large as possible.

Thus R^+ is the smallest worst-case divergence we can guarantee.

13.8 Equivalent minimax problem over weights

The inner maximization over x can be rewritten as a maximization over all distributions on $\mathcal{X}$. For any $q \in \mathcal{P}^{\mathcal{Y}}$,

$$\max_{x \in \mathcal{X}} D(p_Y(\cdot; x) \| q) = \max_{w \in \mathcal{P}^{\mathcal{X}}} \sum_{x \in \mathcal{X}} w(x) D(p_Y(\cdot; x) \| q).$$

since any weighted average of the divergences cannot exceed their maximum, hence the worst-case point mass and the worst-case mixture over parameters give the same value. We can now rewrite the minimax redundancy as

$$R^+ = \min_{q \in \mathcal{P}^{\mathcal{Y}}} \max_{w \in \mathcal{P}^{\mathcal{X}}} \sum_{x \in \mathcal{X}} w(x) D(p_Y(\cdot; x) \| q).$$

This form is important because the objective now has two variables:

- $q \in \mathcal{P}^{\mathcal{Y}}$ is a distribution on $\mathcal{Y}$;
- a distribution w on $\mathcal{X}$.

It will allow us to relate the minimax problem to a maximin problem that is easier to solve.

13.9 Redundancy-capacity theorem says minmax=maxmin

Thrm: the minimax and maximin problems coincide:

$$R^+ = \min_{q \in \mathcal{P}^{\mathcal{Y}}} \max_{w \in \mathcal{P}^{\mathcal{X}}} \sum_{x \in \mathcal{X}} w(x) D(p_Y(\cdot; x) \| q) = \max_{w \in \mathcal{P}^{\mathcal{X}}} \min_{q \in \mathcal{P}^{\mathcal{Y}}} \sum_{x \in \mathcal{X}} w(x) D(p_Y(\cdot; x) \| q) \triangleq R^-.$$

So although the original minimax problem looks hard, we may instead study the maximin version which turns out to have a very explicit inner solution. Proof non-trivial

13.10 Solving the inner minimization for fixed weights

This new maximin formulation is easier to solve than the previous minmax. To solve it, fix weights from the outer player $w \in \mathcal{P}^{\mathcal{X}}$. We seek $q^*(\cdot) \in \mathcal{P}^{\mathcal{Y}}$. first define

$$\varphi_{\text{inner}}(q) \triangleq \sum_{x \in \mathcal{X}} w(x) D(p_Y(\cdot; x) \| q).$$

Expanding KL divergence gives

$$\varphi_{\text{inner}}(q) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} w(x) p_Y(y; x) \log \frac{p_Y(y; x)}{q(y)}.$$

Rearranging,

$$\varphi_{\text{inner}}(q) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} w(x) p_Y(y; x) \log p_Y(y; x) - \sum_{y \in \mathcal{Y}} \left[\sum_{x \in \mathcal{X}} w(x) p_Y(y; x) \right] \log q(y).$$

Define

$$c \triangleq \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} w(x) p_Y(y; x) \log p_Y(y; x),$$

which does not depend on q , and define the mixture distribution

$$q_w(y) \triangleq \sum_{x \in \mathcal{X}} w(x) p_Y(y; x).$$

Then

$$\varphi_{\text{inner}}(q) = c - \mathbb{E}_{q_w} [\log q(y)].$$

Thus minimizing $\varphi_{\text{inner}}(q)$ over q is equivalent to maximizing $\mathbb{E}_{q_w}[\log q(y)]$, which by Gibbs' inequality is achieved at

$$q^*(\cdot) = q_w(\cdot) = \sum_{x \in \mathcal{X}} w(x) p_Y(\cdot; x).$$

Therefore, the optimal model for fixed weights is the mixture model. In particular, for a fixed weight vector w , the minimizing distribution is exactly the mixture model with those weights. In other words, once w is fixed, the best q is not a plug-in model but the corresponding mixture model.

13.11 Solving outer optimization over the weights

After solving the inner minimization, the remaining objective becomes

$$\varphi_{\text{outer}}(w) \triangleq \sum_{x \in \mathcal{X}} w(x) D(p_Y(\cdot; x) \| q_w).$$

which is the cost of using $q \in \mathcal{P}^{\mathcal{Y}}$ in the setting where w is our prior over $\mathcal{X}$. Using the definition of q_w , this can be written as

$$\varphi_{\text{outer}}(w) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} w(x) p_Y(y; x) \log \frac{p_Y(y; x)}{\sum_{x' \in \mathcal{X}} w(x') p_Y(y; x')}.$$

So the full optimization reduces to choosing the weights w over $\mathcal{X}$ to maximize $\varphi_{\text{outer}}(w)$, and then setting the mixture distribution to be

$$q^*(\cdot) = q_{w^*}(\cdot) = \sum_{x \in \mathcal{X}} w^*(x) p_Y(\cdot; x).$$

Main takeaway of the optimization framework

The optimization framework shows that the problem of finding the best universal model reduces to

choose the best weights w^* , then use the mixture q_{w^*} .

This gives a precise optimization justification for modeling via mixtures.

13.12 Bayesian interpretation of the mixture weights

We now interpret the weight function $w \in \mathcal{P}^{\mathcal{X}}$ as a prior on the unknown parameter x , writing

$$w = p_x.$$

Also, write the model family as conditionals

$$p_Y(y; x) = p_{Y|X}(y | x),$$

so the class becomes

$$\mathcal{P} = \{p_{Y|X}(\cdot | x) \in \mathcal{P}^{\mathcal{Y}} : x \in \mathcal{X}\}.$$

Under this interpretation, the mixture model from the previous section is exactly the marginal law of y :

$$q_w(y) = p_Y(y) = \sum_{x \in \mathcal{X}} p_X(x) p_{Y|X}(y | x).$$

So the optimizing mixture model can be viewed as the predictive distribution induced by a prior on x .

13.13 The outer objective is mutual information

Recall the outer objective

$$\varphi_{\text{outer}}(p_x) = \sum_{x \in \mathcal{X}} p_x(x) D(p_{y|x}(\cdot | x) \| p_y).$$

Expanding KL divergence,

$$\varphi_{\text{outer}}(p_x) = \sum_{x \in \mathcal{X}, y \in \mathcal{Y}} p_x(x) p_{y|x}(y | x) \log \frac{p_{y|x}(y | x)}{p_y(y)}.$$

Since we can always expand

$$p_{x,y}(x, y) = p_x(x) p_{y|x}(y | x),$$

the outer objective becomes

$$\varphi_{\text{outer}}(p_x) = \sum_{x \in \mathcal{X}, y \in \mathcal{Y}} p_{x,y}(x, y) \log \frac{p_{x,y}(x, y)}{p_x(x) p_y(y)} = D(p_{x,y} \| p_x p_y) = I(x; y).$$

by definition of mutual information. Thus the maxmin objective reduces to

$$R^- = \max_{p_x \in \mathcal{P}^{\mathcal{X}}} I(x; y).$$

therefore since $R^+ = R^-$, the minimax modeling problem and the maxmin mutual-information problem are the same problem. This is the key conceptual link between universal modeling and information measures.

13.14 Least informative prior and model capacity

Because of this equivalence, the optimizing prior p_x^* is special. Given the family

$$\{p_{y|x}(\cdot | x) : x \in \mathcal{X}\}$$

we say that the least informative prior is defined by

$$p_x^* = \arg \max_{p_x \in \mathcal{P}^{\mathcal{X}}} I(x; y).$$

Rather, it is the prior that makes the induced model family as informative as possible in the worst-case modeling sense. The corresponding optimal value is called the capacity of the model family:

$$C = \max_{p_x \in \mathcal{P}^{\mathcal{X}}} I(x; y).$$

So model capacity measures how rich or expressive the family is. Larger C means the model family can convey more information about x through observations of y . Equivalently, capacity is the optimal worst-case reduction in modeling cost achieved by the best mixture model.

13.15 Basic bounds on capacity

The model capacity satisfies

$$0 \leq C \leq \log |\mathcal{X}|.$$

These bounds have natural interpretations:

- $C = \log |\mathcal{X}|$ in the most informative case, when x can be uniquely recovered from y ; then the optimal prior is uniform on $\mathcal{X}$.

- $C = 0$ in the least informative case, when y reveals nothing about x , meaning

$$p_{y|x}(y | x) = q(y) \quad \text{for all } x \in \mathcal{X}.$$

The upper bound comes from

$$I(x; y) = H(x) - H(x | y) \leq H(x) \leq \log |\mathcal{X}|,$$

and the lower bound from nonnegativity of mutual information.

13.16 Discrete support of the optimal prior and Dependence on the number of observations

Even when the parameter space $\mathcal{X}$ is continuous, if the observation alphabet $\mathcal{Y}$ is discrete, the least informative prior is still discrete, with at most $|\mathcal{Y}|$ mass points. So the optimizer of the capacity problem can be taken to have finite support, making the optimum easier than expected.

If instead of one observation we have a sequence

$$y_1, y_2, \dots, y_N,$$

then both the model capacity C and the optimizing prior p_x^* depend on N . This N is sometimes called the horizon of the problem.

13.17 Equidistance property of the optimal mixture

Thrm: For the model family

$$\mathcal{P} = \{p_y(\cdot; x) \in \mathcal{P}^{\mathcal{Y}} : x \in \mathcal{X}\},$$

the optimum mixture model q^* defining the maxmin redundancy R^- , for which the optimum weights are w^* , is such that

$$D(p_y(\cdot; x) \| q^*) \leq C \quad \text{for all } x \in \mathcal{X},$$

with equality for every x such that

$$w^*(x) > 0.$$

Thus every model in the support of the optimal prior sits at exactly the same KL distance from the optimal approximating distribution q^* . This is why the property is called equidistance. Intuitively, the optimal approximating distribution does not favor one active candidate model over another: all models used by the optimal prior are equally hard in the worst-case sense. Nontrivial proof.

13.18 Computing model capacity as a convex optimization problem

The capacity optimization

$$C = \max_{p_x \in \mathcal{P}^{\mathcal{X}}} I(x; y)$$

is well behaved. Indeed, once can show that for finite $\mathcal{X}$ and $\mathcal{Y}$, mutual information $I(x; y)$ is

- concave in the prior p_x ;
- convex in the conditional law $p_{y|x}$.

Therefore, our objective is concave in the prior p_x , and the feasible set $\mathcal{P}^{\mathcal{X}}$ is convex, so computing capacity is a convex optimization problem. This means standard optimization methods can be used, although the computation may still be expensive when $|\mathcal{X}|$ is large.

Main takeaway: This section identifies the optimal weights in the mixture model with a prior

that maximizes mutual information. As a result:

- the optimal mixture weights are the least informative prior;
- the optimal value is the model capacity;
- the best universal model is characterized by an equidistance property;
- computing the best model reduces to a concave maximization of $I(x; y)$ over priors.

So model capacity is the information-theoretic quantity that exactly captures the difficulty and richness of the universal modeling problem.

14 Information measures for continuous variables

14.1 Differential entropy for continuous random variables

For a continuous random variable $x \in \mathbb{R}$ with density $p_x(\cdot)$, the analogue of entropy (which we previously only had for discrete random variables) is the *differential entropy*, defined by

$$h(x) \triangleq - \int_{-\infty}^{+\infty} p_x(x) \log p_x(x) dx.$$

14.2 Differential entropy can be negative

A simple example is when $x \sim \text{Unif}(0, \Delta)$ for some $\Delta > 0$. Then

$$p_x(x) = \begin{cases} \frac{1}{\Delta}, & x \in (0, \Delta), \\ 0, & \text{otherwise,} \end{cases}$$

so

$$h(x) = - \int_0^\Delta \frac{1}{\Delta} \log \frac{1}{\Delta} dx = \log \Delta.$$

Hence $h(x) < 0$ whenever $\Delta < 1$.

14.3 Conditional differential entropy

Conditional differential entropy is defined analogously. For a conditional density $p_{x|y}(\cdot | y = y)$,

$$h(x | y = y) := - \int_{-\infty}^{+\infty} p_{x|y}(x | y) \log p_{x|y}(x | y) dx.$$

Averaging over y gives

$$h(x | y) := \int_{-\infty}^{+\infty} p_y(y) h(x | y = y) dy.$$

Equivalently,

$$h(x | y) = - \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} p_{x,y}(x, y) \log p_{x|y}(x | y) dx dy.$$

14.4 Mutual information for continuous variables

For continuous random variables, mutual information can still be written in terms of entropies:

$$I(x; y) = h(x) - h(x | y).$$

Although differential entropy itself may be negative, mutual information remains nonnegative. It measures how much uncertainty about x is reduced after observing y .

14.5 Information divergence for continuous distributions

For two densities $p(\cdot)$ and $q(\cdot)$ on the same space, the information divergence is

$$D(p||q) \triangleq \int_{-\infty}^{+\infty} p(x) \log \frac{p(x)}{q(x)} dx.$$

This is the continuous analogue of discrete relative entropy. It measures how different the

density q is from the density p .

14.6 Differential entropy of a Gaussian random variable

If $x \sim \mathcal{N}(\mu, \sigma^2)$, then its differential entropy depends only on the variance:

$$h(x) = \frac{1}{2} \log(2\pi e \sigma^2).$$

Thus, shifting the mean does not change differential entropy.

14.7 Differential entropy of a multivariate Gaussian

If $x \in \mathbb{R}^d$ satisfies

$$x \sim \mathcal{N}(\mu, \Lambda),$$

then

$$h(x) = \frac{1}{2} \log((2\pi e)^d \det(\Lambda)).$$

So for a multivariate Gaussian, the differential entropy is determined by the covariance matrix through its determinant.

14.8 Conditional differential entropy for jointly Gaussian variables

Suppose

$$\begin{bmatrix} x \\ y \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}, \begin{bmatrix} \lambda_x & \lambda_{x,y} \\ \lambda_{x,y} & \lambda_y \end{bmatrix}\right).$$

Then the conditional differential entropy is

$$h(x | y) = h(x | y = y) = \frac{1}{2} \log\left(2\pi e \left(\lambda_x - \frac{\lambda_{x,y}^2}{\lambda_y}\right)\right), \quad y \in \mathbb{R}.$$

The key point is that, in the jointly Gaussian case, the conditional variance does not depend on the realized value y .

14.9 Mutual information of jointly Gaussian variables

If x and y are jointly Gaussian, then

$$I(x; y) = \frac{1}{2} \log\left(\frac{1}{1 - \rho_{x,y}^2}\right),$$

where the Pearson correlation coefficient is

$$\rho_{x,y} \triangleq \frac{\lambda_{x,y}}{\sqrt{\lambda_x} \sqrt{\lambda_y}}.$$

Hence, for jointly Gaussian variables, the mutual information depends only on the correlation coefficient. In particular, stronger linear dependence corresponds to larger mutual information.

14.10 Information divergence between Gaussians with equal variance

Let

$$p_1 \triangleq \mathcal{N}(\mu_1, \sigma^2), \quad p_2 \triangleq \mathcal{N}(\mu_2, \sigma^2).$$

Then

$$D(p_1 \| p_2) = \frac{(\mu_1 - \mu_2)^2}{2\sigma^2}.$$

Thus, when the variances are the same, the divergence is determined by the squared distance between the means, normalized by the variance.

15 Sampling Schemes

15.1 Why approximation methods are needed in inference

Exact implementation of tools are often computationally infeasible, especially in high dimensions. Because of this, approximation methods are essential in practice. A rough taxonomy of approximation tools is:

- stochastic approximation,
- asymptotic approximation,
- variational approximation.

These methods aim to approximate key quantities or computational steps that arise in inference.

15.2 Core computational tasks in inference

Several basic computational problems repeatedly arise in inference. Important examples include:

1. computing the partition function of a distribution from an exponential family,
2. computing marginals of a joint distribution,
3. computing a conditional distribution,
4. computing expectations of functions with respect to a distribution,
5. generating samples from a distribution.

15.3 Direct sampling is often hard due to the normalizing constant

A basic stochastic approximation task is to estimate

$$\mu_f \triangleq \mathbb{E}_{p_x}[f(x)]$$

using i.i.d. samples $x_1, \dots, x_N \sim p_x$. The natural plug-in estimator is

$$\hat{\mu}_f(x) \triangleq \frac{1}{N} \sum_{n=1}^N f(x_n).$$

In many problems the target distribution has the form

$$p_x(x) = \frac{p_o(x)}{Z_p}, \quad Z_p = \sum_{x \in \mathcal{X}} p_o(x),$$

where $p_o(x)$ is easy to evaluate but Z_p is hard to compute. Since evaluating $p_x(x)$ requires Z_p , direct sampling from p_x is often infeasible. This is the main motivation for methods such as importance sampling and rejection sampling.

15.4 Importance sampling uses reweighted samples from an easy distribution

Importance sampling replaces direct sampling from p_x by sampling from an easier distribution

$$q_x(x) = \frac{q_o(x)}{Z_q}, \quad Z_q = \sum_{x \in \mathcal{X}} q_o(x).$$

Given i.i.d. samples from this easier distribution $x_1, \dots, x_N \sim q_x$, we then define the importance weight

$$w(x) \triangleq \frac{p_o(x)}{q_o(x)}.$$

Under these weights, the self-normalized importance sampling estimator for μ_f is

$$\hat{\mu}_f(x) \triangleq \frac{\frac{1}{N} \sum_{n=1}^N w(x_n) f(x_n)}{\frac{1}{N} \sum_{n=1}^N w(x_n)}.$$

Thus, samples drawn from q_x are reweighted to mimic sampling from p_x . One can show that by LLN

$$\hat{\mu}_f(x) \rightarrow \mu_f \quad \text{with probability one.}$$

So importance sampling is consistent. However, importance sampling works only if q_x places enough mass in the regions that matter for the target expectation, e.g. where $|f(x)p_o(x)|$ is large. Therefore, success depends less on asymptotic correctness, which is automatic, and much more on choosing a good sampler distribution.

15.5 Rejection sampling: main idea

Rejection sampling aims to generate exact i.i.d. samples directly from the target distribution p_x . We first assume that we have a proposal distribution $q_o(\cdot)$ with a constant c such that

$$c q_o(x) \geq p_o(x) \quad \text{for all } x \in \mathcal{X}.$$

The function $c q_o(\cdot)$ is an envelope above $p_o(\cdot)$. To generate one target sample from p_x :

1. Draw $x \sim q_x$.

2. Draw

$$u \sim \text{Unif}([0, c q_o(x)]).$$

3. Accept x if

$$u \leq p_o(x);$$

otherwise reject it and repeat.

The accepted samples are i.i.d. from p_x .

For geometric intuition, view x as horizontal and u as vertical. The algorithm generates points uniformly under the curve $c q_o(\cdot)$, then keeps only those points lying below $p_o(\cdot)$. Conditioning on acceptance leaves points uniformly distributed under the graph of $p_o(\cdot)$. Therefore, the marginal distribution of accepted x is proportional to $p_o(x)$, namely

$$p_x(x) = \frac{p_o(x)}{Z_p}.$$

So rejection sampling gives exact target samples, not just weighted approximations. However, the method is efficient only when the envelope $c q_o(\cdot)$ is close to $p_o(\cdot)$. A large c means a large gap between the two curves, so most proposed samples are rejected. The wasted mass is quantified by

$$c Z_q - Z_p.$$

In high dimensions, the required constant c often grows rapidly with dimension. Then the acceptance probability becomes extremely small, so the number of generated samples needed for one accepted draw becomes huge. This makes rejection sampling exact but often computationally useless in high-dimensional problems.

15.6 Gaussian example

Let

$$p \sim \mathcal{N}(0, \sigma_p^2 \mathbf{I}), \quad q \sim \mathcal{N}(0, \sigma_q^2 \mathbf{I}),$$

in d dimensions. The smallest envelope constant is

$$c = \frac{(2\pi\sigma_q^2)^{d/2}}{(2\pi\sigma_p^2)^{d/2}}.$$

If

$$\sigma_q^2 = (1 + \epsilon)\sigma_p^2 \quad \text{for some } \epsilon > 0,$$

then

$$c = (1 + \epsilon)^{d/2}.$$

So even if ϵ is tiny,

$$c \rightarrow \infty \quad \text{as } d \rightarrow \infty.$$

Hence acceptance deteriorates exponentially in dimension. Therefore, even a proposal distribution that looks very close to the target can be unusable in high dimensions.

15.7 Why MCMC is needed

For large state spaces $\mathcal{X}$, direct sampling methods such as rejection sampling and importance sampling often become ineffective. Markov Chain Monte Carlo avoids this by constructing a Markov chain whose stationary distribution is the target distribution p_x . Instead of producing i.i.d. samples, MCMC produces correlated samples.

15.8 Finite-state homogeneous Markov chains

A finite-state homogeneous Markov chain is a sequence $x_1, x_2, \dots$ taking values called states in a finite set $x \in \mathcal{X}$ such that:

- the future states depends on the past only through the present state,
- the transition rule does not change with time, hence time homogeneous.

Such a chain is fully specified by:

- an initial distribution p_{x_1} ,
- a transition distribution (which is the same for all n)

$$w(x | x') = p_{x_{n+1}|x_n}(x | x'),$$

15.9 Stationary distributions and how to verify a distribution is stationary

A distribution p_x on $\mathcal{X}$ is stationary if applying one transition step leaves it unchanged. There are two ways to check: if discrete

$$p_x(x) = \sum_{x' \in \mathcal{X}} w(x | x') p_x(x'), \quad \text{for all } x \in \mathcal{X}.$$

If continuous,

$$p_x(x) = \int_{\mathcal{X}} w(x | x') p_x(x') dx', \quad \text{for all } x \in \mathcal{X}.$$

Here $w(x | x')$ is the transition density from x' to x . Intuitively, the total probability mass flowing into x after one step equals the amount of probability mass that was already at x .

The second way to show that p_x is stationary is by checking the balance equations:

$$w(x' | x) p_x(x) = w(x | x') p_x(x'), \quad \text{for all } x, x' \in \mathcal{X}.$$

This condition is called detailed balance. It says that, under p_x , the probability flow from x to x' exactly equals the probability flow from x' to x . Detailed balance implies stationarity because, in the discrete case,

$$\sum_{x' \in \mathcal{X}} w(x | x') p_x(x') = \sum_{x' \in \mathcal{X}} w(x' | x) p_x(x) = p_x(x) \sum_{x' \in \mathcal{X}} w(x' | x) = p_x(x),$$

and analogously, in the continuous case,

$$\int_{\mathcal{X}} w(x | x') p_x(x') dx' = \int_{\mathcal{X}} w(x' | x) p_x(x) dx' = p_x(x) \int_{\mathcal{X}} w(x' | x) dx' = p_x(x).$$

15.10 Irreducible chains $\implies$ existence and uniqueness of stationary distribution

A chain is *fully communicating* if every state can eventually reach every other state with positive probability, meaning that chain behaves as one connected system rather than splitting into disconnected pieces.

For finite-state homogeneous Markov chains, full communication implies that the stationary distribution exists and is unique.

15.11 If irreducible, then aperiodicity $\implies$ convergence to stationary

A fully communicating/irreducible chain is aperiodic if it does not get trapped in a rigid cycle. This condition is what upgrades uniqueness of the stationary distribution into convergence to it. In particular, if a finite-state homogeneous Markov chain is fully communicating and aperiodic, then for any initial distribution,

$$\lim_{n \rightarrow \infty} p_{x_n}(x) = p_x(x), \quad \text{for all } x \in \mathcal{X},$$

where p_x is the stationary distribution.

15.12 MCMC recipe

Thus, the practical MCMC recipe is:

- construct a chain whose stationary distribution is the target,
- ensure the chain is fully communicating and aperiodic,
- run it long enough so that its law is close to stationarity.

15.13 Mixing time is determined by eigenvalues of transition matrix

The convergence theorem says that p_{x_n} approaches p_x , but it does not say how fast. The rate of convergence is governed by the mixing time. For a finite-state homogeneous Markov chain, let $\mathbf{W}$ be the transition matrix. Its largest eigenvalue is

$$\lambda_1 = 1,$$

corresponding to the stationary distribution. The second-largest eigenvalue λ_2 controls convergence speed: if λ_2 is close to 1, the chain mixes slowly; if it is well below 1, the chain mixes

more quickly. So the spectral gap,

$$1 - \lambda_2,$$

is the key quantity: larger gap means faster mixing. The main intuition is that repeated multiplication by $\mathbf{W}$ damps out all components orthogonal to stationarity, and λ_2 determines how fast the slowest such component decays.

15.14 Distance to stationary is monotonic

A useful general fact is that Markov updates cannot move a distribution farther from a stationary distribution in KL divergence. If p_x is stationary and p_{x_n} is the law at time n , then

$$D(p_{x_{n+1}} \| p_x) \leq D(p_{x_n} \| p_x).$$

This is a direct consequence of the divergence form of the data processing inequality.

16 The Metropolis–Hastings algorithm

The preceding sections tell us that we can use a Markov chain whose stationary distribution is the target distribution to obtain arbitrarily good sample approximations. We now develop the Metropolis–Hastings (MH) MCMC method, which is a way to construct a Markov chain with this property.

Suppose we want to sample from a target distribution

$$p(x) = \frac{p_o(x)}{Z_p}, \quad x \in \mathcal{X},$$

where $p_o(x)$ is known, but the partition function Z_p is unknown/expensive. The key idea of Metropolis–Hastings (MH) is to build a Markov chain whose stationary distribution is p , while only using ratios of p_o , so that Z_p cancels out.

16.1 The Metropolis–Hastings algorithm

We choose a proposal kernel $v(x' | x)$, where $v(x' | x)$ is the probability of proposing state x' when the current state is x . Given $x_n = x$, the update is as follows.

1. Propose $x' \sim v(\cdot | x)$.
2. Compute the acceptance probability

$$a(x \rightarrow x') = \min \left\{ 1, \frac{p_o(x') v(x | x')}{p_o(x) v(x' | x)} \right\}.$$

3. With probability $a(x \rightarrow x')$, accept the proposal and set $x_{n+1} = x'$. Otherwise reject the proposal and set $x_{n+1} = x$.

Thus, the proposal kernel suggests possible moves, while the acceptance probability corrects for both the target density and any asymmetry in the proposal. The resulting transition kernel w is

$$w(x' | x) = v(x' | x) a(x \rightarrow x'), \quad x' \neq x,$$

with the remaining probability assigned to staying put:

$$w(x | x) = 1 - \sum_{y \neq x} w(y | x).$$

16.2 Why the stationary distribution is p

Recall that a standard way to prove that p is stationary is to verify the detailed balance equations hold under the desired target distribution p , namely that:

$$p(x) w(x' | x) = p(x') w(x | x') \quad \text{for all } x, x' \in \mathcal{X}.$$

For $x' \neq x$, we have $w(x' | x) = v(x' | x) a(x \rightarrow x')$, so detailed balance becomes

$$p(x) v(x' | x) a(x \rightarrow x') = p(x') v(x | x') a(x' \rightarrow x).$$

The Metropolis–Hastings acceptance rule is

$$a(x \rightarrow x') = \min \left\{ 1, \frac{p(x') v(x | x')}{p(x) v(x' | x)} \right\}.$$

Let

$$R(x, x') = \frac{p(x')v(x | x')}{p(x)v(x' | x)}.$$

Then

$$a(x \rightarrow x') = \min\{1, R(x, x')\}, \quad a(x' \rightarrow x) = \min\{1, R(x, x')^{-1}\}.$$

Therefore

$$p(x)v(x' | x)a(x \rightarrow x') = \min\{p(x)v(x' | x), p(x')v(x | x')\},$$

and similarly

$$p(x')v(x | x')a(x' \rightarrow x) = \min\{p(x')v(x | x'), p(x)v(x' | x)\}.$$

These two expressions are equal, so detailed balance holds. Hence the MH chain is reversible with respect to p , and therefore p is a stationary distribution.

16.3 Convergence, burn-in, and proposal requirements

Having p as a stationary distribution does not automatically mean that the chain will reach p . For the law of x_n to converge to p , we need irreducible/fully communicating and aperiodic.

For $x' \neq x$, MH uses

$$w(x' | x) = v(x' | x)a(x \rightarrow x'),$$

so a move can occur under w only if it was possible under v and has positive acceptance probability. Thus regarding irreducible and aperiodic:

1. if the proposal chain v does not connect the state space, MH cannot fix this;
2. if a proposed move has no possible reverse move, MH rejects it with probability 1;
3. rejections create self-loops, which often help make the chain aperiodic.

Finally, if we know that the chain has stationary distribution p and is fully communicating and aperiodic, then if the chain is initialized at some arbitrary x_0 then the distribution of x_n is generally only approximately p for large n . In practice, one often discards an initial segment of the chain before using the remaining states. This initial discarded segment is called the burn-in period.

16.4 Examples of proposal kernels

For a finite cyclic state space

$$\mathcal{X} = \{0, 1, \dots, L-1\},$$

a natural proposal is the nearest-neighbor random walk

$$v(x' | x) = \begin{cases} 1/2, & x' = (x+1) \bmod L, \\ 1/2, & x' = (x-1) \bmod L. \end{cases}$$

This proposal is symmetric:

$$v(x' | x) = v(x | x').$$

Therefore the proposal terms cancel in the acceptance ratio, and

$$a(x \rightarrow x') = \min\left\{1, \frac{p(x')}{p(x)}\right\} = \min\left\{1, \frac{p_o(x')}{p_o(x)}\right\}.$$

By contrast, consider the one-way proposal

$$v(x' | x) = \begin{cases} 1/2, & x' = (x + 1) \bmod L, \\ 1/2, & x' = x. \end{cases}$$

This proposal moves only forward around the cycle or stays put. For some $x \neq x'$,

$$v(x' | x) > 0 \quad \text{but} \quad v(x | x') = 0.$$

For a positive target p , this gives

$$a(x \rightarrow x') = 0$$

on such one-way moves. Thus MH eliminates those moves entirely. This illustrates why proposal kernels should not contain one-way edges: detailed balance requires possible reverse movement. There is also a subtler issue. Suppose v is already reversible with stationary distribution p , but is periodic. Then the MH acceptance probability is 1 on every allowed move, so MH leaves the proposal chain unchanged. The resulting chain is still periodic. Thus having the correct stationary distribution is not enough; the chain also needs good convergence properties.

16.5 A rejection-sampling perspective

Metropolis–Hastings can be viewed as a local version of rejection sampling owing to the rejection probability, rather than one global constant. This explains both why MH preserves the target distribution and why it can be much more efficient than global rejection sampling.